HEALEY ALS Platform Trial

Regulatory Sponsor:	Merit Cudkowicz, MD, MSc Healey Center for ALS at Mass General
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PROTOCOL APPROVAL SIGNATURE PAGE

HEALEY ALS Platform Trial

The undersigned accept the content of this protocol in accordance with the appropriate regulations and agree to adhere to it throughout the execution of the study.

Merit Cudkowicz, MD, MSc Principal Investigator Director, Healey Center for ALS at Mass General	**Signature**	**Date**
Sabrina Paganoni, MD, PhD Co-Principal Investigator Healey Center for ALS at Mass General	**Signature**	**Date**
Marianne Kearney Chase Director of Research Operations, Healey Center for ALS at Mass General	**Signature**	**Date**

STATEMENT OF COMPLIANCE

This study will be conducted in compliance with the protocol, International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), Guideline for Good Clinical Practice (GCP), and applicable regulatory requirements, including United States Code of Federal Regulations (CFR) Title 45 CFR Part 46 and Title 21 CFR Parts 50, 56, and 312.

SIGNATURE PAGE

I have read the attached ALS Platform Trial protocol entitled, "HEALEY ALS **Platform Trial**" dated **12/15/2022** (Version **5.0**) and agree to abide by all described protocol procedures. I agree to comply with the ICH Guideline on GCP, applicable FDA regulations and guidelines, including those identified in Title 21 CFR Parts 11, 50, 54, and 312, single Institutional Review Board (IRB) guidelines and policies, and the Health Insurance Portability and Accountability Act (HIPAA).

By signing the protocol, I agree to keep all information provided in strict confidence and to request the same from my staff. Study documents will be stored appropriately to ensure their confidentiality. I will not disclose such information to others without authorization, except to the extent necessary to conduct the study.

Site Name: ______________________________

Site Investigator: ______________________________

Signed: ____________________ Date: ____________

LIST OF ABBREVIATIONS

ADR	Adverse Drug Reaction
AE	Adverse Event
ALS	Amyotrophic Lateral Sclerosis
ALSFRS-R	ALS Functional Rating Scale-Revised
ATE	Active Treatment Extension
CC	Coordination Center
CDC	Centers for Disease Control and Prevention
CFR	Code of Federal Regulations
CRF	Case Report Form
CSF	Cerebrospinal Fluid
C-SSRS	Columbia-Suicide Severity Rating Scale
DCC	Data Coordination Center
DSMB	Data and Safety Monitoring Board
DNA	Deoxyribonucleic Acid
eCRF	Electronic Case Report Form
ECG	Electrocardiogram
EDC	Electronic Data Capture
FVC	Forced Vital Capacity
FDA	Food and Drug Administration
GCP	Good Clinical Practice
GMP	Good Manufacturing Practices
HHD	Hand Held Dynamometry
HIPAA	Health Insurance Portability and Accountability Act
ICF	Informed Consent Form
ICH	International Council for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use
IND	Investigational New Drug Application
IRB	Institutional Review Board
IRT	Interactive Response Technology
ISF	Investigator Site File
ITT	Intent-to-Treat
LFTs	Liver Function Tests
LP	Lumbar Puncture
MGB	Mass General Brigham
MM	Medical Monitor
MoA	Mechanism of Action
MOP	Manual of Procedures
N	Number (typically refers to number of participants)

NCRI	Neurological Clinical Research Institute
NEALS	Northeast ALS Consortium
NeuroGUID	Neurological Global Unique Identifier
NfL	Neurofilament Light Chain
NIH	National Institutes of Health
OHRP	Office for Human Research Protections
PAV	Permanent Assisted Ventilation
PHI	Protected Health Information
PI	Principal Investigator
PK	Pharmacokinetic
RA	Reliance Agreement
RSA	Regimen-Specific Appendix
SAE	Serious Adverse Event
SDTs	Source Document Templates
SI	Site Investigator
sIRB	Single Institutional Review Board
SOA	Schedule of Activities
SOP	Standard Operating Procedure
SUSAR	Serious, Unexpected, Suspected Adverse Drug Reaction
SVC	Slow Vital Capacity
TEC	Therapy Evaluation Committee
TMF	Trial Master File
US	United States
VC	Vital Capacity
WOCBP	Women of Childbearing Potential

PROTOCOL SUMMARY

Study Title

HEALEY ALS Platform Trial

Study Indication

Amyotrophic Lateral Sclerosis (ALS)

Phase of Development

Phase 2/3

Rationale and Study Design

The HEALEY ALS Platform Trial is a perpetual multi-center, multi-regimen clinical trial evaluating the safety and efficacy of investigational products for the treatment of ALS.

The trial is designed as a perpetual platform trial. This means that there is a single Master Protocol dictating the conduct of the trial.

The Master Protocol describes the overall framework of the platform trial, including the target population, inclusion and exclusion criteria, regimen assignment and randomization schemes, study endpoints, schedule of assessments, trial design, the mechanism for adding and removing interventions, and the statistical methodology and recommended statistical methods for evaluating interventions.

Interventions (i.e., investigational products) are tested in trial regimens. Each trial regimen is described in its own Regimen-Specific Appendix (RSA) to the Master Protocol. The RSA will describe the nature of the intervention and its mechanism of action (MoA) including the mode and frequency of administration, dosage, the specific target population (to be selected within the pre-defined subsets of the Master Protocol), additional enrollment criteria (if any), sample size, and other specific intervention-related information and assessments (safety or other assessments that may be in addition to those outlined in the Master Protocol).

Allocation to Regimens

Participants that provide consent to the Master Protocol will be screened for Master Protocol-level inclusion and exclusion criteria. Those determined eligible will be randomly assigned with equal probability among all active regimens. This perpetual platform trial will continue enrolling research participants if there are regimens that are enrolling. As soon as pre-defined criteria for futility or success (if applicable) are met, or the target number of randomized participants in a regimen has been reached, enrollment will stop in that regimen.

Number of Planned Participants and Treatment Arms

The sample size enrolled in each regimen will be determined based on the specifics of the intervention, anticipated effect size, the expected variability in the enrolling population, and number of treatment arms (e.g., dosage within a regimen, if applicable). Those participants that meet the Master Protocol inclusion and exclusion criteria will be randomly assigned to a regimen. Within each regimen, participants will then be randomized a second time in a 3:1 ratio to active treatment or matching placebo.

Planned Number of Sites

Research participants will be enrolled from up to approximately 80 centers in the US.

Treatment Duration

Treatment duration of the placebo-controlled period is a maximum of 24-weeks for each regimen. The duration of the Active Treatment Extension (ATE) period will be described in the respective RSAs.

Post-treatment Follow-up Duration

Will be described in the RSAs.

Study Objectives and Endpoints

Default Primary Efficacy Objective:

To evaluate the efficacy of multiple investigational products, as compared to placebo, on ALS disease progression.

Secondary Efficacy Objective:

- To evaluate the effect of multiple investigational products on selected secondary measures of disease progression.

Safety Objective:

- To evaluate the safety of multiple investigational products for people with ALS.

Exploratory Efficacy Objective:

- To evaluate the effect of multiple investigational products on selected biomarkers and endpoints.

Default Primary Efficacy Endpoint:

Change from baseline through week 24 in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) total score and survival.

Secondary Efficacy Endpoints:

- Change from baseline through week 24 in respiratory function as assessed by slow vital capacity (SVC).
- Change from baseline through week 24 in muscle strength as measured isometrically using hand-held dynamometry (HHD) and grip strength.
- Survival.

Safety Endpoints:

- Treatment-emergent adverse and serious adverse events.
- Changes in laboratory values and treatment-emergent and clinically significant laboratory abnormalities.
- Changes in ECG parameters and treatment-emergent and clinically significant ECG abnormalities.
- Treatment-emergent suicidal ideation and suicidal behavior.

Exploratory Efficacy Endpoints:

- Changes in biofluid biomarkers.
- Changes in patient reported outcomes.

Investigational Products

Investigational products will be tested at different times (in parallel and sequentially) as described in this Master Protocol. This Master Protocol describes the common framework of the study. Each investigational product will have its own RSA to the protocol.

SCHEDULE OF ACTIVITIES

The Master Protocol sets out the minimum visit and assessment requirements that must be incorporated into each RSA. As per the Schedule of Activities (SOA) below, visits occur every 4 weeks and will be clinic-, phone-, or telemedicine-based, as applicable. Additional assessments may be added and will be described in each RSA. The Master Protocol allows a 24-week duration of placebo-controlled treatment for an intervention and an additional Active Treatment Extension (ATE) period, as described in each RSA.

Activity	**Master Protocol Screening[1]**	**Regimen Specific Screening[1]**	**Baseline**	**Week 2**	**Week 4[13]**	**Week 8[13]**	**Week 12**	**Week 16[13]**	**Week 20**	**Week 24**	**Follow-Up Safety Call**
	Clinic	Clinic	Clinic	Phone	Clinic	Clinic	Phone	Clinic	Phone	Clinic	**Phone**
	-42 to -1 Days	Refer to RSA	Day 0	Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day 168 ±7	Refer to RSA
Written Informed Consent[2]	X	X									
Inclusion/Exclusion Review	X	X[3]									
Regimen Specific Screening procedure(s)		X									
ALS & Medical History	X										
Demographics	X										
Physical Examination	X										
Neurological Exam	X										
Vital Signs[4]	X		X		X	X		X		X	
Slow Vital Capacity	X[12]		X			X		X		X	
Muscle Strength Assessment			X			X		X		X	
ALSFRS-R	X		X		X	X	X	X	X	X	
Patient Reported Outcomes[10]			X			X		X		X	
12-Lead ECG	X									X	
Clinical Safety Labs[5]	X		X		X	X		X		X	
Concomitant Medication Review	X	X	X		X	X	X	X	X	X	X
Adverse Event Review[6]	X	X	X	X	X	X	X	X	X	X	X

Activity	Master Protocol Screening[1]	Regimen Specific Screening[1]	Baseline	Week 2	Week 4[13]	Week 8[13]	Week 12	Week 16[13]	Week 20	Week 24	Follow-Up Safety Call
	Clinic	Clinic	Clinic	Phone	Clinic	Clinic	Phone	Clinic	Phone	Clinic	**Phone**
	-42 to -1 Days	Refer to RSA	Day 0	Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day 168 ±7	Refer to RSA
Columbia-Suicide Severity Rating Scale			X		X	X		X		X	
Biomarker Blood Collection			X			X		X		X	
Biomarker Urine Collection			X			X		X		X	
DNA Collection[7] (optional)			X								
CSF Collection[11]			X					X			
Assignment to a regimen	X										
Randomization within a regimen		X									
Administer/Dispense Study Drug			X[8]			X		X			
Study Drug Accountability/Compliance				X[14]	X	X	X[14]	X	X[14]	X	
Exit Questionnaire										X	
Vital Status Determination										X[9]	

[1] Master Protocol Screening procedures must be completed within 42 days to 1 day prior to the Baseline Visit. Refer to RSA for details about RSA Specific Screening procedures, if any, should be performed.
[2] During the Master Protocol Screening Visit, participants will be consented via the Master Protocol informed consent form (ICF). After a participant is assigned to a regimen, participants will be consented a second time via the regimen-specific ICF.
[3] At the Regimen Specific Screening Visit, participants will have regimen-specific inclusion and exclusion criteria assessed, if applicable.
[4] Vital signs include weight, systolic and diastolic pressure, respiratory rate, heart rate and temperature. Height in cm measured at Master Protocol Screening Visit only.
[5] Clinical safety labs include hematology (CBC with differential), complete chemistry panel, liver function tests, thyroid function and urinalysis. Coagulation panel will occur at the Master Protocol Screening and Week 16 Visits or as in the RSA. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during the study. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.
[6] Adverse events that occur after signing Master Protocol consent form will be recorded.
[7] The DNA sample can be collected after the Baseline Visit if a baseline sample is not obtained or the sample is not usable.
[8] Administer first dose of study drug only after Baseline Visit procedures are completed
[9] Vital status, defined as a determination of date of death or death equivalent or date last known alive, will be determined for each randomized participant at the end of the placebo-controlled period of their follow-up (generally the Week 24 Visit, as indicated). If at that time the participant is alive, his or her vital status should be determined again at the time of the last participant's last visit (LPLV) of the placebo-controlled period of a given regimen. We may also ascertain vital status at later time points by using publicly available data sources as described in section 8.15.b

[10] Each RSA will detail which PRO is collected at each Visit. Specific PRO collection may differ from this Schedule of Activities.
[11] CSF collection may differ from the Schedule of Activities, and will be performed as described in each RSA.
[12] If required due to pandemic-related or other restrictions, Forced Vital Capacity (FVC) performed by a Pulmonary Function Laboratory evaluator may be used for eligibility (Master Protocol Screening ONLY).
[13] Visit may be conducted via phone or telemedicine with remote services instead of in-person if needed to protect the safety of the participant due to a pandemic, disability or other medical reason.
[14] Drug accountability will not be done at phone visits. A drug compliance check in will be conducted during phone visits to ensure participant is taking drug per dose regimen and to note any report of missed doses.

STUDY WORKFLOW

Figure 1. Example Master Protocol Schema

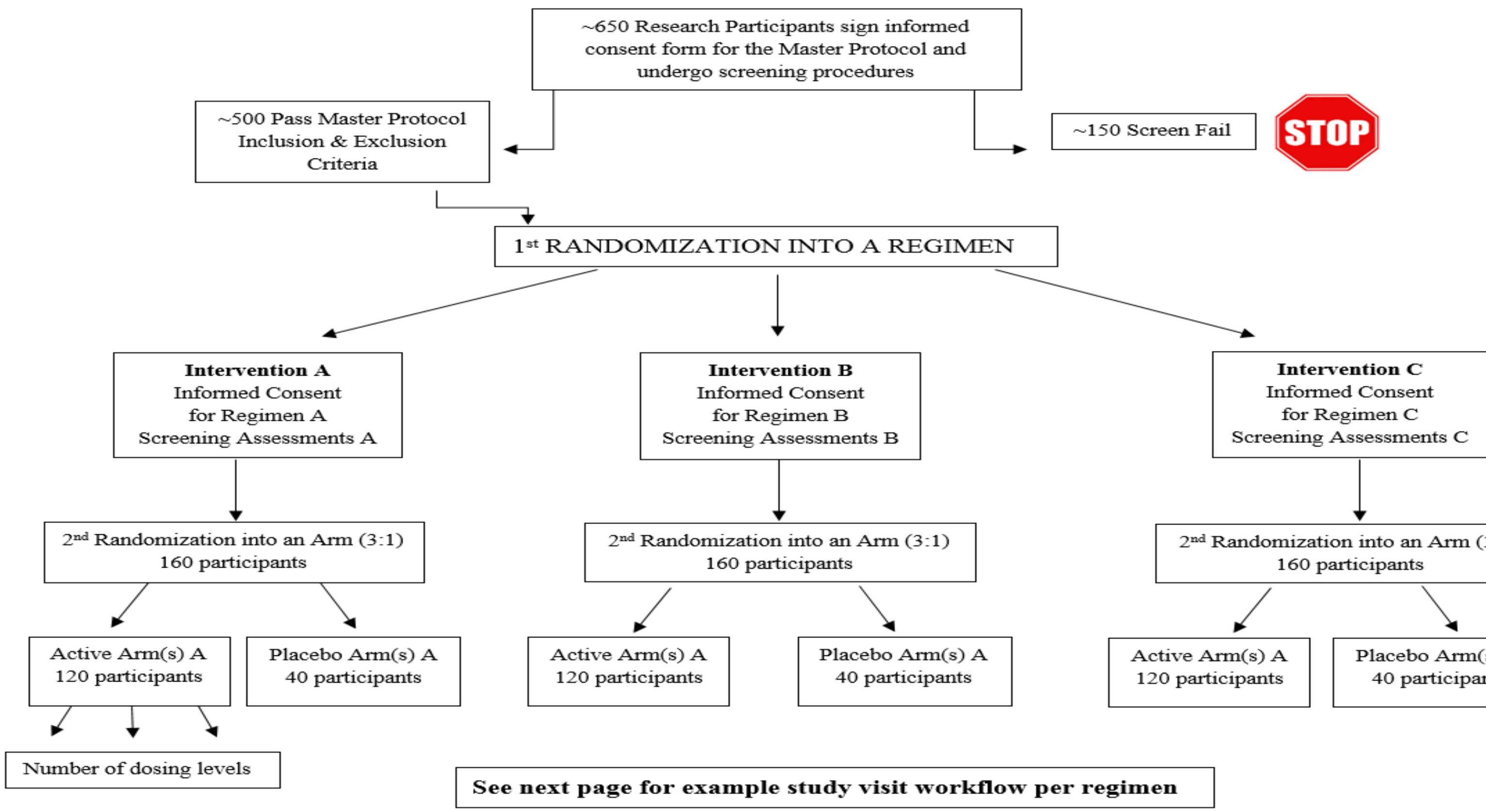

Figure 1 represents an example of the study schema assuming three concurrently enrolling regimens. The Master Protocol schema will change depending on the number of regimens that are enrolling at any given time. Intervention A investigates an active treatment with multiple dosing levels. In this example, regimen sample size is 160 participants, actual regimen sample size will be specified in each RSA.

Figure 2. Example Study Visit Workflow per Regimen

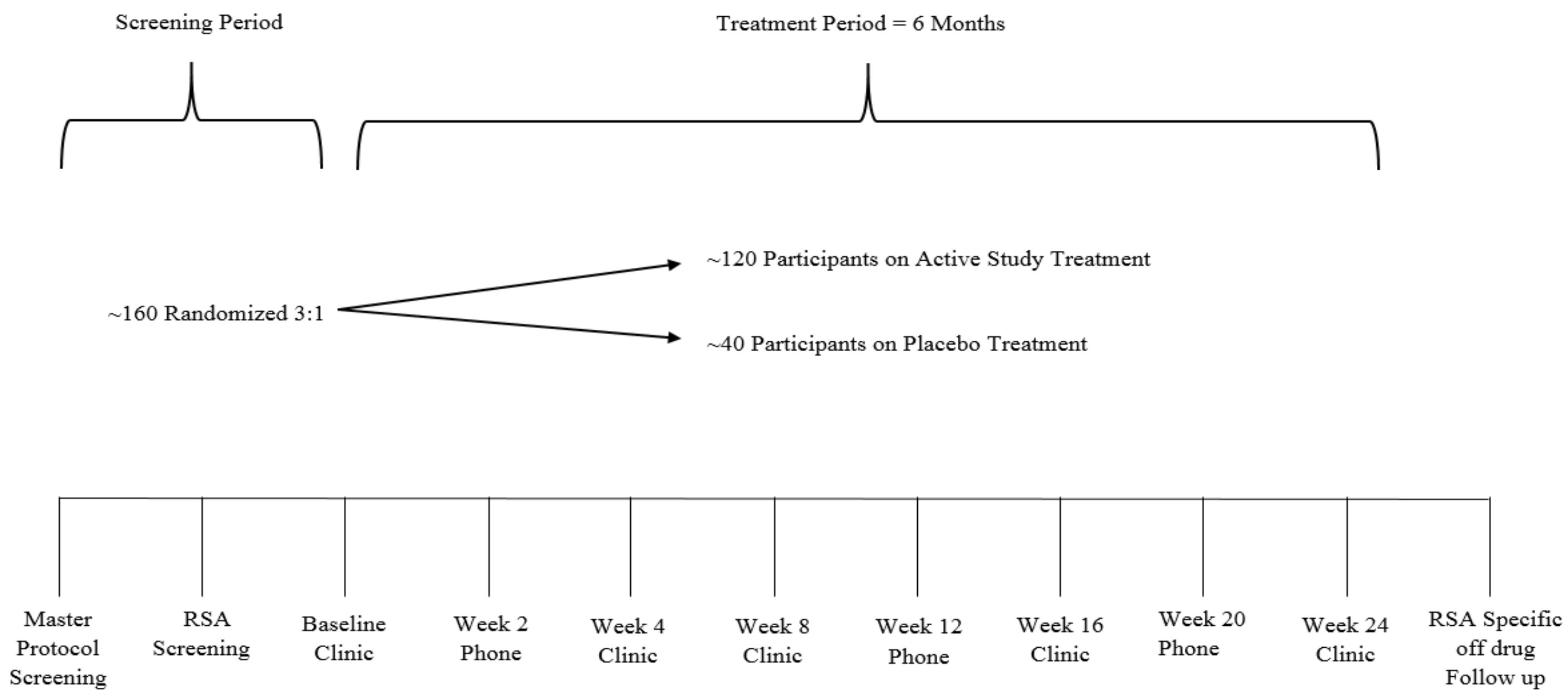

Figure 2 represents an example of the study visits that will occur over 24 weeks for all participants that are assigned to a regimen. In this example, regimen sample size is 160 participants, actual regimen sample size will be specified in each RSA.

1 ETHICS/PROTECTION OF HUMAN PARTICIPANTS

1.1 Institutional Review Board

This Master Protocol will be conducted in compliance with current GCP and Title 21 Part 56 of the United States of America CFR relating to IRBs. The Mass General Brigham (MGB) IRBs located in Boston, MA, collectively known as the MGB Human Research System (MGB HRS), have been selected by the Northeast ALS Consortium (NEALS) to serve as the single IRB (sIRB) for the Master Protocol. NEALS is a non-profit group of researcher institutions, each of which is a NEALS Member, who collaboratively conduct clinical research in Amyotrophic Lateral Sclerosis and other motor neuron diseases. Sites participating in this Master Protocol must have an executed sIRB Authorization Agreement (Reliance Agreement or RA) to rely on the MGB HRS to participate.

1.2 Ethical Conduct of Study

The study will be conducted in accordance with GCP defined by the ICH and the ethical principles of the Declaration of Helsinki.

The study will be conducted in compliance with the protocol. The protocol and any amendments as well as the participant informed consent will receive sIRB approval prior to initiation of the study.

Study personnel involved in conducting this study will be qualified to perform their respective task(s) as confirmed by the site and collection of required documentation.

This study will not use the services of study personnel where sanctions have been invoked or where there has been scientific misconduct or fraud (e.g., loss of medical licensure, debarment).

1.3 Participant Information and Consent

This study will be conducted in compliance with Title 21 Part 50 of the United States of America CFR, Federal Regulations and ICH Guidance Documents pertaining to informed consent. At the first visit, prior to initiation of any study-related procedures not performed as clinic visit standard of care (e.g., vital signs, weight), participants will be informed about the nature and purpose of the Master Protocol, participation/termination conditions, and risks and benefits. Participants will be given adequate time to ask questions and become familiar with the Master Protocol prior to providing consent to participate. Participants will give their documented informed consent to participate in the Master Protocol and will be provided with a copy of the fully executed Master Protocol consent form for their records.

Participants meeting Master Protocol eligibility criteria will be randomly assigned to a regimen.

Participants will then be informed about the intervention specific to their assigned regimen and any participation/termination conditions or risks and benefits specific to that regimen. Participants will give their written consent to participate in their assigned regimen and will be provided with a copy of the fully executed regimen-specific consent form for their records. Participants meeting additional eligibility criteria required by their assigned regimen, if any, will be randomized in a 3:1 ratio to receive either active treatment or matching placebo in that regimen.

In some situations, an individual may be re-assigned to multiple different regimens, if eligible. The procedures detailing the re-assignment process can be found in Section 8.1.2 Screen Failures and Re-Assignment.

1.4 Changes in Conduct of the Study

1.4.1 Protocol Amendments

Any change to the Master Protocol will be documented in a protocol amendment, issued by the Sponsor. Master Protocol amendments will be submitted for approval to the sIRB prior to implementation. Written informed re-consent for continued participation in the study may be required by participants already enrolled in the Master Protocol.

As regimens are added to the Platform Trial, they will be submitted for approval to the sIRB prior to implementation. Each regimen will be added as an appendix to the Master Protocol prior to any participant being assigned to that regimen. Addition of regimens will not require re-consent. Regimens that are stopped for futility or ended early due to success (if applicable) will not result in an amendment to the Master Protocol.

1.4.2 Premature Termination of Study Sites

The Sponsor reserves the right to terminate the participation of individual study sites. Conditions that may warrant termination include, but are not limited to, insufficient adherence to protocol requirements and failure to enter participants at an acceptable rate.

1.5 Protocol Adherence

Each Site Investigator (SI) must adhere to the Master Protocol detailed in this document and agree that any changes to the protocol must be approved by the sIRB. Each SI will be responsible for enrolling into the Master Protocol only those study participants who have met all Master Protocol eligibility criteria, and for enrolling into a Regimen only those study participants who have met the additional eligibility criteria (if any) of the corresponding RSA.

2 INTRODUCTION

2.1 Background Information and Rationale

ALS is a progressive, fatal neurodegenerative disease. ALS is characterized by motor neuron loss resulting in muscle weakness and atrophy, disability, and eventually death from failure of the ventilatory muscles[1]. The median age of onset is 55 years and average survival is 3-5 years after onset of first symptoms[2]. While the incidence of ALS is comparable to that of multiple sclerosis (approximately 2/100,000), its prevalence is much lower because of its rapid progression (about 5/100,000)[3]. The only FDA-approved disease modifying medications, riluzole, edaravone, and Relyvrio confer only a modest survival benefit[4].

While therapies for ALS remain limited, basic and translational ALS research has resulted in numerous influential discoveries in recent years, including breakthroughs in genetics and progress in our understanding of disease mechanisms, therapeutic targets, and biomarkers. These discoveries have led to a large pipeline of potential therapies that await testing in clinical trials, suggesting that we are at a time of great opportunity to translate advances in the understanding of ALS into meaningful treatments for people.

Given the poor prognosis and dearth of effective treatments, clinical trials are of primary importance for people with ALS, their families, their providers, and the entire ALS clinical and scientific community[5]. Clinical trials, however, can be complex, time-consuming, and expensive. <u>Challenges to ALS clinical trials are both operational and scientific,</u> and innovative clinical trials are needed to accelerate the path to effective treatments[6-9].

Operational challenges include inefficiencies in institutional approvals and trial management that may lead to delays in study start-up, slow execution times, and sub-optimal recruitment and retention[10]. Barriers to trial access are so substantial that less than 10% of the ALS population is estimated to participate in clinical research[11, 12]. Fortunately, experience in selected disease-specific networks and federally-funded trial networks demonstrates that logistical trial challenges are surmountable with an infrastructure designed for efficiency and collaboration. In fact, clinical trial networks can be very productive if built with appropriate infrastructure, leadership, and cooperation. Several of these networks leverage efficient infrastructure models, such as the use of a single IRB, Master Clinical Trial Site Agreements, and centralized recruitment and retention strategies[13].

Scientific challenges to ALS therapy development include disease rarity, patient heterogeneity, lack of validated surrogate endpoints, and a relatively rapid disease course. To overcome some of these barriers, the ALS trial community has stepped up biomarker discovery efforts and devoted substantial resources to the development of novel outcome measures for use in early phase clinical trials[14, 15]. Routine incorporation of these novel endpoints into clinical trials is a critical step forward and holds the promise to improve the chances of success of the drug development pipeline. Yet, progress is hindered by the lack of coordination among different biomarker discovery and outcome measure development efforts[15]. Several previous and current research projects target different patient populations and analyze novel biomarkers/outcome measures using different assays and procedures, and also include variable schedules of assessments. A major, coordinated collaboration that includes academia, industry, and patient-advocacy organizations is needed to innovate the ALS clinical trial landscape[6-9].

2.2 Platform Trials: an efficient strategy to accelerate drug development and scientific discovery

Traditionally, clinical trials are designed by a single sponsor to answer only one question: does a single investigational product work? These trials include a pre-specified number of treatment arms (generally limited to two or three arms) and have a finite duration based on the time required to answer the trial question. Moreover, each trial requires an expensive, ad hoc trial infrastructure that is dismantled at the end of the trial.

In contrast, platform trials are designed to investigate multiple investigational products in parallel and sequentially with the capability to adapt over time. Platform trials do not have a pre-specified end date: the platform remains open long-term and is available to evaluate new investigational products as they become available[16]. The trial infrastructure is built at the beginning and is shared across different treatments, leading to operational efficiencies. Investigational products are compared to a shared placebo group. Sharing of placebo participants is possible because all investigational products are evaluated using a common Master Protocol[17]. The Master Protocol describes the framework of the study, including the trial population, inclusion and exclusion criteria, regimen assignment and randomization schemes, a schedule of assessments, primary, secondary and exploratory outcomes, statistical methodology, and planned analyses that are common for all investigational products to be tested.

Platform trials are an ideal setting to advance the scientific understanding of the disease and novel endpoints with a coordinated strategy. Thanks to its coordinated structure and common data and sample acquisition processes, the platform trial can serve as a natural history registry and bio-repository from placebo participants[17]. With the testing of multiple investigational products using a common protocol and uniform data and sample acquisition processes, the platform is designed to answer multiple scientific questions and to serve as a source of data that can be used to enhance the design of other research projects. Further, accumulated learnings about endpoint behavior can be leveraged to adapt the Master Protocol so that future investigational products will be studied using more efficient biomarkers, outcome measures, and analyses[17, 18].

2.3 Investigational Product Profile

In this trial, multiple investigational products for ALS will be tested simultaneously or sequentially. These investigational products will be provided by different partners, either pharmaceutical companies or academic groups.

This trial is designed as a perpetual platform trial. This means that there is a single Master Protocol dictating the conduct of the trial. The additional details that govern the testing of each investigational product will be summarized in separate RSAs.

During the trial, a regimen may be discontinued early due to futility or safety concerns. Additionally, a regimen may be added to the trial based on the ability of the trial to accommodate their incorporation.

Selection of interventions will be done by the Therapy Evaluation Committee (TEC) based on evidence supporting the intended mechanism of action, target engagement, previous pre-clinical data and Phase I or other clinical data (including safety data available for each compound), and compatibility with the Master Protocol.

Each RSA that is part of the HEALEY ALS Platform Trial will describe:

1. Specific additional inclusion or exclusion criteria for that investigational product, if any. The RSA will include a summary of the product characteristics, including mechanism of action and pharmacology background, drug profile including a summary of available preclinical and clinical data including pharmacokinetic (PK)/pharmacodynamic, target of action, biomarker data, drug-drug interaction, safety data, prohibited medications (as applicable) and benefit-risk assessment. For a more detailed description of the investigational product profile, the RSA will refer to the current Investigator's Brochure.

2. Specific biomarker assessments not performed as part of the Master Protocol, if any, which are specific to the intervention and proposed mechanism of action.

3. Specific safety assessments not performed as part of the Master Protocol, if any, which are specific to known toxicology or potential safety concerns associated with the intervention.

4. Specific information on study methodology, such as the primary endpoint, maximum sample size, number of arms (e.g., to accommodate multiple dosages), and details for interim analyses for early futility and early success (if applicable).

3 Objectives and Endpoints

OBJECTIVES	ENDPOINTS	JUSTIFICATION FOR ENDPOINTS
Primary		
To evaluate the efficacy of multiple investigational products on ALS disease progression.	Change in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) total score and survival	*The ALSFRS-R measures function in daily activities and is an established scale for monitoring disease progression in ALS.*
Secondary		
To test the effect of multiple investigational products on selected secondary measures of disease progression.	• Change in respiratory function as assessed by slow vital capacity (SVC). • Change in muscle strength as measured isometrically using hand-held dynamometry (HHD) and grip strength. • Survival.	*Decline in respiratory function is a direct result of the known pathophysiology of the ALS and demonstration of a treatment benefit on respiratory endpoints may also provide evidence of effectiveness.* *Loss of strength is a hallmark of disease progression in ALS and meaningful differences in muscle strength should be supportive of an effect on measures of function in activities of daily living.*
Safety		
	• Treatment-emergent adverse and serious adverse events.	*Toxicities associated with investigational products may*

OBJECTIVES	ENDPOINTS	JUSTIFICATION FOR ENDPOINTS
To evaluate the safety of multiple investigational products for people with ALS.	• Changes in laboratory values and treatment-emergent and clinically significant laboratory abnormalities. • Changes in ECG parameters and treatment-emergent and clinically significant ECG abnormalities. • Treatment-emergent suicidal ideation and suicidal behavior.	*manifest as clinical signs and symptoms, new diagnoses, laboratory or cardiac changes and abnormalities, or suicidality.*
Exploratory		
To test the effect of multiple investigational products on selected biomarkers and endpoints.	• Changes in biofluid biomarkers. • Changes in patient reported outcomes.	*These endpoints have been chosen to provide greater understanding of ALS and may provide opportunities for identification of surrogate endpoints that are reasonably likely to predict clinical benefit and that might serve as a basis for accelerated approval in future trials.*

4 STUDY DESIGN

4.1 Overall Study Design and Plan

This is a perpetual, multicenter, multi-regimen, randomized, placebo-controlled, adaptive platform clinical trial evaluating the safety and efficacy of multiple investigational products simultaneously or sequentially in ALS.

There will be multiple interventional regimens, each consisting of the research participants receiving either the active treatment or its matching placebo. Research participants, investigators and site staff will not be blinded to the regimen assignment, but they will be blinded to active product or matching placebo assignment and this blind will be maintained throughout the ATE period of the study. Regimens may start at different time points during the trial.

This Master Protocol describes the framework for the trial design population and the minimum inclusion and exclusion criteria for all regimens. Each RSA will contain intervention-specific information and will define any additional trial elements and regimen-specific eligibility criteria that may exist.

The number of research participants, the duration of treatment for each intervention, and the follow-up period will all be described in the RSA. The primary endpoint will be assessed for all interventions every 4 weeks, either on-site or via phone.

Interim analyses for regimens will begin when at least one regimen has 40 randomized participants within that regimen who have had the opportunity to complete at least 24 weeks of follow-up. Interim analyses will continue to occur simultaneously for all actively enrolling regimens every 12 weeks. A regimen is eligible to stop early for futility (all regimens) or for success (for applicable regimens only) at any interim analysis once there are 40 randomized participants within that regimen who have had the opportunity to complete at least 24 weeks of follow-up. The default design for a regimen will include stopping early only for futility.

At each interim analysis, a set of possible changes or adaptions that an intervention is eligible for will be checked.

- An intervention can be found to have failed to favorably modify the progression of disease and be deemed futile or unsafe. Upon meeting the criterion for futility, enrollment in that regimen would be discontinued and end of study procedures would be initiated. Research participants in that regimen may be re-screened for entry into the Master Protocol after the mandatory wash-out period, if eligible.

- Accrual and maximum exposure of enrolled participants may be reached for an intervention. In this case, this is the end of that regimen and all research participants in that regimen would have completed follow-up through week 24 of the placebo-controlled period. Research participants will continue in the ATE period or may be re-screened for entry into the Master Protocol after the mandatory wash-out period, if eligible.

- If a regimen includes more than one active arm to allow for multiple dosing levels or other variations in treatment, then rules specifying the allocation to these arms can be included. This will be detailed in the RSA.

5 STUDY POPULATION

5.1 Number of Research Participants

Research participants that meet the Master Protocol inclusion and exclusion criteria listed below will be further screened and randomly assigned to a regimen.

The sample size enrolled in each regimen will be determined based on the specifics of the intervention, anticipated effect size, and number of treatment arms evaluated within that regimen.

5.2 Master Protocol Inclusion and Exclusion Criteria

5.2.1 Master Protocol Inclusion Criteria

These are the inclusion criteria that research participants must meet to be eligible to enter the Master Protocol and are common for all regimens:

1. Sporadic or familial ALS diagnosed as clinically possible, probable, lab-supported probable, or definite ALS defined by revised El Escorial criteria (Appendix I)**.**
2. Age 18 years or older.
3. Capable of providing informed consent and complying with study procedures, in the SI's opinion.
4. Time since onset of weakness due to ALS ≤ 36 months at the time of the Master Protocol Screening Visit.
5. Vital Capacity ≥ 50% of predicted capacity at the time of the Master Protocol Screening Visit measured by Slow Vital Capacity (SVC), or, if required due to pandemic-related restrictions, Forced Vital Capacity (FVC) measured in person.
6. Participants must either not take riluzole or be on a stable dose of riluzole for ≥ 30 days prior to the Master Protocol Screening Visit.
7. Participants must either not take edaravone or have completed at least one cycle (typically 14 days) of edaravone prior to the Master Protocol Screening Visit.
8. Participants must have the ability to swallow pills and liquids at the time of the Master Protocol Screening Visit and, in the SI's opinion, have the ability to swallow for the duration of the study.
9. Geographically accessible to the site.
10. Participants must either not take Relyvrio/ Albrioza or have started Relyvrio/ Albrioza ≥ 30 days prior to the Master Protocol Screening Visit.

5.2.2 Master Protocol Exclusion Criteria

These following exclusion criteria are common for all regimens:

1. Clinically significant unstable medical condition (other than ALS) that would pose a risk to the participant, according to SI's judgment (e.g., cardiovascular instability, systemic infection, untreated thyroid dysfunction), or clinically significant laboratory abnormality or EKG changes.

   Clinically significant abnormal liver or kidney function is exclusionary. The following values [alanine aminotransferase (ALT) or aspartate aminotransferase (AST) > 3 times the upper limit of normal (ULN) or estimated Glomerular Filtration Rate (eGFR) < 30 mL/min/1.73m^2] are exclusionary regardless of clinical symptoms.

2. Presence of unstable psychiatric disease, cognitive impairment, dementia or substance abuse that would impair ability of the participant to provide informed consent, in the SI's opinion.
3. Active cancer or history of cancer, except for the following: basal cell carcinoma or successfully treated squamous cell carcinoma of the skin, cervical carcinoma in situ, prostatic carcinoma in situ, or other malignancies curatively treated and with no evidence of disease recurrence for at least 3 years.
4. Use of investigational treatments for ALS (off-label use or active participation in a clinical trial) within 5 half-lives (if known) or 30 days (whichever is longer) prior to the Master Protocol Screening Visit. (*Please refer to the Manual of Procedures (MOP) for current list of experimental therapies*)
5. Exposure at any time to any gene therapies under investigation for the treatment of ALS (off-label use or investigational).
6. If female, breastfeeding, known to be pregnant, planning to become pregnant during the study, or of child-bearing potential and unwilling to use effective contraception, for the duration of the trial and for 3 months, or as specified in each RSA, after discontinuing study treatment.
7. If male of reproductive capacity, unwilling to use effective contraception for the duration of the trial and for 3 months, or as specified in each RSA, after discontinuing study treatment.
8. Anything that would place the participant at increased risk or preclude the participant's full compliance with or completion of the study, in the SI's opinion.
9. If a participant is being re-screened, the disqualifying condition has not been resolved, or the mandatory wash-out duration has not occurred.

If justified based on regimen's specific investigational product, additional inclusion or exclusion criteria (e.g., contraindications, laboratory parameters) will be listed in the RSA.

Riluzole. Participants taking concomitant riluzole at study entry must be on a stable dose for 30 days prior to the Master Protocol Screening Visit and must intend to continue taking the same stable dosage throughout the study.

Edaravone. Participants taking concomitant edaravone at study entry must have completed at least one cycle of treatment (typically 14 days) prior to the Master Protocol Screening Visit and must intend to continue taking the same stable dosage throughout the study.

Relyvrio/ Albrioza. Participants taking concomitant Relyvrio/ Albrioza at study entry must have started ≥ 30 days prior to the Master Protocol Screening Visit

If the SI determines that riluzole, edaravone or Relyvrio/ Albrioza should be discontinued or dose-adjusted for medical reasons, this will not be considered a protocol deviation. If extenuating circumstances cause doses to be missed for reasons not intended or medically indicated (scheduling, drug availability, etc.), this will not be considered a protocol deviation. If riluzole, edaravone or Relyvrio/ Albrioza is stopped for any non-medical reason (participant preference), this will be recorded as a minor protocol deviation. If initiation of riluzole, edaravone or Relyvrio/ Albrioza therapy begins at any point during participation, this will be recorded as a minor protocol deviation

***Date of ALS Symptom Onset*.** For the purposes of this study, the date of symptom onset will be defined as the date the participant first had symptoms of muscle weakness. To be eligible for this study, the date of symptom onset must be no greater than 36 months prior to the Master Protocol Screening Visit date.

6 PARTICIPANT SELECTION AND ENROLLMENT

6.1 Identifying Participants

This study will be conducted at selected ALS centers in the US. Sites participating in this Master Protocol must be members of the Northeast ALS Consortium (NEALS). NEALS is a non-profit group of research institutions, each of which is a NEALS Member, who collaboratively conduct clinical research in ALS and other motor neuron diseases. Research participants will be recruited from these centers, after approval by the sIRB.

Information about all research participants enrolled into the trial will be recorded in the Electronic Data Capture (EDC) system. This information will include reasons for screen failure for participants not assigned to a regimen, as well as a log of all participants randomized within a regimen, irrespective of whether they have been treated with the intervention or not.

6.2 Consenting Participants

Written informed consent will be obtained from participants before any study procedures or assessments, not performed as clinic visit standard of care (e.g., vital signs, weight), are done and after the study objectives and endpoints, methods, anticipated benefits, and potential hazards are explained. The willingness of the participant to participate in the study will be documented in writing on a consent form approved by the sIRB, which will be signed and dated by the participant. The SI will keep the original consent forms and a copy will be given to the participant. The participant will also be informed that they may refuse entry into the trial and are free to withdraw from the trial at any time without prejudice to future treatment.

Informed consent will be obtained by the site investigator or other licensed staff who are trained on the study and authorized to obtain consent. The SI, or other licensed staff, will be involved in explaining the study and will play a role during the consent process, answering any questions in conjunction with the coordinator and ensuring that the process is carried out correctly. All participants will be offered the opportunity to discuss participation with a licensed physician Investigator and the participant’s decision to accept or decline this opportunity will be documented in the research file. Site study staff performing consent must provide participants adequate information to allow for an informed decision about participation, facilitate the potential participant's understanding of the information, and provide ample time and opportunity to inquire about details of the trial. The SI, nor the site study staff, should coerce or unduly influence a potential participant to participate or to continue to participate in the study. Site study staff must continuously provide information as the clinical investigation progresses or as the participant or situation requires.

Informed consent will be conducted for the Master Protocol and subsequently for the applicable RSA to which that participant was assigned. Those participants who are not found eligible for the Master Protocol will not be consented for any RSA.

Participants may be required to reconsent to the Master Protocol if new procedures or information is added in the future. Should a participant need to reconsent, this should occur during the participant's next in-person visit. If the participant's next in-person visit is conducted remotely, reconsent may also be completed remotely using the following procedures:

1. The site staff provides a copy of the informed consent form to the participant.
2. The participant reads through the consent form but does not sign.
3. The Site Investigator, or other study staff member approved and delegated to obtain informed consent, contacts the participant and reviews the informed consent form with the participant over the phone or via telemedicine.
4. The participant signs the informed consent form and provides the original signed consent form back to the site.

Once received at the site, the individual who consented the participant via phone or telemedicine signs the informed consent form.

6.3 Ineligible Participants

All sites will be required to collect demographic information including age, gender, race, and ethnicity, and reasons for ineligibility for the Master Protocol for participants that have signed the Master Protocol consent form and are deemed ineligible for the Master Protocol.
Once assigned to a regimen, participant eligibility will be confirmed using regimen-specific inclusion and exclusion criteria, if any, during the Regimen-Specific Screening Visit. Reasons for ineligibility for the regimen must be documented.

6.4 Randomization

6.4.1 Randomization Procedures

Randomization will be conducted using an interactive response technology (IRT) system. The randomization for the Master Protocol consists of a tiered randomization as described below and depicted in the Study Workflow.

1. Randomization equally to a regimen among all available regimens to which the participant has not previously been randomized.
2. Randomization to placebo or an active intervention in the assigned regimen, and randomization to the set of active sub-arms, if applicable.

In the first stage, after a participant is determined eligible according to the Master Protocol inclusion and exclusion criteria, a research participant will be randomized equally among all available regimens to which the participant has not previously been randomized. In the second stage, if a participant has been determined eligible by meeting all inclusion and no exclusion criteria specific to their assigned regimen, the research participant will be randomized 3:1 to the active intervention or the placebo for the regimen. If the regimen has multiple arms, a participant will also be randomized according to the allocation scheme for that regimen. The allocation scheme for regimens with multiple arms will be detailed in the respective RSA.

6.4.2 Treatment Allocation

Treatment group within a regimen will be assigned via the IRT system after a participant has been determined eligible for any inclusion and exclusion criteria specific to their assigned regimen.

6.4.3 Blinding

In this trial, investigational products may have a different mode (e.g., oral, subcutaneous) or frequency of administration and may be introduced at different time points. Each intervention arm will have its own concurrent, randomized placebo control arm.

Research participants, SIs, and everyone involved in the conduct of a given trial regimen will not be blinded to regimen assignments but will be blinded to the randomized treatment assignments within a regimen, and this blind will be maintained through the ATE period of the study, as outlined in the HEALEY ALS Platform Trial Bliniding and Unblinding Plan. In the event of a medical emergency that necessitates the unblinding of the Medical Monitor (MM) or a SI to safely provide care for a participant, emergency unblinding of that single participant may be undertaken as outlined below.

6.4.4 Emergency Unblinding Plan

Emergency unblinding for a research participant will only be undertaken when it is essential to treat the participant safely. It must only be used in an emergency when the identity of the treatment arm must be known to the SI to provide appropriate medical treatment or otherwise ensure the safety of research participants or others exposed to investigational products. In most cases, investigational product discontinuation and knowledge of the possible treatment assignments will be sufficient to treat a study participant who presents with an emergency condition. However, if unblinding is necessary, the blind may be broken only for that participant.

6.5 Discontinuation of Treatment and Terminations

6.5.1 Discontinuation of Investigational Product

If the SI or designee is concerned about the use of a prohibited medication or other safety issues (including lack of safety data collected), then the investigational product may need to be discontinued.

If the investigational product for an RSA is administered orally, and a participant loses the ability to swallow the investigational product, the possibility of alternative modes of administration (e.g., via G-tube) will be detailed in the RSA.

A research participant may choose to discontinue the investigational product at any time for any reason. However, the SI or designee will encourage the research participant to follow the study protocol under the intent-to-treat principle (ITT). These research participants will be encouraged to follow the study visits, off treatment, up to the week 24 visit and through the ATE period of the study, following the Schedule of Activities. At a minimum, collection of the ALSFRS-R, AEs, Concomitant Medications, CSSRS, and other outcome measures should be encouraged. Loss to follow-up should be prevented whenever possible. For participants followed under the ITT principle, an in-person Early Termination Visit will be completed after they discontinue follow up under the ITT principle. A Follow-Up Safety Call at the time study drug is discontinued may not be required, as long as the participant is followed under the ITT principle for a minimum of 28 days.

Upon discontinuation of investigational product, the research participant should return any unused investigational product.

For all research participants, the reason for permanent discontinuation of investigational product must be recorded in the Case Report Form (CRF). These data will be included in the trial database and reported.

6.5.2 Termination of Individual Research Participants

A SI or designee may terminate a research participant if a medical condition or other situation occurs such that continued participation would not be in the best interest of the participant.

A research participant may withdraw consent for study participation. No justification is required for the decision.

Upon termination, the research participant should return any unused investigational product.

6.5.3 Termination of the Master Protocol or a Regimen by the Investigational New Drug (IND) Holder/ Sponsor

The IND-holder/Sponsor reserves the right to terminate the overall Master Protocol or any individual regimen at any time or during an interim analysis.

If there are regimen-specific criteria for terminating a regimen, they will be detailed in the RSA.

7 INVESTIGATIONAL PRODUCT AND PLACEBO

7.1 Investigational Product

Multiple investigational products (i.e., interventions, or active agents, from different regimen partners) will be tested in this Platform Trial.

Each investigational product will have an RSA in which the complete description of the tested product can be found. Each intervention may have multiple arms, such as different dosages or frequencies or routes of administration.

Each active agent will have a matching placebo. All regimens will be compliant with the Master Protocol, which outlines the majority of all clinical, biomarker, and safety assessments, making a shared placebo group both desirable and feasible.

7.1.1 Investigational Product Manufacturer

Details identifying the investigational product manufacturer will be included in the corresponding RSA.

7.1.2 Labeling and Packaging

Investigational product for each regimen will be provided by the regimen partner and will be described in the corresponding RSA. Packaging and labeling will follow Good Manufacturing Practices (GMP) regulations.

Samples of labels will be kept with the Trial Master File (TMF).

The participants will be instructed to return all remaining investigational product including empty package material, if applicable, to the study site. Drug accountability will be performed as described in each RSA.

Details for packaging, labeling, and re-supply will be described in the corresponding RSA.

7.1.3 Acquisition and Storage

Investigational product for all regimens will be received at the study site by designated study staff, handled and stored safely and properly at the site pharmacy or other designated location, and kept in a secure location to which only the trial pharmacist and designated pharmacy staff, SI, and clinical staff have access. Upon receipt, the investigational product will be stored according to the instructions specified on the labels. Storage conditions will be adequately monitored and temperature in the area in which the investigational product is stored will be controlled, monitored and recorded, at a minimum daily.

In accordance with local regulatory requirements, the SI, designated site staff, or head of the medical institution (where applicable) at each site must document the amount of investigational product dispensed and/or administered to study participants, the amount received from the Central Pharmacy, and the amount destroyed upon completion of the study. An investigator is responsible for ensuring product accountability records are maintained throughout the course of the study. The research pharmacist or designated study staff will be responsible for maintaining an accurate record of the shipment and dispensing of investigational product in a drug accountability log.

Additional investigational product-specific details will be provided in the RSA.

7.1.4 Destruction of Investigational Product

Details on how the investigational product will be destroyed, who is responsible for the destruction, and how long documentation will be retained at sites, will be provided in the RSA or defer to site specific policies.

7.1.5 Investigator's Brochure

An Investigator's Brochure will be provided for each regimen as a separate document to this protocol.

7.2 Dosage Changes

Any changes in dosage, along with the circumstances surrounding those for such, will be described in the RSA.

7.3 Participant Compliance

Compliance for each regimen will be defined in the RSA. In cases of non-compliance, the participant will be reminded of the importance of taking the investigational product per protocol.

7.4 Overdose

Details on the actions to take in the event of an overdose will be included in the RSA.

7.5 Other Medications

Throughout the study, participants may be prescribed concomitant medications deemed necessary to provide adequate care. Participants should not receive other investigational products during the study. This includes marketed agents at experimental dosages that are being tested for the treatment of ALS. All concomitant medications and significant non-drug therapies, including supplements, received by a participant should be recorded on the appropriate source document and electronic Case Report Form (eCRF).

7.5.1 Prohibited Medications

Details on medications that may not be taken during the trial will be included in the RSA.

8 STUDY SCHEDULE

This section describes trial procedures that are common to all regimens tested in the HEALEY ALS Platform Trial as detailed in the Master Protocol. Each regimen will follow these procedures and may add additional procedures as described in the corresponding RSA.

No study procedures should be performed prior to the signing of the Master Protocol ICF. All participants will sign the Master Protocol ICF prior to undergoing any study tests or procedures.

Study Visit Recommendations:

1. Visit windows are consecutive calendar days and the target visit dates are calculated from the Baseline Visit – Day 0 (first day of investigational product administration).
2. Assessment of vital capacity should be performed at the start of a visit so as not to fatigue the participant with other testing.
3. The remaining clinical assessments should be administered in the same sequence as listed in the sections below and at approximately the same time of the day.
    a. Any additional assessments that are included in an RSA for a given visit should be completed once the Master Protocol SOA for that visit is complete, unless otherwise specified in the RSA.
4. Outcomes for individual participants should be measured by the same evaluator throughout their participation in a regimen. Evaluators must be qualified, trained, and must fulfill certification criteria defined by the Sponsor. Training documentation must be filed in the Investigator Site File (ISF).
5. In-person visits may be conducted in-clinic, or by phone or telemedicine, or in-home.

If needed to protect the safety of the participant due to a pandemic, disability or other medical reason, designated visits in the Schedule of Activities (i.e. Week 4, Week 8, and Week 16) may be conducted via telemedicine (or phone if telemedicine is not available) with remote services instead of in-person. If a planned in-clinic visit is conducted via telemedicine (or phone if telemedicine is not available) with remote services, only selected procedures will be performed. Instructions on how to document missed procedures are included in the study Manual of Procedures (MOP). The following procedures should be completed in the suggested order and as per the Schedule of Activities: ALSFRS-R,AE Review, Columbia-Suicide Severity Rating Scale, Concomitant Medications, and IP Accountability/Compliance; each RSA may describe additional activities. Procedures for collecting IP Accountability/Compliance remotely are described in the study MOPs. Additionally, selected vital signs and laboratory evaluations may be performed by a home health agency as described in the study MOP. Vital signs include: systolic and diastolic pressure, respiratory rate, heart rate, and temperature. Laboratory evaluation includes hematology (CBC with differential), coagulation parameters, complete chemistry panel and thyroid function, and urinalysis. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.

Vital Capacity: the default and preferred method for assessing Vital Capacity within the Master Protocol Screening period for eligibility is SVC performed by NEALS certified evaluators using a study-approved portable spirometer. If needed to protect the safety of the participants and/or site staff due to pandemic-related restrictions, Vital Capacity can be measured by a Pulmonary Function Laboratory evaluator to assess eligibility. At all subsequent visits to Master Protocol Screening, if SVC cannot be obtained due to to a pandemic, disability or other medical reason, this outcome measure should be skipped, recorded as such in the applicable source documentation, and a protocol deviation reported.

Clinical Safety Labs: the default and preferred method for collecting clinical safety labs is by using the study specific central laboratory as described in the study MOP. If needed to protect the safety of the participant due to a pandemic, disability or other medical reason, clinical safety labs can be collected using one of the following methods:

- Home health agency as described in the study MOP, or
- Local site laboratory resources, or
- Independent labs offering such services as described in this protocol and/or local to the participant (e.g. primary care physician)

8.1 Master Protocol Screening Visit

The following procedures will be performed in-person, after Master Protocol informed consent is obtained, to determine the participant's eligibility for the Master Protocol and are listed in preferred order to be completed:

Assess Master Protocol inclusion and exclusion criteria
Obtain ALS and medical history
Obtain demographics
Perform Vital Capacity
Perform physical examination
Perform neurological examination
Measure vital signs including height and weight
Administer ALSFRS-R questionnaire
Perform 12-lead electrocardiogram (ECG)
Collect blood samples for clinical screening assessments (including coagulation parameters) and, for women of child bearing potential (WOCBP), for pregnancy test
Collect urine sample for urinalysis
Review and document concomitant medications

Schedule the Regimen Specific Screening Visit if the participant passes screening for the Master Protocol

8.1.1 Regimen Assignment and Within-regimen Randomiation

If the participant meets all Master Protocol eligibility criteria, the participant will be randomly assigned to a regimen and will undergo informed consent for that regimen at their scheduled Regimen-Specific Screening Visit. If eligible, the participant will then be randomized in a 3:1 ratio to active treatment or matching placebo for that regimen. If the intervention has multiple arms (for example, multiple dosages), a research participant will also be randomized according to the allocation scheme specific for that regimen.

8.1.2 Screen Failures and Re-Assignment

Any participant who signs the Master Protocol consent form will be considered enrolled in the Platform Trial. If a participant fails the Master Protocol screening, *at a minimum* the following information should be captured and entered into the EDC system:

- Demographics (including the participant's NeuroSTAMP)
- Reason for screen failure
- Any eligibility criteria that were assessed prior to when the research participant failed to meet an inclusion criterion or met an exclusion criterion.

If a research participant fails to meet a time-dependent Master Protocol eligibility criterion (e.g., duration of stable medication prior to enrollment, interval following last use of an investigational treatment, or other criteria as applicable), then that participant will be considered to have screen failed, and may be re-screened for Master Protocol eligibility immediately after resolution of the disqualifying condition. Safety labs required for the site investigator to determine eligibility, can be repeated during the Master Protocol screening period, if necessary. If a participant is re-screened for Master Protocol eligibility, they must complete a full Master Protocol Screening Visit as outlined in section 8.1.

If a research participant meets Master Protocol eligibility, is assigned to a specific regimen, but does not meet additional regimen-specific inclusion and exclusion criteria for that regimen, then the research participant may be re-assigned immediately to a different regimen.

If a research participant meets Master Protocol eligibility, is assigned to a regimen, and is randomized within that regimen, but their participation in that regimen is discontinued, that participant may have the opportunity to be re-screened for the Master Protocol. The research participant must provide written informed re-consent to the Master Protocol and must still meet the Master Protocol inclusion and exclusion eligibility criteria in order to be re-assigned into a different regimen. Re-screening for the Master Protocol and potential re-assignment into a different regimen, if available, may occur due to the following situations:

- A regimen is stopped due to futility or success
    - The research participant may be re-screened after 30 days or 5 half-lives (if known), whichever is longer
- A research participant completes their Week 24 Visit in the placebo-controlled period and terminates their participation in the study
    - The research participant may be re-screened after 30 days or 5 half-lives (if known), whichever is longer
- A research participant discontinues participation in a regimen for personal reasons (e.g., inconvenience of the intervention, mode of administration, required assessments) or due to an adverse event, or is discontinued by the SI
    - The research participant may not be re-screened until after the date that their 24-Week Visit would have occurred, had they completed participation in that regimen.

8.2 Regimen Specific Screening Visit

This visit will take place in-person after random assignment to a regimen. This visit may be combined with the Baseline Visit, please refer to the RSA for applicability. The following procedures will be performed and are listed in preferred order to be completed:

Obtain written regimen-specific informed consent from participant
Assess regimen-specific inclusion and exclusion criteria
Concomitant medication review
Assess and document AEs

8.3 Baseline Visit

This visit will take place in-person after the regimen-specific Screening Visit. This visit may be combined with the Regimen Specific Screening Visit, please refer to the RSA for applicability. The following procedures will be performed and are listed in preferred order to be completed:

Randomize within the regimen
Collect vital signs
Perform SVC
Perform muscle strength testing
Administer ALSFRS-R questionnaire
Administer Patient Reported Outcomes
Collect blood samples for Clinical Safety Labs, biomarkers, and optional DNA sequencing
Collect urine sample for biomarkers
Perform LP for CSF collection (as described in the RSA)
Review and document concomitant medications
Assess and document AEs

Administer the C-SSRS Baseline questionnaire
Administer first dose of investigational product for the regimen, per the instructions in the RSA.
Dispense investigational product to participant, per instructions in the RSA

8.4 Week 2 Telephone Visit

This visit will take place 14 ± 3 days after the Baseline Visit via telephone. The following procedures will be performed and documented and are listed in preferred order to be completed:

Assess and document AEs
Drug Compliance Check-In
Remind participant to bring in investigational product to the Week 4 Visit

8.5 Week 4 Visit

This visit will take place in-person 28 ± 7 days after the Baseline Visit. The following procedures will be performed and are listed in the preferred order to be completed:

Collect vital signs
Administer ALSFRS-R questionnaire
Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
Review and document concomitant medications
Assess and document AEs
Administer the C-SSRS Since Last Visit questionnaire
Collect blood for optional DNA sequencing (if not done at Baseline)
Perform investigational product accountability and compliance
Schedule next Study Visit
Remind participant to bring in investigational product to the Week 8 Visit

8.6 Week 8 Visit

This visit will take place in-person 56 ± 7 days after the Baseline Visit. The following procedures will be performed and are listed in the preferred order to be completed:

Collect vital signs including weight
Perform SVC
Perform muscle strength testing
Administer ALSFRS-R questionnaire
Ask participant to complete Patient Reported Outcomes
Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
Collect urine sample biomarker analyses

Collect blood sample for biomarker analyses
Collect blood for optional DNA sequencing (if not done previously)
Review and document concomitant medications
Assess and document AEs
Administer the C-SSRS Since Last Visit questionnaire
Perform investigational product accountability and compliance

8.7 Week 12 Telephone Visit

This visit will take place 84 ± 3 days after the Baseline Visit via telephone. The following procedures will be performed and are listed in the preferred order to be completed:

Administer ALSFRS-R questionnaire
Review and document concomitant medications
Assess and document AEs
Drug Compliance Check-In
Schedule next Study Visit
Remind participant to bring in investigational product to the Week 16 Visit

8.8 Week 16 Visit

This visit will take place in-person 112 ± 7 days after the Baseline Visit. The following procedures will be performed and are listed in the preferred order to be completed:

Collect vital signs including weight
Perform SVC
Perform muscle strength testing
Administer ALSFRS-R questionnaire
Ask participant to complete Patient Reported Outcomes
Collect blood samples for Clinical Safety Labs (including coagulation parameters) and, for WOCBP, for pregnancy test if applicable
Collect urine sample biomarker analyses
Collect blood sample for biomarker analyses
Collect blood for optional DNA sequencing (if not done previously)
Review and document concomitant medications
Assess and document AEs
Administer the C-SSRS Since Last Visit questionnaire
Perform LP for CSF collection (as described in the RSA)
Perform investigational product accountability and compliance

8.9 Week 20 Telephone Visit

This visit will take place 140 ± 3 days after the Baseline Visit via telephone. The following procedures will be performed and are listed in preferred order to be completed:

Administer ALSFRS-R questionnaire
Review and document concomitant medications
Assess and document AEs
Drug Compliance Check-In
Schedule next Study Visit

8.10 Week 24 Visit

This visit will take place in-person 168 ± 7 days after the Baseline Visit. The following procedures will be performed and are listed in preferred order to be completed:

Collect vital signs including weight
Perform SVC
Perform muscle strength testing
Administer ALSFRS-R questionnaire
Ask participant to complete Patient Reported Outcomes
Perform 12-lead electrocardiogram (ECG)
Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
Collect urine sample biomarker analyses
Collect blood sample for biomarker analyses
Collect blood for optional DNA sequencing (if not done previously)
Review and document concomitant medications,
Assess and document AEs
Administer the C-SSRS Since Last Visit questionnaire
Perform investigational product accountability and compliance
Ask participant to complete the Exit Questionnaire

8.11 Follow-Up Safety Call

Participants will have a Follow-Up Safety after their last dose of study drug. Additional details, including the timing of this call, can be found in the RSA. The following procedure will be performed at a minimum:

- Assess and document AEs
- Review and document concomitant medications

8.12 Early Termination Visit & Follow-up Safety Call for Early Terminations

Participants who terminate their participation in the study early will be asked to be seen for an in-person Early Termination Visit and complete a Follow-Up Safety Call.

The outcome measures that are required to be collected for each regimen, and the timing of and activities for the Early Termination Visit and Follow-Up Safety Call, will be detailed in the RSA.

8.13 Protocol Deviations

A protocol deviation is any noncompliance with the sIRB approved clinical trial protocol. The noncompliance may be either on the part of the participant, the SI, or the study site staff. Patient reported outcomes (e.g. ALSAQ-40 and CNS-BFS) not performed due a participant language barrier, and procedures that were attempted but failed will not be reported as protocol deviations. A minor protocol deviation will be considered a departure from the protocol of relatively minor degree (such as study visit a day or two out of window due to participant being sick). A major deviation will be considered any significant deviation from the protocol that the Sponsor determines negatively affect: (1) the rights or welfare of the participant; (2), risk benefit assessment; or (3) the integrity of the data (ability to draw valid conclusions from the study data). As a result of deviations, corrective and preventative actions are to be developed by the site with review by the CC and Study Monitors and implemented promptly.

All deviations from the protocol will be recorded in the participant's source documentation. Protocol deviations must be entered in the Protocol Deviations Log in the EDC system. Deviations will be reported to the sIRB per their guidelines.

8.14 Missed Visits and Procedures

Missed visits and any procedures not performed (not attempted) for reasons other than illness, injury or progressive disability (i.e., participant is physically unable to perform test) will be reported as protocol deviations.

Procedures or visits not performed due to illness, injury or disability, including procedures that were attempted but failed (e.g., blood samples unable to be drawn after multiple attempts, or weight unable to be obtained due to participant immobility) will not be reported as protocol deviations.

Investigational product compliance that is outside the limits set in the RSA will be reported as a protocol deviation (see RSA for further details).

Details and specific instructions regarding protocol deviations, including any exceptions to this standard procedure, are found in the study Manual of Procedures.

Missed assessments due to pandemic-related restrictions must be documented as such in the source documentation and in the EDC.

8.15 Recording Deaths and Vital Status Determination

Information on whether a participant has died may be obtained from the participant's family, from clinic notes, or from a publicly available data source like the Centers for Disease Control and Prevention (CDC) National Death Index or the Social Security Death Index.

Vital status, defined as a determination of date of death or PAV status or date last known alive and date last known to be PAV-free, will be determined by site study staff for each randomized participant throughout the placebo-controlled portion of their follow-up (generally the Week 24 visit, as indicated in the SOA). At approximately the time of the last patient last visit (LPLV) of the placebo-controlled period of a given regimen, a second vital status check will be completed by site study staff for each randomized participant. When prompted by the Coordination Center, sites will contact all randomized participants to assess vital status.

We may also ascertain vital status at later time points. An outside vendor will be used to ascertain death or date of last known alive for all randomized participants by using publicly available data sources. When prompted by the Coordination Center, sites will provide demographic information (e.g. participant name, date of birth, last known address) to the vendor using a secure method.

9 CLINICAL ASSESSMENTS AND OUTCOME MEASURES

9.1 Clinical Variables

Assessments will be performed at designated time-points throughout the study for clinical evaluation. In addition to the assessments evaluated below, participants will provide information on their demographics, past medical history, including ALS, as well as concomitant medication usage.

Details of any additional specific clinical assessments required for a regimen will be reported in the corresponding RSA.

9.1.1 Vital Signs and Anthropometrics

Vital signs include weight in kg or lbs, systolic and diastolic blood pressure in mmHg, pulse rate (radial artery)/minute, respiratory rate/minute, and temperature in °C or °F. **Height in cm or in will be measured and recorded at the Master Protocol Screening Visit (Visit 1) only.** This one-time height measurement must be used throughout the duration of the study.

9.1.2 Clinical Safety Laboratory Tests

The following laboratory tests will be performed for safety at the Master Protocol Screening to determine eligibility and at the following visits: Screening/Baseline as described in the RSA, Week 4, Week 8, Week 16, and Week 24/ET

- Hematology with differential panel: complete blood count with differential (hematocrit, hemoglobin, platelet count, RBC indices, Total RBC, Total WBC, and WBC & differential)
- Blood chemistry panel/Liver function tests (LFTs): alanine aminotransferase (ALT (SGPT)), aspartate aminotransferase (AST (SGOT)), albumin, alkaline phosphatase, bicarbonate, blood urea nitrogen, calcium, chloride, creatinine (estimated Glomerular Filtration Rate [eGFR] will be calculated using the MDRD equation and creatinine clearance will be calculated using the Cockcroft-Gault equation), glucose, magnesium, phosphate, potassium, sodium, thyroid-stimulating hormone (TSH), total bilirubin and total protein
- Urinalysis: albumin, bilirubin, blood, clarity, color, glucose, ketones, nitrate, pH, protein, specific gravity, urobilinogen and WBC screen
- Serum human chorionic gonadotrophin (hCG) for WOCBP (collected only at Master Protocol Screening Visit, and as necessary throughout course of study)

Additionally, coagulation parameters (Prothrombin time, International normalized ratio, Activated partial thromboplastin time) will be collected for all participants at the Master Protocol Screening and Week 16 Visits.

All participants will have clinical safety laboratory tests at the designated visits outlined in the schedule of activities. All laboratory samples will be analyzed at a central laboratory, unless needed to protect the safety of the participant due to a pandemic, disability or other medical reason, they are collected by a home health agency, local site laboratory or independent laboratory. The SI may order additional local testing or a retest at the central laboratory, if needed, to further assess an AE, or if there is any suspicion that a participant may be pregnant, or if a result could not be obtained throughout the course of the study. If additional testing is required to obtain results, the participant may be asked to return to repeat blood collection for such missing results.

9.1.3 12-Lead ECG

A standard 12-lead ECG will be performed for safety at the screening visit to determine eligibility and for safety as outlined in the schedule of activities. Tracings will be reviewed by a central ECG reader and a copy of the tracings will be kept on site as part of the source documents.

9.1.4 Physical Examination

A physical examination will be performed and recorded. The following systems will be examined: head/neck, eyes, ears, nose and throat, cardiovascular, lungs, abdomen, musculoskeletal, central nervous system, extremities, and skin.

9.1.5 Neurological Examination

A neurological examination will be performed and recorded. Examination will include assessment of mental status, motor and sensory function, reflexes, and coordination/cerebellar function.

9.1.6 Columbia-Suicide Severity Rating Scale

The US FDA recommends the use of a suicidality assessment instrument that maps to the Columbia Classification Algorithm for Suicide Assessment (C-CASA). The C-CASA was developed to assist the FDA in coding suicidality data accumulated during the conduct of clinical trials of antidepressant drugs. One such assessment instrument is the C-SSRS. The C-SSRS involves a series of probing questions to inquire about possible suicidal thinking and behavior.

At the Baseline Visit, the C-SSRS Baseline version will be administered. This version is used to assess suicidality over the participant’s lifetime.

At all other visits, the Since Last Visit version of the C-SSRS will be administered. This version of the scale assesses suicidality since the participant's last visit.

If there is a positive response to question 4 or 5 on the severity of ideation subscale or any positive response on the suicidal behavior subscale of suicide attempt or suicidal ideation by the participant during the administration of the C-SSRS during the treatment period, the appropriately qualified clinician will be notified during the study visit to determine the appropriate actions required to ensure the participant's safety. The site must ensure that the participant is seen by a licensed physician (or other qualified individual as required by local institutional policy) before leaving the study site. The SI will determine whether the participant should remain on study drug. Reference to the Clinical Triage Guidelines Using the C-SSRS can be found here https://cssrs.columbia.edu/wp-content/uploads/C-SSRStriageexampleguidelines.pdf.

It is recommended that a medically licensed physician, nurse, nurse practitioner, or physician assistant to assess the C-SSRS. All evaluators must be certified to perform the C-SSRS; specific certification requirements are outlined in the study operations manual. Certification is required prior to performing the C-SSRS.

9.2 Endpoints

Details of any additional specific endpoints required for a regimen will be described in the corresponding RSA.

9.2.1 ALS Functional Rating Scale - Revised

The ALSFRS-R is a quickly administered (5 minutes) ordinal rating scale used to determine participants' assessment of their capability and independence in 12 functional activities. Each functional activity is rated 0-4 for a total score that ranges from 0 to 48. Higher scores indicate better function. Initial validity in ALS patients was established by documenting that, change in ALSFRS-R scores correlated with change in strength over time, was closely associated with quality of life measures, and predicted survival. The test-retest reliability is greater than 0.88 for all test items. The advantages of the ALSFRS-R are that all 12 functional activities are relevant to ALS, it is a sensitive and reliable tool for assessing activities of daily living function in those with ALS, and it is quickly administered. With appropriate training the ALSFRS-R can be administered with high inter-rater reliability and test-retest reliability. The ALSFRS-R can be administered by phone with good inter-rater and test-retest reliability. The equivalency of phone versus in-person testing, and the equivalency of study participant versus caregiver responses have also been established. Additionally, the ALSFRS-R can also be obtained using a web-based interface with good concordance with in-person assessment. All ALSFRS-R evaluators must be NEALS certified.

9.2.2 Slow Vital Capacity

The vital capacity (VC) will be determined using the upright slow VC (SVC) method. All evaluators performing SVC, must be NEALS certified. The SVC will be measured using the study-approved portable spirometer, and assessments will be performed using a face mask. A printout from the spirometer of all VC trials will be retained. Three VC trials are required for each testing session, however up to 5 trials may be performed if the variability between the highest and second highest VC is 10% or greater for the first 3 trials. Only the 3 best trials are recorded on the CRF. The highest VC recorded is utilized for eligibility. At least 3 measurable VC trials must be completed to score VC for all visits after screening. Predicted VC values and percent-predicted VC values will be calculated using the Quanjer Global Lung Initiative equations.

9.2.3 Measures of Muscle Strength

Hand Held Dynamometry: HHD will be used as a quantitative measure of muscle strength for this study. Six proximal muscle groups will be examined bilaterally in both upper and lower extremities (shoulder flexion, elbow flexion, elbow extension, hip flexion, knee flexion, and knee extension), all of which have been validated against maximum voluntary isometric contraction (MVIC) testing[19]. In addition, wrist extension, abductor pollicis brevis, abductor digiti minimi, first dorsal interosseous contraction and ankle dorsiflexion will be measured bilaterally; these muscles are often affected in ALS.

Bilateral Hand Grip: Bilateral hand grip will be measured using a Jamar hand dynamometer to test the maximum isometric strength of the hand and forearm muscles, measured in pounds.

9.2.4 Training and Validation

All evaluators must be NEALS trained and certified to perform the ALSFRS-R, SVC, and HHD assessments; specific certification requirements are outlined in the study MOP. Certification and training will be provided by the Barrow Outcome Center. Certification occurs via a formal evaluation of reliability and accuracy of performance of these measures. Repeat certification will be required on an annual basis for ALSFRS-R and every two years for SVC and HHD outcome measures. It is strongly preferred that a single evaluator performs all measures throughout the study, as much as possible.

9.2.5 Patient Reported Outcomes

Patient reported outcomes will be assessed in this trial and will be defined in the RSA.

9.3 Biofluid Collection

All samples will be labeled with a code. The code will not include any identifiable information but will be linked to a specific participant, visit, and sample by data entry of the code into the EDC. There is no scheduled date on which the samples will be destroyed. Samples may be stored for research until they are used, damaged, decayed or otherwise unfit for analysis. Participants have the option of declining participation in this portion of the study at any time by withdrawing their consent to have their sample used. However, it will not be possible to destroy samples that may have already been used.

Specific instructions regarding the collection, processing, storage, and shipment of these samples will be provided in the study Laboratory Manual.

Details of any additional specific biofluid samples required for a regimen will be described in the corresponding RSA.

9.3.1 DNA Collection

All participants consenting to DNA collection will provide a blood sample for deoxyribonucleic acid (DNA) extraction for genome sequencing during the Baseline Visit. The DNA sample can be collected after the Baseline Visit if a baseline sample is not obtained for any reason or the sample is not usable.

9.3.2 Blood Biomarker Sample Collection

All participants will provide blood samples for biomarker assessments.

9.3.3 Urine Collection

Urine samples will be collected at Baseline, Week 8, Week 16 and Week 24 study visits. Up to 10mL of urine may be collected.

9.3.4 Lumbar Puncture

The lumber puncture (LP) procedure will be performed to collect CSF, as described in each RSA.

Study staff should document the time of study drug dose administered.
Coagulation parameters (Prothrombin time, International normalized ratio, Activated partial thromboplastin time) will be collected for all participants at the Master Protocol Screening and Week 16 Visits.
The SI will discuss all potential LP risks to the subjects including:

- Local pain at injection site
- Reaction to anesthetic agents

- Bleeding at needle entrance site
- Infection at needle entrance site
- Post-LP low-pressure headache

Extensive experience with research LP in Alzheimer's disease reveals a very low incidence of complication, including the incidence of post-LP headache (Zetterberg et al., 2010). Fewer than 2.6% of patients in a memory disorder clinic developed post-LP headache, and only a single patient in a cohort of over 1,000 had a headache lasting more than 5 days (Zetterberg et al., 2010). No other local or generalized complications occurred.

The procedure must be performed by the SI or another licensed practitioner with experience and training in performing LPs,. When possible, LPs should be performed with an atraumatic Sprotte needle to reduce the risk of post-LP headache.

10 SAFETY AND ADVERSE EVENTS

The AE definitions and reporting procedures provided in this protocol comply with all applicable United States Food and Drug Administration (FDA) regulations and ICH guidelines. The SI will carefully monitor each participant throughout the study for possible adverse events. All AEs will be documented on CRFs designed specifically for this purpose. It is also important to report all AEs, especially those that result in permanent discontinuation of the investigational product being studied, whether serious or non-serious.

10.1 Definitions of AEs, Suspected Adverse Drug Reactions & SAEs

10.1.1 Adverse Event and Suspected Adverse Drug Reactions

An AE is any unfavorable and unintended sign (including a clinically significant abnormal laboratory finding, for example), symptom, or disease temporally associated with a study, use of a drug product or device whether or not considered related to the drug product or device.

Adverse drug reactions (ADR) are all noxious and unintended responses to a medicinal product related to any dose. The phrase “responses to a medicinal product” means that a causal relationship between a medicinal product and an adverse event is at least a reasonable possibility, i.e., the relationship cannot be ruled out. Therefore, a subset of AEs can be classified as suspected ADRs, if there is a causal relationship to the medicinal product.

Examples of adverse events include: new conditions, worsening of pre-existing conditions, clinically significant abnormal physical examination signs (e.g., skin rash, peripheral edema, etc.), or clinically significant abnormal test results (e.g., lab values or vital signs), with the exception of outcome measure results, which are not being recorded as adverse events in this trial (they are being collected, but analyzed separately). Stable chronic conditions (e.g., diabetes, arthritis) that are present prior to the start of the study and do not worsen during the trial are NOT considered adverse events. Chronic conditions that occur more frequently (for intermittent conditions) or with greater severity, would be considered as worsened and therefore would be recorded as adverse events.

Adverse events are generally detected in two ways:

1. Clinical: symptoms reported by the participant or signs detected on examination.
2. Ancillary Tests: abnormalities of vital signs, laboratory tests, and other diagnostic procedures (other than the outcome measures, the results of which are not being captured as AEs).

For the purposes of this study, symptoms of progression/worsening of ALS, including ‘normal’ progression, will be recorded as adverse events.

The following measures of disease progression will not be recorded as adverse events even if they worsen (they are being recorded and analyzed separately): SVC results, ALSFRS-R results, and muscle strength results.

If discernible at the time of completing the AE log, a specific disease or syndrome rather than individual associated signs and symptoms should be identified by the SI and recorded on the AE log. However, if an observed or reported sign, symptom, or clinically significant laboratory anomaly is not considered by the SI to be a component of a specific disease or syndrome, then it should be recorded as a separate AE on the AE log. Clinically significant laboratory abnormalities, such as those that require intervention, are those that are identified as such by the SI.

Participants will be monitored for adverse events from the time they sign consent for the Master Protocol until completion of their participation as defined in the RSA.

An unexpected adverse event is any adverse event, the specificity or severity of which is not consistent with the current regimen applicable Investigator’s Brochure. An unexpected, suspected adverse drug reaction is any unexpected adverse event that for which, in the opinion of the SI or Sponsor, there is a reasonable possibility that the investigational product caused the event.

10.1.2 Serious Adverse Events

A SAE is defined as an adverse event that meets any of the following criteria:

1. Results in death.
2. Is life threatening: that is, poses an immediate risk of death as the event occurred.
    a. This serious criterion applies if the study participant, in the view of the SI or Sponsor, is at immediate risk of death from the AE <u>as it occurs</u>. It does not apply if an AE that might hypothetically have caused death if it were more severe.
3. Requires inpatient hospitalization or prolongation of existing hospitalization.
    a. Hospitalization for an elective procedure (including elective PEG tube/g-tube/feeding tube placement) or a routinely scheduled treatment is not an SAE by this criterion because an elective or scheduled “procedure” or a “treatment” is not an untoward medical occurrence.
4. Results in persistent or significant disability or incapacity.
    a. This criterion applies if the “disability” caused by the reported AE results in a substantial disruption of the participant’s ability to carry out normal life functions.
5. Results in congenital anomaly or birth defect in the offspring of the participant (whether the participant is male or female).

6. Necessitates medical or surgical intervention to preclude permanent impairment of a body function or permanent damage to a body structure.
7. Important medical events that may not result in death, are not life-threatening, or do not require hospitalization may also be considered SAEs when, based upon appropriate medical judgment, they may jeopardize the participant and may require medical or surgical intervention to prevent one of the outcomes listed in this definition. Examples of such medical events include blood dyscrasias or convulsions that do not result in inpatient hospitalization, or the development of drug dependency or drug abuse.

An inpatient hospital admission in the absence of a precipitating, treatment-emergent, clinical adverse event may meet criteria for "seriousness" but is not an adverse experience, and will therefore, not be considered an SAE. An example of this would include a social admission (participant admitted for other reasons than medical, e.g., lives far from the hospital, has no place to sleep).

A serious, suspected adverse drug reaction is an SAE that, in the opinion of the SI or Sponsor, there is a reasonable possibility that the investigational product caused the event.

The SI is responsible for classifying adverse events as serious or non-serious.

10.2 Assessment and Recording of Adverse Events

The SI will carefully monitor each participant throughout the study for possible AEs. All AEs will be documented on source and CRFs designed specifically for this purpose. All AEs will be collected and reported in the EDC system and compiled into reports for monthly reviewing by the MM for the Master Protocol. The Master Protocol MM shall promptly review all information relevant to the safety of the investigational product, including all SAEs. Special attention will be paid to those that result in permanent discontinuation of the investigational product(s) being studied, whether serious or non-serious.

10.2.1 Assessment of Adverse Events

At each visit (including telephone interviews), the participant will be asked if they have had any problems or symptoms since their last visit in order to determine the occurrence of adverse events. If the participant reports an adverse event, site staff will probe further to determine:

1. Type of event
2. Date of onset and resolution (duration)
3. Severity (mild, moderate, severe)
4. Seriousness (does the event meet the above definition for an SAE)
5. Causality, relation to investigational product and disease

6. Action taken regarding investigational product
7. Outcome

Severity of Event
The following guidelines will be used to describe severity.

- **Mild** – Events require minimal or no treatment and do not interfere with the participant's daily activities.
- **Moderate** – Events result in a low level of inconvenience or concern with the therapeutic measures. Moderate events may cause some interference with functioning.
- **Severe** – Events interrupt a participant's usual daily activity and may require systemic drug therapy or other treatment. Severe events are usually potentially life-threatening or incapacitating. Of note, the term "severe" does not necessarily equate to "serious".

10.2.2 Relatedness of Adverse Event to Investigational Product

The relationship of the AE to the investigational product should be specified by the SI based on temporal relationship and his/her clinical judgment, using the following definitions:

- **Related** – The AE is known to occur with the study intervention, there is a reasonable possibility that the study intervention caused the AE, or there is a temporal relationship between the study intervention and event. <u>Reasonable possibility means that there is evidence to suggest a causal relationship between the study intervention and the AE</u>.
- **Not Related** – There is not a reasonable possibility that the administration of the study intervention caused the event, there is no temporal relationship between the study intervention and event onset, or an alternate etiology has been established.

10.2.3 Recording of Adverse Events

All clinical Adverse Events and Key Study Events (e.g., Mortality, Pregnancy, PAV, and Tracheostomy) are recorded in the participant's records. The site should enter the AE and Key Study Event information into the EDC system as soon as possible after learning of the event or receiving an update on an existing event.

Entries on the AE Log (and into the EDC system) will include the following: description of the event, severity, seriousness, date of onset, date of resolution, relationship to investigational product, action taken, and primary outcome of event.

10.3 Adverse Events and Serious Adverse Events - Reportable Events

The following are considered reportable events and must be reported to the CC within 24 hours of the site being notified of the event: all events that meet the above criteria for SAEs, or any unexpected AE that is deemed related to the study drug, which would have implications for the conduct of the study (e.g., requiring a significant and usually safety-related change to the protocol such as revising inclusion/exclusion criteria or including a new monitoring requirement, informed consent or investigator's brochure)

The timelines for reporting dosage changes are described in each RSA as applicable.

The MM also reviews AE reports, compiled by Data Management, as described in the safety monitoring plan. The MM will review blinded study data on enrollment, abnormal laboratory results and protocol deviations. These reports will collectively be known as the Medical Monitor Report.

The MM communicates with the IND Holder / Sponsor, the DSMB, and the CC as needed for reporting of SAEs to the FDA within the required timeframe per FDA investigational new drug application (IND) regulations.

All AEs that meet the criteria for a serious, unexpected, suspected adverse drug reaction (SUSAR), for which there is a reasonable possibility that the investigational product caused the event, in the opinion of the IND Holder / Sponsor, will be submitted to FDA in an expedited fashion.

Death, respiratory failure and hospitalization for routine procedures (i.e., g-tube placement) will not be reported individually in an expedited manner because they are anticipated to occur in the study population at some frequency independent of drug expose. The DSMB will review aggregate analysis of these events and the sponsor will submit an IND safety report if the aggregate analysis indicates these events occur more frequently in any of the active drug treatment groups, as compared to the concurrent placebo group.

11 STATISTICAL CONSIDERATIONS

The statistical considerations specified here describe the overall structure and statistical design of the platform trial and propose recommended statistical details for evaluating individual regimens. Additional details regarding the master protocol recommended design are specified in Appendix I (ALS Master Protocol Recommended Design and Simulation Report) to the Master Protocol Statistical Analysis Plan (M-SAP). Regimen-specific deviations from the Master Protocol recommended statistical design are allowed and full statistical details are provided in separate regimen specific statistical analysis plans (R-SAPs) for each regimen. In the case of any deviation between this protocol and Appendix I of the M-SAP, Appendix I of the M-SAP takes precedence. In case of any deviation with an R-SAP, the order of precedence is specified in the R-SAP.

Interim analyses will be performed by a distinct team of unblinded statisticians. All other analyses will be performed either on blinded data or after database lock for a given regimen.

11.1 Statistical Design Overview

The primary aim of the trial is to compare the effectiveness of novel/emerging therapies to placebo for participants with ALS. The master protocol will include multiple regimens where a regimen is defined as both an active treatment and the corresponding randomized placebo.

The recommended primary effecicay endpoint is the change from baseline through week 24 in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) total score and survival. The primary analysis is a Bayesian shared parameter analysis of ALSFRS-R and survival that compares each regimen-specific active treatment to a shared placebo group that is common across all regimens, as specified in Appendix I of the M-SAP, Section 2.2 (ALS Master Protocol Recommended Design and Simulation Report Index). The primary survival endpoint is the composite event of death or a death equivalent of progression to PAV. PAV is defined as use of noninvasive or invasive mechanical ventilation for more than 22 hours per day for more than seven consecutive days. The date of PAV initiation is the first day of the consecutive days.

Participants in the master protocol will be randomized in two stages; first to one of the actively enrolling regimens after the participant is determined eligible based on Master Protocol criteria; and then 3:1 to active treatment or placebo within the regimen after the participant is determined eligible based on any regimen-specific eligibility criteria. The master protocol recommended design is 160 participants randomized within each regimen.

New regimens are launched and added in parallel to existing regimens already being investigated in the platform. Regimens exit the platform either after completing all planned accrual and follow-up or by meeting criteria for success (if applicable) or futility at pre-defined interim analyses. The recommended early stopping criteria for success (if applicable) and futility are defined in Appendix I of the M-SAP. Early stopping criteria are non-binding.

Early stopping criteria will be evaluated for active regimens that have sufficient data, defined as having at least 40 randomized participants within that regimen who have had the opportunity to complete at least 24 weeks of follow-up. Interim analyses will occur globally across the platform every 12 weeks. If a regimen is not stopped early, then the regimen-specific final analysis will take place when all its randomized participants have had the opportunity to complete 24-week-placebo-controlled follow-up.

11.2 Analysis Plan

11.2.1 Primary Analysis Population and Shared Control

The primary analysis dataset uses all available longitudinal data (ALSFRS-R total score assessments at 0, 4, 8, 12, 16, 20 and 24 weeks,and assessments during the placebo-controlled follow up period that substitute for missed scheduled visits, including early termination visits) as well as survival within 24 weeks for participants within the primary analysis population.

The primary analysis population for each analysis regimen is based on the intent to treat (ITT) principle and includes all participants randomized to the active treatment arm within the analysis regimen, all participants concurrently randomized to the control arm within the analysis regimen, and control participants from all contributing regimens. The shared control group includes all participants concurrently randomized to the control arm within the analysis regimen, participants randomized to the control arm across other actively enrolling regimens (concurrent shared controls) and participants randomized to the control arm in previously completed regimens (non-concurrent shared controls). Concurrent shared controls are defined as participants randomized to placebo control who were randomized within a 6-month (defined as 180 days) window of the first and last randomization within the analysis regimen.

The Master Protocol allows for additional and more restrictive eligibility criteria specified in a regimen specific appendix (RSA). These additional eligibility criteria will be limited and will be allowed only to ensure participant safety or based on mechanism of action. Major additional regimen-specific eligibility criteria are those that have the potential to lead to significant differences in the patient population with respect to the rate of disease progression. If these criteria can be assessed for all participants entering the platform, then they may be used to subset the shared control to include only those participants that meet these additional regimen-specific eligibility criteria in the primary analysis. If a subset of the shared control will be used in the primary analysis due to major additional regimen-specific eligibility criteria, this will be pre-specified within each RSA.

11.2.2 Primary Efficacy Analysis

Unless otherwise specified in the RSA or R-SAP, the primary analysis for each analysis regimen is a Bayesian shared parameter model of function and survival. The functional component is a repeated measures ALSFRS-R model that measures the slowing in the mean rate of progression for ALSFRS-R in active treatment participants relative to control participants and uses all available longitudinal data for all participants within the primary analysis population. The survival component is an exponential proportional hazards model. The functional and survival components are connected through a shared common treatment effect (slowing in rate of progression of the disease). The shared treatment effect between the functional and survival components allows the participants who were loss to follow-up due to mortality to inform treatment effect estimates beyond their censored ALSFRS-R longitudinal data. The degree to which treatment effects on mortality inform the shared treatment effect parameter depends on the mortality rate within the study.

The ALSFRS-R repeated measures model is based on a linear rate of progression in ALSFRS-R for placebo participants and a proportional slowing in the rate of progression for active treatment participants at each time point. The treatment effect is quantified by the disease rate ratio (DRR), which represents the ration of the rate of decline of disease progression (rate of ALSFRS-R and mortality rate) of active treatement participant relative to placebo participants (i.e. a proportional slowing). The model incorporates participant-level random effects in the baseline value of the ALSFRS-R (intercept) and in the rate of progression (slope); covariate effects to account for covariate-explainable differences in rates of progression based on participant-specific baseline covariates; and regimen-specific random effects in the rate of progression (slope). Regimen-specific random effects are used to account for potential differences in the shared control group not explained by the included covariates. Full specification of the primary efficacy analysis is described in Appendix I of the M-SAP.

11.2.3 Handling of Missing Data

The Bayesian repeated-measures model of ALSFRS-R naturally accommodates differential length of follow-up and utilizes all available longitudinal data for each participant. The model also accommodates missing data due to death (via included mortality component), as well as data missing at random (i.e. due to reasons unrelated to disease progression). No imputation is planned for the primary efficacy analysis. However, planned sensitivity analyses will investigate the potential effect of missing data on inference and estimates from the primary analysis.

11.2.4 Handling of Re-Screened and Repeat Participants

Upon completion of a regimen, a research participant may be re-screened into the Master Protocol and assigned to another regimen after 30 days or 5 half-lives (if known), whichever is longer. Those participants who have been re-screened and meet eligibility criteria for a subsequent randomization will be analyzed as new participants with new baseline visits and sets of observations for each regimen to which they were assigned. The baseline visit for each regimen is defined as the visit after assignment to and randomization within that regimen. Within the primary analysis, sets of observations from a repeat participant across multiple regimens will be treated as independent conditional on baseline covariates within each regimen. After the required washout period and adjusting for progression during previous regimens in the primary analysis (pre-baseline ALSFRS-R slope), it is not expected that future ALSFRS-R progression will be associated with active treatment or placebo received in the previous regimen.

11.2.5 Inclusion of Multiple Doses

If an RSA includes multiple doses under investigation, the recommended primary analysis will be a pooled analysis of all active doses compared to the shared control. Additionally, secondary analyses will be specified within the R-SAP to investigate each active dose separately.

11.3 Interim Analyses and Trial Adaptations

Multiple interim analyses will allow each regimen to stop randomization early for success (if applicable) or futility, with interims occurring simultaneously for all actively enrolling regimens with sufficient data. Interims will begin when at least one regimen within the master protocol has 40 randomized participants (30 active and 10 control) with the opportunity to complete at least 24 weeks of follow-up. Subsequent interims will occur every 12 weeks. If a regimen is not stopped early for success (if applicable) or futility at an interim analysis, the regimen-specific final analysis of the placebo-controlled period will take place after all participants randomized to that regimen have had the opportunity to complete 24-weeks of follow-up. The default design for a regimen will include stopping early only for futility.

11.3.1 Regimen Success

The default design for a regimen will not include early stopping for success. If early stopping for success is included in a regimen, success will be declared based on the posterior probability that the active treatment is superior to the placebo group. The posterior probability for declaring success is specified in Appendix I of the M-SAP and determined based on clinical trial simulations to achieve an overall one-sided Type I error of 2.5%.

11.3.2 Regimen Futility

Futility will be declared for a regimen at an interim analysis if the posterior probability that active treatment slows disease progression relative to placebo by at least 10% is less than 5%. Early stopping for futility in non-binding.

11.4 Secondary, Exploratory, and Sensitivity Analyses

11.4.1 Secondary Efficacy Analyses

As a secondary analysis, a joint-rank test of function and survival will be performed. The analysis will rank each pair of participants based first on survival time during the 24-week placebo-controlled period and second, if they both survived or one was censored prior to the observed survival endpoint of the other, on the change in ALSFRS-R total score from baseline to the last jointly observed visit up to 24-weeks[20-22]. The summed ranks will be analyzed by linear regression to adjust for important covariates.

11.4.2 Secondary Endpoints

Secondary endpoints and analyses will be detailed within each RSA and R-SAP.

11.4.3 Exploratory Analyses

Exploratory endpoints and their corresponding analyses will be detailed within each RSA and R-SAP. Such analyses may evaluate alternative functional endpoints and biofluid biomarkers for monitoring disease progression.

11.4.4 Sensitivity Analyses

Additional sensitivity analyses of the primary efficacy analysis will be conducted for various definitions of the analysis population (e.g. depending on definition of shared controls), and to evaluate the potential effect of modeling assumptions and missing data. Further details are provided in Appendix I of the M-SAP.

11.5 Safety Analyses

Safety data, including deaths and death-equivalents, treatment-emergent adverse events (TEAE), and changes in vital signs, ECGs, and clinical laboratory test results, will be summarized for each treatment group among all participants who initiate treatment. The frequency and type of any observed suicidal ideation or suicidal behavior will be described. Descriptive statistics of continuous measures will be provided for the observed data and for the change from baseline at each measured time point. Time-to-death or death equivalent will be estimated by Kaplan-Meier product-limit estimates. Adverse events will be summarized by treatment group as counts of events and as the proportion of participants experiencing any given type of TEAE as classified by MedDRA system organ class, high level term, and preferred term and classified according to severity, relatedness, and outcome. Laboratory test results will be classified by Common Terminology Criteria for Adverse Events (CTCAE) grad and as below the lower limit of normal, within normal limits and above the upper limit of normal. Shift tables will be used to summarize changes from baseline to each visit by treatment group. Clinically significant out-of-range laboratory tests are recorded as adverse events and will be documented in the TEAE summaries. Additional safety assessments may be specified in the R-SAP.

11.6 Sample Size Justification

Clinical trial simulation is used to quantify operating characteristics for each regimen. In particular, virtual participant outcomes are created under different assumptions for key design parameters including placebo rates of progression for ALSFRS-R, measurement error for ALSFRS-R, mortality rates, treatment effects on ALSFRS-R, treatment effects on mortality, dropout rates and accrual rates. For each set of simulation assumptions (i.e. a scenario), many trials are simulated and virtually executed, including all interim analyses for early success and early futility. Trial operating characteristics are summarized across all simulated trials for each scenario. Complete details and results are provided in Appendix I of the M-SAP.

Simulations outlined in Appendix I of the M-SAP, provide example operating characteristics of the recommended design for the first 3 regimens enrolling in the platform across a range of simulation assumptions. Under the null scenario there was an overall one-sided Type I error rate of 2.4% per regimen. The average duration of the regimen was 14 months and there was a 28% chance that the regimen would meet the criterion for early stopping for futility.

The Master Protocol recommended design has 61%, 77%, and 88% power to detect a slowing in the rate of ALSFRS-R progression of 25%, 30% and 35% respectively when function and survival have a common treatment effect. When there is no benefit for survival and a 30% slowing in ALSFRS-R progression the power drops to 72%. When there is a negative effect on survival and a 30% slowing in ALSFRS-R progression, the power drops further to 68%. There is less than a 1% chance in these scenarios that the regimen would meet the criterion for early stopping for futility.

12 DATA COLLECTION, MANAGEMENT, MONITORING, AND REPORTING

The EDC system will be developed to facilitate the collection, management, monitoring, and reporting of study data for the Master Protocol.

12.1 Quality Assurance

Protocol procedures are reviewed with the SI and associated personnel prior to the study to ensure the accuracy and reliability of data. Each SI must adhere to the protocol detailed in this document and agree that any changes to the protocol must be approved by the CC prior to implementing the change unless it is to address immediate participant safety.

12.2 Role of Data Management

Data Management (DM) is the development, execution and supervision of plans, policies, programs, and practices that control, protect, deliver, and enhance the value of data and information assets. The Neurological Clinical Research Institute (NCRI) will serve as the Data Coordination Center (DCC) for the Master Protocol and is responsible for developing, testing, and managing the clinical data systems and management activities. All data will be managed in compliance with NCRI policies and applicable regulatory requirements.

12.2.1 Data Collection

Site personnel will collect data onto paper source documents as appropriate and into electronic source documents (e.g., voice sample data, EMR data, etc). Values from paper source documents will be transcribed into the corresponding eCRFs in the EDC system by site personnel; electronically captured data will be transmitted to their respective vendors for consolidation and reconciliation with the study EDC system. Clinical sites will be monitored to ensure compliance with the study protocol, data management requirements, and GCP.

12.2.2 Data Entry and Edit Checks

Site personnel are instructed to enter collected data into the EDC system shortly after a visit. Please note: SAEs must be entered into the EDC system and reported to the CC within 24 hours of site awareness of the SAE. Data collection is the responsibility of the staff at the site under the supervision of the SI as specified in the delegation log for that site. During the study, the SI must maintain complete and accurate documentation for the study.

To ensure accuracy and completeness of the data set, logic and range checks as well as in-form rules will be built into the EDC system, and electronic queries will be created to track potential data issues. The sites will only have access to the queries concerning their own participants.

12.2.3 Data Lock Process

The EDC system will have the ability to lock each trial regimen independently to prevent any modification of data once a trial regimen is closed. Once this option is activated, every user will have read-only access to the data until all access is revoked following NCRI DM procedures. The specific database lock procedures are detailed in the DM Standard Operating Procedures (SOP).

12.3 Role of Study Monitors

All aspects of the trial, including the individual regimens, will be monitored by qualified individuals designated by the Sponsor. Monitoring will be conducted according to GCP and applicable government regulations. The SIs will agree to allow monitors access to the clinical supplies, dispensing and storage areas, and to the clinical files of the study participants, and, if requested, agree to assist the monitors. Monitoring visits will be conducted according to a pre-defined Monitoring Plan.

12.3.1 Clinical Monitoring

Study Monitors will visit each study site during the course of the study to review source documentation, ICF, and the clinical facilities to ensure the study is conducted in accordance with the study protocol and compliance with ICH/GCP and regulatory guidelines. Investigators are responsible for allowing access to all source documents and medical records related to the subject's participation in the study. Monitoring visits will occur at defined intervals per the Study Monitoring Plan. Study monitor(s) will identify study non-compliance and if appropriate, the study monitor(s) will assist the site with developing a corrective and preventative action plan. All significant noncompliance will be communicated to the Sponsor. Regimen specific monitoring activities will be specified in the appendices of the Study Monitoring Plan.

12.3.2 Monitoring Report

The Study Monitors will provide monitoring reports to the Sponsor and, if requested, and when required, will provide reports of protocol compliance to the Sponsor and other applicable parties as detailed in the Study Monitoring Plan.

12.4 Data and Safety Monitoring Board

An independent DSMB will be assembled for the trial. A DSMB Charter will detail the processes of this group. The DSMB will receive blinded and unblinded summary reports of the frequency of all clinical adverse events and safety laboratory tests for each trial regimens at planned periodic meetings throughout the study as specified in the DSMB Charter. The DSMB will receive separate reports per trial regimen to review independently. Placebo data from all trial regimens will be compiled into a single report for review. Meetings will be held in-person or via teleconference.

Summaries of serious adverse events and enrollment per trial regimen will be provided to the DSMB by the Study Biostatisticians as specified in the DSMB Charter. AE(s) of Special Interest events occurring within 24 hours of dosing for any given treatment arm, and any severe unexpected serious adverse events for a treatment arm are considered events of interest and will be reported in real-time (within 1 business day of CC awareness) to the DSMB. The DSMB can ask to receive the SAE reports more frequently. As necessary, the DSMB can review the frequencies of clinical and laboratory abnormalities. Recommendations for modification or termination of the trial based on safety data will be made by the DSMB to the Sponsor. The DSMB will review safety data throughout the trial and may stop enrollment into a treatment arm for safety if they determine that there is a significant difference in the rate of a particular adverse event that would indicate a risk that is greater than the possible benefit of the investigational product. A notable increase in the frequency of any adverse event should be examined by the DSMB although it may not lead to a recommendation by the DSMB.

Prior to each DSMB meeting, the CC will provide an update to the DSMB on enrollment, data quality (missing data), and protocol adherence for each treatment arm. The CC will be responsible for communication with the DSMB.

Complete information can be found in the DSMB Charter.

12.5 Neurological Global Unique Identifier

A participant Neurological Global Unique Identifier (NeuroGUID), or its derivitives', will be used as the identifier for participant's samples participating in this study. The NeuroGUID is an 11-character string that is generated using encryption technology and algorithms licensed by the NCRI from the NIH in 2013.

The NeuroGUID is generated on a secure website that utilizes 128-bit Secure Socket Layer by using an irreversible encryption algorithm that accepts ten identifying data elements, (e.g., last name at birth, first name at birth, sex at birth, day, month and year of birth, city and country of birth, etc.), and creates a series of coded strings ("hashes") that are encrypted and sent to a secure server. The server produces a unique randomly generated alphanumeric string or NeuroGUID. No identifying information is stored in the system; it is simply used to guarantee the uniqueness of a NeuroGUID. If the same information is entered again, the same NeuroGUID will be returned.

12.6 Data Handling and Record Keeping

The SI is responsible for ensuring the accuracy, completeness, legibility, and timeliness of the data reported. All source documents should be completed in a neat, legible manner to ensure accurate interpretation of data. Dark ink is required to ensure clarity of reproduced copies. When making changes or corrections, cross out the original entry with a single line, and initial and date the change. Do not erase, overwrite, or use correction fluid or tape on the original.

Source document templates (SDTs) will be provided for use and maintained for recording data for each participant enrolled in the study. Data reported in the eCRF derived from source documents should be consistent with the source documents and discrepancies should be explained.

12.6.1 Confidentiality

Study participant medical information obtained by this study is confidential, and disclosure to third parties other than those noted below is prohibited. Upon the participant's permission, medical information may be given to his or her personal physician or other appropriate medical personnel responsible for his or her welfare. All local and federal guidelines and regulations regarding maintaining study participant confidentiality of data will be adhered to.

Data generated by this study must be available for inspection by representatives of the US FDA, the Office for Human Research Protections (OHRP), the sponsor, all pertinent national and local health and regulatory authorities, the CC or their representative, Study Monitoring personnel, and the sIRB.

12.6.2 Retention of Records

US FDA regulations (21 CFR 312.62[c]) require that records and documents pertaining to the conduct of this study and the distribution of investigational drugs, including CRFs (if applicable), consent forms, laboratory test results, and medical inventory records, must be retained by the SI for two years after marketing application approval. If no application is filed, these records must be kept for two years after the investigation is discontinued and the US FDA and the applicable national and local health authorities are notified. The CC or their representative will notify the SIs of these events. The SIs should retain all study documents and records until they are notified in writing by the Sponsor or their representative.

No records may be disposed of without the written approval of the Sponsor. Written notification should be provided to the Sponsor prior to transferring any records to another party or moving them to another location.

12.7 Reporting, Publications, and Notification of Results

12.7.1 Publication Policy

The HEALEY ALS Platform Trial Executive Committee will set the policy for publication of results from the HEALEY ALS Platform Trial and all RSAs. No data or results generated from the HEALEY ALS Platform Trial or its RSAs may be published without written agreement from the HEALEY ALS Platform Trial Executive Committee.

The responsibilities of the HEALEY ALS Platform Trial Executive Committee are outlined in the HEALEY ALS Platform Trial Governance Plan.

12.7.2 Data and Sample Sharing

Data and sample sharing policies will be specified in the legal agreements governing the trial.

12.7.3 Trial Registration

The Master Protocol and each RSA will be registered on www.ClinicalTrials.gov. Results will be posted to ClinicalTrials.gov according to applicable regulations.

13 APPENDICES

Appendix I: El Escorial World Federation of Neurology Criteria for the Diagnosis of ALS

Information obtained from the web site: ***www.wfnals.org***.

The diagnosis of Amyotrophic Lateral Sclerosis [ALS] requires:

A - The presence of:

(A:1) evidence of lower motor neuron (LMN) degeneration by clinical, electrophysiology cal or neuropathologic examination,

(A:2) evidence of upper motor neuron (UMN) degeneration by clinical examination, and

(A:3) progressive spread of symptoms or signs within a region or to other regions, as determined by history or examination, together with

B - The absence of:

(B:1) electrophysiological and pathological evidence of other disease processes that might explain the signs of LMN and/or UMN degeneration, and

(B:2) neuroimaging evidence of other disease processes that might explain the observed clinical and electrophysiological signs.

CLINICAL STUDIES IN THE DIAGNOSIS OF ALS

A careful history, physical and neurological examination must search for clinical evidence of UMN and LMN signs in four regions [brainstem, cervical, thoracic, or lumbosacral spinal cord] (see Table 1) of the central nervous system [CNS]. Ancillary tests should be reasonably applied, as clinically indicated, to exclude other disease processes. These should include electrodiagnostic, neurophysiological, neuroimaging and clinical laboratory studies. Clinical evidence of LMN and UMN degeneration is required for the diagnosis of ALS. The clinical diagnosis of ALS, without pathological confirmation, may be categorized into various levels of certainty by clinical assessment alone depending on the presence of UMN and LMN signs together in the same topographical anatomic region in either the brainstem [bulbar cranial motor neurons], cervical, thoracic, or lumbosacral spinal cord [anterior horn motor neurons]. The terms Clinical Definite ALS and Clinically Probable ALS are used to describe these categories of clinical diagnostic certainty on clinical criteria alone:

A. Clinically Definite ALS is defined on clinical evidence alone by the presence of UMN, as well as LMN signs, in three regions.

B. Clinically Probable ALS is defined on clinical evidence alone by UMN and LMN signs in at least two regions with some UMN signs necessarily rostral to (above) the LMN signs.

C. Clinically Probable ALS - Laboratory-supported is defined when clinical signs of UMN and LMN dysfunction are in only one region, or when UMN signs alone are present in one region, and LMN signs defined by EMG criteria are present in at least two limbs, with proper application of neuroimaging and clinical laboratory protocols to exclude other causes.

D. Clinically Possible ALS is defined when clinical signs of UMN and LMN dysfunction are found together in only one region or UMN signs are found alone in two or more regions; or LMN signs are found rostral to UMN signs and the diagnosis of Clinically Probable - Laboratory-supported ALS cannot be proven by evidence on clinical grounds in conjunction with electrodiagnostic, neurophysiologic, neuroimaging or clinical laboratory studies. Other diagnoses must have been excluded to accept a diagnosis of Clinically Possible ALS.

Table 1

	Brainstem	Cervical	Thoracic	Lumbosacral
Lower motor neuron signs weakness, atrophy, fasciculations	jaw, face, palate, tongue, larynx	neck, arm, hand, diaphragm	back, abdomen	back, abdomen, leg, foot
Upper motor neuron signs pathologic spread of reflexes, clonus, etc.	clonic jaw gag reflex exaggerated snout reflex pseudobulbar features forced yawning pathologic DTRs spastic tone	clonic DTRs Hoffman reflex pathologic DTRs spastic tone preserved reflex in weak wasted limb	loss of superficial abdominal reflexes pathologic DTRs spastic tone	clonic DTRs - extensor plantar response pathologic DTRs spastic tone preserved reflex in weak wasted limb

Appendix II: ALS Functional Rating Scale – Revised (ALSFRS-R)

ALSFRS-R

QUESTIONS: **SCORE:**

1. Speech
   4 = Normal speech processes
   3 = Detectable speech disturbances
   2 = Intelligible with repeating
   1 = Speech combined with nonvocal communication
   0 = Loss of useful speech

2. Salivation
   4 = Normal
   3 = Slight but definite excess of saliva in mouth; may have nighttime drooling
   2 = Moderately excessive saliva; may have minimal drooling
   1 = Marked excess of saliva with some drooling
   0 = Marked drooling; requires constant tissue or handkerchief

3. Swallowing
   4 = Normal eating habits
   3 = Early eating problems – occasional choking
   2 = Dietary consistency changes
   1 = Needs supplemental tube feeding
   0 = NPO (exclusively parenteral or enteral feeding)

4. Handwriting
   4 = Normal
   3 = Slow or sloppy; all words are legible
   2 = Not all words are legible
   1 = No words are legible but can still grip a pen
   0 = Unable to grip pen

5a. Cutting Food and Handling Utensils (patients without gastrostomy)
   4 = Normal
   3 = Somewhat slow and clumsy, but no help needed

2 = Can cut most foods, although clumsy and slow; some help needed
1 = Food must be cut by someone, but can still feed slowly
0 = Needs to be fed

5b. Cutting Food and Handling Utensils (alternate scale for patients with gastrostomy)
4 = Normal
3 = Clumsy, but able to perform all manipulations independently
2 = Some help needed with closures and fasteners
1 = Provides minimal assistance to caregivers
0 = Unable to perform any aspect of task

6. Dressing and Hygiene
4 = Normal function
3 = Independent, can complete self-care with effort or decreased efficiency
2 = Intermittent assistance or substitute methods
1 = Needs attendant for self-care
0 = Total dependence

7. Turning in Bed and Adjusting Bed Clothes
4 = Normal function
3 = Somewhat slow and clumsy, but no help needed
2 = Can turn alone, or adjust sheets, but with great difficulty
1 = Can initiate, but not turn or adjust sheets alone
0 = Helpless

8. Walking
4 = Normal
3 = Early ambulation difficulties
2 = Walks with assistance
1 = Nonambulatory functional movement only
0 = No purposeful leg movement

9. Climbing Stairs
4 = Normal
3 = Slow
2 = Mild unsteadiness or fatigue
1 = Needs assistance

0 = Cannot do

R-1. Dyspnea

4 = None

3 = Occurs when walking

2 = Occurs with one or more of the following: eating, bathing, dressing

1 = Occurs at rest, difficulty breathing when either sitting or lying

0 = Significant difficulty, considering using mechanical respiratory support

R-2 Orthopnea

4 = None

3 = Some difficulty sleeping at night due to shortness of breath, does not routinely use more than two pillows

2 = Needs extra pillow in order to sleep (more than two)

1 = Can only sleep sitting up

0 = Unable to sleep without mechanical assistance

R-3 Respiratory Insufficiency

4 = None

3 = Intermittent use of BiPAP

2 = Continuous use of BiPAP during the night

1 = Continuous use of BiPAP during the night and day

0 = Invasive mechanical ventilation by intubation or tracheostomy

Evaluator’s Initials: _____ _____ ______

Appendix III: Columbia-Suicide Severity Rating Scale (C-SSRS) Baseline Version

Information obtained from: http://www.cssrs.columbia.edu/

SUICIDAL IDEATION	
Ask questions 1 and 2. If both are negative, proceed to "Suicidal Behavior" section. If the answer to question 2 is "yes", ask questions 3, 4 and 5. If the answer to question 1 and/or 2 is "yes", complete "Intensity of Ideation" section below.	Lifetime: Time He/She Felt Most Suicidal
1. Wish to be Dead Subject endorses thoughts about a wish to be dead or not alive anymore, or wish to fall asleep and not wake up. ***Have you wished you were dead or wished you could go to sleep and not wake up?*** If yes, describe:	**Yes No** ☐ ☐
2. Non-Specific Active Suicidal Thoughts General, non-specific thoughts of wanting to end one's life/commit suicide (e.g., *"I've thought about killing myself"*) without thoughts of ways to kill oneself/associated methods, intent, or plan. ***Have you actually had any thoughts of killing yourself?*** If yes, describe:	**Yes No** ☐ ☐
3. Active Suicidal Ideation with Any Methods (Not Plan) without Intent to Act Subject endorses thoughts of suicide and has thought of at least one method during the assessment period. This is different than a specific plan with time, place or method details worked out (e.g., thought of method to kill self but not a specific plan). Includes person who would say, *"I thought about taking an overdose but I never made a specific plan as to when, where or how I would actually do it…and I would never go through with it."* ***Have you been thinking about how you might do this?*** If yes, describe:	**Yes No** ☐ ☐
4. Active Suicidal Ideation with Some Intent to Act, without Specific Plan Active suicidal thoughts of killing oneself and subject reports having <u>some intent to act on such thoughts</u>, as opposed to *"I have the thoughts but I definitely will not do anything about them."* ***Have you had these thoughts and had some intention of acting on them?*** If yes, describe:	**Yes No** ☐ ☐
5. Active Suicidal Ideation with Specific Plan and Intent Thoughts of killing oneself with details of plan fully or partially worked out and subject has some intent to carry it out. ***Have you started to work out or worked out the details of how to kill yourself? Do you intend to carry out this plan?*** If yes, describe:	**Yes No** ☐ ☐
INTENSITY OF IDEATION	
The following features should be rated with respect to the most severe type of ideation (i.e., 1-5 from above, with 1 being the least severe and 5 being the most severe). Ask about time he/she was feeling the most suicidal. _____ ____ ***Most Severe Ideation:*** ***Type # (1-5) Description of Ideation***	Most Severe
Frequency ***How many times have you had these thoughts?*** (1) Less than once a week (2) Once a week (3) 2-5 times in week (4) Daily or almost daily (5) Many times each day	____
Duration ***When you have the thoughts, how long do they last?*** (1) Fleeting - few seconds or minutes (4) 4-8 hours/most of day (2) Less than 1 hour/some of the time (5) More than 8 hours/persistent or continuous (3) 1-4 hours/a lot of time	____

Controllability ***Could/can you stop thinking about killing yourself or wanting to die if you want to?*** (1) Easily able to control thoughts (4) Can control thoughts with a lot of difficulty (2) Can control thoughts with little difficulty (5) Unable to control thoughts (3) Can control thoughts with some difficulty (0) Does not attempt to control thoughts	____
Deterrents ***Are there things - anyone or anything (e.g., family, religion, pain of death) - that stopped you from wanting to die or acting on thoughts of committing suicide?*** (1) Deterrents definitely stopped you from attempting suicide (4) Deterrents most likely did not stop you (2) Deterrents probably stopped you (5) Deterrents definitely did not stop you (3) Uncertain that deterrents stopped you (0) Does not apply	____
Reasons for Ideation ***What sort of reasons did you have for thinking about wanting to die or killing yourself? Was it to end the pain or stop the way you were feeling (in other words you couldn't go on living with this pain or how you were feeling) or was it to get attention, revenge or a reaction from others? Or both?*** (1) Completely to get attention, revenge or a reaction from others (4) Mostly to end or stop the pain (you couldn't go on (2) Mostly to get attention, revenge or a reaction from others living with the pain or how you were feeling) (3) Equally to get attention, revenge or a reaction from others (5) Completely to end or stop the pain (you couldn't go on and to end/stop the pain. living with the pain or how you were feeling) (0) Does not apply	____

SUICIDAL BEHAVIOR *(Check all that apply, so long as these are separate events; must ask about all types)*	**Lifetime**
Actual Attempt: A potentially self-injurious act committed with at least some wish to die, *as a result of act.* Behavior was in part thought of as method to kill oneself. Intent does not have to be 100%. If there is ***any*** intent/desire to die associated with the act, then it can be considered an actual suicide attempt. ***There does not have to be any injury or harm***, just the potential for injury or harm. If person pulls trigger while gun is in mouth but gun is broken so no injury results, this is considered an attempt. Inferring Intent: Even if an individual denies intent/wish to die, it may be inferred clinically from the behavior or circumstances. For example, a highly lethal act that is clearly not an accident so no other intent but suicide can be inferred (e.g., gunshot to head, jumping from window of a high floor/story). Also, if someone denies intent to die, but they thought that what they did could be lethal, intent may be inferred. ***Have you made a suicide attempt?*** ***Have you done anything to harm yourself?*** ***Have you done anything dangerous where you could have died?*** ***What did you do?*** ***Did you______ as a way to end your life?*** ***Did you want to die (even a little) when you_____?*** ***Were you trying to end your life when you _____?*** ***Or did you think it was possible you could have died from_____?*** ***Or did you do it purely for other reasons / without ANY intention of killing yourself (like to relieve stress, feel better, get sympathy, or get something else to happen)?*** *(*Self-Injurious Behavior without suicidal intent) If yes, describe: **Has subject engaged in Non-Suicidal Self-Injurious Behavior?**	**Yes No** ☐ ☐ Total # of Attempts ______ **Yes No** ☐ ☐

<table>
<tr><td colspan="3">Interrupted Attempt:
When the person is interrupted (by an outside circumstance) from starting the potentially self-injurious act (if not for that, actual attempt would have occurred).
Overdose: Person has pills in hand but is stopped from ingesting. Once they ingest any pills, this becomes an attempt rather than an interrupted attempt. Shooting: Person has gun pointed toward self, gun is taken away by someone else, or is somehow prevented from pulling trigger. Once they pull the trigger, even if the gun fails to fire, it is an attempt. Jumping: Person is poised to jump, is grabbed and taken down from ledge. Hanging: Person has noose around neck but has not yet started to hang - is stopped from doing so.
Has there been a time when you started to do something to end your life but someone or something stopped you before you actually did anything?
If yes, describe:</td><td>Yes No
☐ ☐

Total # of interrupted
______</td></tr>
<tr><td colspan="3">Aborted Attempt:
When person begins to take steps toward making a suicide attempt, but stops themselves before they actually have engaged in any self-destructive behavior. Examples are similar to interrupted attempts, except that the individual stops him/herself, instead of being stopped by something else.
Has there been a time when you started to do something to try to end your life but you stopped yourself before you actually did anything?
If yes, describe:</td><td>Yes No
☐ ☐

Total # of aborted
______</td></tr>
<tr><td colspan="3">Preparatory Acts or Behavior:
Acts or preparation towards imminently making a suicide attempt. This can include anything beyond a verbalization or thought, such as assembling a specific method (e.g., buying pills, purchasing a gun) or preparing for one's death by suicide (e.g., giving things away, writing a suicide note).
Have you taken any steps towards making a suicide attempt or preparing to kill yourself (such as collecting pills, getting a gun, giving valuables away or writing a suicide note)?
If yes, describe:</td><td>Yes No
☐ ☐</td></tr>
<tr><td colspan="3">Suicidal Behavior:
Suicidal behavior was present during the assessment period?</td><td>Yes No
☐ ☐</td></tr>
<tr><td>Answer for Actual Attempts Only</td><td>Most Recent Attempt Date:</td><td>Most Lethal Attempt Date:</td><td>Initial/First Attempt Date:</td></tr>
<tr><td>Actual Lethality/Medical Damage:
0. No physical damage or very minor physical damage (e.g., surface scratches).
1. Minor physical damage (e.g., lethargic speech; first-degree burns; mild bleeding; sprains).
2. Moderate physical damage; medical attention needed (e.g., conscious but sleepy, somewhat responsive; second-degree burns; bleeding of major vessel).
3. Moderately severe physical damage; medical hospitalization and likely intensive care required (e.g., comatose with reflexes intact; third-degree burns less than 20% of body; extensive blood loss but can recover; major fractures).
4. Severe physical damage; medical hospitalization with intensive care required (e.g., comatose without reflexes; third-degree burns over 20% of body; extensive blood loss with unstable vital signs; major damage to a vital area).
5. Death</td><td>Enter Code
______</td><td>Enter Code
______</td><td>Enter Code
______</td></tr>
<tr><td>Potential Lethality: Only Answer if Actual Lethality=0
Likely lethality of actual attempt if no medical damage (the following examples, while having no actual medical damage, had potential for very serious lethality: put gun in mouth and pulled the trigger but gun fails to fire so no medical damage; laying on train tracks with oncoming train but pulled away before run over).
0 = Behavior not likely to result in injury
1 = Behavior likely to result in injury but not likely to cause death
2 = Behavior likely to result in death despite available medical care</td><td>Enter Code
______</td><td>Enter Code
______</td><td>Enter Code
______</td></tr>
</table>

Appendix IV: Columbia-Suicide Severity Rating Scale (C-SSRS) Since Last Visit Version

Information obtained from: http://www.cssrs.columbia.edu/

SUICIDAL IDEATION	
Ask questions 1 and 2. If both are negative, proceed to "Suicidal Behavior" section. If the answer to question 2 is "yes", ask questions 3, 4 and 5. If the answer to question 1 and/or 2 is "yes", complete "Intensity of Ideation" section below.	Since Last Visit
1. Wish to be Dead Subject endorses thoughts about a wish to be dead or not alive anymore, or wish to fall asleep and not wake up. ***Have you wished you were dead or wished you could go to sleep and not wake up?*** If yes, describe:	**Yes No** □ □
2. Non-Specific Active Suicidal Thoughts General, non-specific thoughts of wanting to end one's life/commit suicide (e.g., *"I've thought about killing myself"*) without thoughts of ways to kill oneself/associated methods, intent, or plan during the assessment period. ***Have you actually had any thoughts of killing yourself?*** If yes, describe:	**Yes No** □ □
3. Active Suicidal Ideation with Any Methods (Not Plan) without Intent to Act Subject endorses thoughts of suicide and has thought of at least one method during the assessment period. This is different than a specific plan with time, place or method details worked out (e.g., thought of method to kill self but not a specific plan). Includes person who would say, *"I thought about taking an overdose but I never made a specific plan as to when, where or how I would actually do it…and I would never go through with it."* ***Have you been thinking about how you might do this?*** If yes, describe:	**Yes No** □ □
4. Active Suicidal Ideation with Some Intent to Act, without Specific Plan Active suicidal thoughts of killing oneself and subject reports having some intent to act on such thoughts, as opposed to *"I have the thoughts but I definitely will not do anything about them."* ***Have you had these thoughts and had some intention of acting on them?*** If yes, describe:	**Yes No** □ □
5. Active Suicidal Ideation with Specific Plan and Intent Thoughts of killing oneself with details of plan fully or partially worked out and subject has some intent to carry it out. ***Have you started to work out or worked out the details of how to kill yourself? Do you intend to carry out this plan?*** If yes, describe:	**Yes No** □ □
INTENSITY OF IDEATION	
The following features should be rated with respect to the most severe type of ideation (i.e., 1-5 from above, with 1 being the least severe and 5 being the most severe). ***Most Severe Ideation: _____ ____*** ***Type # (1-5) Description of Ideation***	Most Severe
Frequency ***How many times have you had these thoughts?*** (1) Less than once a week (2) Once a week (3) 2-5 times in week (4) Daily or almost daily (5) Many times each day	____
Duration ***When you have the thoughts, how long do they last?*** (1) Fleeting - few seconds or minutes (4) 4-8 hours/most of day (2) Less than 1 hour/some of the time (5) More than 8 hours/persistent or continuous (3) 1-4 hours/a lot of time	____
Controllability ***Could/can you stop thinking about killing yourself or wanting to die if you want to?*** (1) Easily able to control thoughts (4) Can control thoughts with a lot of difficulty (2) Can control thoughts with little difficulty (5) Unable to control thoughts (3) Can control thoughts with some difficulty (0) Does not attempt to control thoughts	____

Deterrents ***Are there things - anyone or anything (e.g., family, religion, pain of death) - that stopped you from wanting to die or acting on thoughts of committing suicide?*** (1) Deterrents definitely stopped you from attempting suicide (4) Deterrents most likely did not stop you (2) Deterrents probably stopped you (5) Deterrents definitely did not stop you (3) Uncertain that deterrents stopped you (0) Does not apply	____
Reasons for Ideation ***What sort of reasons did you have for thinking about wanting to die or killing yourself? Was it to end the pain or stop the way you were feeling (in other words you couldn't go on living with this pain or how you were feeling) or was it to get attention, revenge or a reaction from others? Or both?*** (1) Completely to get attention, revenge or a reaction from others (4) Mostly to end or stop the pain (you couldn't go on (2) Mostly to get attention, revenge or a reaction from others living with the pain or how you were feeling) (3) Equally to get attention, revenge or a reaction from others (5) Completely to end or stop the pain (you couldn't go on and to end/stop the pain living with the pain or how you were feeling) (0) Does not apply	____

SUICIDAL BEHAVIOR *(Check all that apply, so long as these are separate events; must ask about all types)*	Since Last Visit
Actual Attempt: A potentially self-injurious act committed with at least some wish to die, *as a result of act.* Behavior was in part thought of as method to kill oneself. Intent does not have to be 100%. If there is ***any*** intent/desire to die associated with the act, then it can be considered an actual suicide attempt. ***There does not have to be any injury or harm***, just the potential for injury or harm. If person pulls trigger while gun is in mouth but gun is broken so no injury results, this is considered an attempt. Inferring Intent: Even if an individual denies intent/wish to die, it may be inferred clinically from the behavior or circumstances. For example, a highly lethal act that is clearly not an accident so no other intent but suicide can be inferred (e.g., gunshot to head, jumping from window of a high floor/story). Also, if someone denies intent to die, but they thought that what they did could be lethal, intent may be inferred. ***Have you made a suicide attempt?*** ***Have you done anything to harm yourself?*** ***Have you done anything dangerous where you could have died?*** ***What did you do?*** ***Did you______ as a way to end your life?*** ***Did you want to die (even a little) when you_____?*** ***Were you trying to end your life when you _____?*** ***Or did you think it was possible you could have died from_____?*** ***Or did you do it purely for other reasons / without ANY intention of killing yourself (like to relieve stress, feel better, get sympathy, or get something else to happen)?*** (Self-Injurious Behavior without suicidal intent) If yes, describe: **Has subject engaged in Non-Suicidal Self-Injurious Behavior?**	**Yes No** ☐ ☐ Total # of Attempts ______ **Yes No** ☐ ☐
Interrupted Attempt: When the person is interrupted (by an outside circumstance) from starting the potentially self-injurious act *(if not for that, actual attempt would have occurred).* Overdose: Person has pills in hand but is stopped from ingesting. Once they ingest any pills, this becomes an attempt rather than an interrupted attempt. Shooting: Person has gun pointed toward self, gun is taken away by someone else, or is somehow prevented from pulling trigger. Once they pull the trigger, even if the gun fails to fire, it is an attempt. Jumping: Person is poised to jump, is grabbed and taken down from ledge. Hanging: Person has noose around neck but has not yet started to hang - is stopped from doing so. ***Has there been a time when you started to do something to end your life but someone or something stopped you before you actually did anything?*** If yes, describe:	**Yes No** ☐ ☐ Total # of interrupted ______

Aborted Attempt: When person begins to take steps toward making a suicide attempt, but stops themselves before they actually have engaged in any self-destructive behavior. Examples are similar to interrupted attempts, except that the individual stops him/herself, instead of being stopped by something else. ***Has there been a time when you started to do something to try to end your life but you stopped yourself before you actually did anything?*** If yes, describe:	**Yes No** ☐ ☐ Total # of aborted ______
Preparatory Acts or Behavior: Acts or preparation towards imminently making a suicide attempt. This can include anything beyond a verbalization or thought, such as assembling a specific method (e.g., buying pills, purchasing a gun) or preparing for one's death by suicide (e.g., giving things away, writing a suicide note). ***Have you taken any steps towards making a suicide attempt or preparing to kill yourself (such as collecting pills, getting a gun, giving valuables away or writing a suicide note)?*** If yes, describe:	**Yes No** ☐ ☐
Suicidal Behavior: Suicidal behavior was present during the assessment period?	**Yes No** ☐ ☐
Suicide:	**Yes No** ☐ ☐
Answer for Actual Attempts Only	Most Lethal Attempt Date:
Actual Lethality/Medical Damage: 0. No physical damage or very minor physical damage (e.g., surface scratches). 1. Minor physical damage (e.g., lethargic speech; first-degree burns; mild bleeding; sprains). 2. Moderate physical damage; medical attention needed (e.g., conscious but sleepy, somewhat responsive; second-degree burns; bleeding of major vessel). 3. Moderately severe physical damage; *medical* hospitalization and likely intensive care required (e.g., comatose with reflexes intact; third-degree burns less than 20% of body; extensive blood loss but can recover; major fractures). 4. Severe physical damage; *medical* hospitalization with intensive care required (e.g., comatose without reflexes; third-degree burns over 20% of body; extensive blood loss with unstable vital signs; major damage to a vital area). 5. Death	*Enter Code* _______
Potential Lethality: Only Answer if Actual Lethality=0 Likely lethality of actual attempt if no medical damage (the following examples, while having no actual medical damage, had potential for very serious lethality: put gun in mouth and pulled the trigger but gun fails to fire so no medical damage; laying on train tracks with oncoming train but pulled away before run over). 0 = Behavior not likely to result in injury 1 = Behavior likely to result in injury but not likely to cause death 2 = Behavior likely to result in death despite available medical care	*Enter Code* _______

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REGIMEN-SPECIFIC APPENDIX C

FOR CNM-Au8

Regimen-Specific Appendix Date: 6/3/2020

Version Number: 3.0

TABLE OF CONTENTS

SIGNATURE PAGE

I have read the attached Regimen-Specific Appendix (RSA) entitled, “REGIMEN C: CNM-Au8” dated June 3, 2020 (Version 3.0) and agree to abide by all described RSA procedures. I agree to comply with the International Conference on Harmonisation Tripartite Guideline on Good Clinical Practice, applicable FDA regulations and guidelines identified in 21 CFR Parts 11, 50, 54, and 312, central Institutional Review Board (IRB) guidelines and policies, and the Health Insurance Portability and Accountability Act (HIPAA).

By signing the RSA, I agree to keep all information provided in strict confidence and to request the same from my staff. Study documents will be stored appropriately to ensure their confidentiality. I will not disclose such information to others without authorization, except to the extent necessary to conduct the study.

Site Name: ______________________________

Site Investigator: ______________________________

Signed: ____________________ Date: ____________

LIST OF ABBREVIATIONS

Abbreviation	Definition
ALSAQ-40	Amyotrophic Lateral Sclerosis Assessment Questionnaire-40
ASTM	American Society for Testing and Materials
ATP	Adenosine Trinucleotide Phosphate
Au	Gold
AUC	Area under the curve
AUC(0-24)	Area under the curve for 24 hours
CNM-Au8	Aqueous suspension of clean surfaced nanocrystals consisting of gold atoms self-organized into crystals of various faceted, geometrical shapes
CNS	Central nervous system
CNS-BFS	Center for Neurologic Study Bulbar Function Scale
CSF	Cerebrospinal fluid or cerebral spinal fluid
fALS	Familial ALS
FVC	Forced Vital Capacity
HED	Human equivalent dose
HDPE	High density polyethylene
hr	Hour
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
IP	Investigational Product
K2EDTA	Dipotassium ethylenediaminetetraacetic acid
kg	Kilogram
m	meter
MAD	Multiple ascending dose
mg	Milligram
mL	Milliliter
MRS	Magnetic Resonance Spectroscopy
mRNA	Messenger Ribonucleic acid (RNA)
MTD	Maximum tolerated dose
NAD+	Oxidized form of nicotinamide adenine dinucleotide
NADH	Reduced form of nicotinamide adenine dinucleotide
NADPH	Reduced form of nicotinamide adenine dinucleotide phosphate
NaHCO3	Sodium bicarbonate (baking soda)

Abbreviation	Definition
ng	nanogram
nm	nanometer
NOAEL	No observed adverse effect level
OLE	Open Label Extension
PD	Pharmacodynamic
PPP	Pentose phosphate pathway
RSA	Regimen-Specific Appendix
SAE	Serious adverse event
SOD1	Superoxide dismutase
SVC	Slow Vital Capacity
TEM	Transmission electron microscopy
TK	Toxicokinetic
USP	United States Pharmacopeia
VC	Vital Capacity

REGIMEN-SPECIFIC APPENDIX SUMMARY

Regimen-Specific Appendix C
For CNM-Au8

Rationale and RSA Design

CNM-Au8 is a suspension of clean-surfaced, faceted gold nanocrystals whose catalytic activity independently impacts energetic, metabolic, and redox pathways, resulting in significant neuroprotection from oxidative and excitotoxic insults.

CNM-Au8 is a nanocatalyst that is administered orally, penetrates the blood brain barrier, and has a unique mechanism of action involving: (1) the catalytic production of nicotinamide adenine dinucleotide (NAD), a key co-factor in the ATP-generating pathways of all cells, (2) catalytic anti-oxidative activity resulting in cellular protection from oxidative stress, [redacted] Importantly, CNM-Au8 demonstrated significant neuroprotection of healthy human motor neurons from cell death induced by toxic ALS-participant derived astrocytes, and of primary rat motor neurons from glutamatergic excitotoxicity. Oral delivery of CNM-Au8 to a mouse model of ALS resulted in the recovery of functional behaviors and significant extension of lifespan.

Given the involvement of cellular bioenergetic failure in the pathogenesis of ALS, the development of a disease modifying therapeutic that addresses the energetic dysregulation underlying the progressive accumulation of oxidative stress [redacted] is a rational therapeutic strategy.

This clinical protocol will test CNM-Au8 at doses of 30 mg and 60 mg versus placebo in a randomized, double-blind manner, to determine if CNM-Au8 can demonstrate sufficient efficacy and safety on accepted clinical and paraclinical outcomes for the treatment of ALS. The investigation of two doses of the study drug is based on counsel from the FDA recommending including a dose finding strategy in the study. The 30 mg and 60 mg doses of CNM-Au8 selected for the study achieve at least comparable AUC exposure in humans as those shown to demonstrate neuroprotection efficacy in nonclinical animals studies, and demonstrated minimal treatment emergent adverse events in the Phase 1 First-In-Human study in healthy volunteers.

Allocation to Treatment Regimens
Participants must first be screened under the Master Protocol before they are randomized to an RSA.

As soon as pre-defined criteria for futility for the RSA are met, or the target number of randomized participants has been reached, enrollment will stop in the RSA.

Number of Planned Participants and Treatment Groups

The number of planned participants for this regimen is 160.

There are 2 treatment groups for this regimen, active and placebo. Participants will be randomized in a 3:1 ratio to active treatment or placebo (i.e., 120 active: 40 placebo). Half of the active treatment group will receive CNM-Au8 30mg/day and the other half will receive 60mg/day (i.e., 60 at 30mg/day: 60 at 60mg/day).

Planned Number of Sites

Research participants will be enrolled from approximately 60 centers in the US.

Treatment Duration

The maximum duration of the placebo-controlled treatment portion is 24 weeks.

Follow-up Duration

At the conclusion of the 24-week placebo-controlled Treatment Period of the study, all participants will either schedule a 28-day follow up phone call and end their participation in the regimen or have the option to receive CNM-Au8 in the Open Label Extension phase of the study. The duration of the Open Label Extension phase is planned for a minimum of 52 weeks.

Total Planned Trial Duration

For participants completing the placebo-controlled Treatment Period of the Study, the planned amount of time for a participant in the trial is up to 34 weeks, or about 8 months. This duration assumes a 6-week screening window, a 24-week placebo-controlled treatment period, and a 4-week safety follow-up period for those participants who do not enter the Open Label Extension. Participants will complete approximately 10 study visits during the placebo-controlled treatment period of the study.

If the participant opts into the subsequent Open Label Extension, the total planned amount of time for a participant in the trial is approximately 86 weeks. This duration assumes a 6-week screening window, 24-week placebo-controlled treatment period, a 52-week Open Label Extension period, and a 4-week safety follow-up period. Participants will complete 10 study visits across the planned 52-week Open Label Extension phase of the study.

SCHEDULE OF ACTIVITIES – PLACEBO-CONTROLLED PERIOD[14]

As per the Schedule of Activities (SOA) below, visits must occur every 4 weeks and will be alternatively clinic-, phone-, or telemedicine-based, as applicable. There is a maximum 24-week duration of placebo-controlled treatment for a Regimen.

Activity (page 1 of 2)	Master Protocol or Regimen-Specific	Master Protocol Screening[1]	Regimen Specific Screening[1]	Baseline	Week 2	Week 4[16, 17]	Week 8[16, 17]	Week 12	Week 16[17]	Week 20	Week 24 or Early Term. Visit[14, 16]	Follow-Up Safety Call[11, 14]
		Clinic	Clinic	Clinic	Phone	Clinic	Clinic	Phone	Clinic	Phone	Clinic	Phone
		-42 to -1 Days	-41 to 0 Day	Day 0	Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day 168 ±7	28 days after last dose ±3 days
Written Informed Consent[2]	Master	X	X									
Written Informed Consent - OLE	Master								X			
Inclusion/Exclusion Review	Master	X	X[3]									
ALS & Medical History	Master	X										
Demographics	Master	X										
Physical Examination	Master	X										
Neurological Exam	Master	X										
Vital Signs[4]	Master	X		X		X	X		X		X	
Slow Vital Capacity	Master	X[18]		X			X		X		X	
Muscle Strength Assessment	Master			X			X		X		X	
ALSFRS-R	Master	X		X		X	X	X	X	X	X	
ALSAQ-40	Regimen			X							X	
CNS Bulbar Function Scale	Regimen			X			X		X		X	
12-Lead ECG	Master	X									X	
Clinical Safety Labs[5]	Master	X		X		X	X		X		X	
PK/PD Biomarker Blood Collection[12]	Regimen			X		X	X				X	
Biomarker PD Urine Collection	Regimen			X							X	
Biomarker Blood Collection	Master			X			X		X		X	

Activity (page 2 of 2)	Master Protocol or Regimen-Specific	Master Protocol Screening[1]	Regimen Specific Screening[1]	Baseline	Week 2	Week 4[16, 17]	Week 8[16, 17]	Week 12	Week 16[17]	Week 20	Week 24 or Early Term. Visit[14, 16]	Follow-Up Safety Call[11, 14]
		Clinic	Clinic	Clinic	Phone	Clinic	Clinic	Phone	Clinic	Phone	Clinic	Phone
		-42 to -1 Days	-41 to 0 Day	Day 0	Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day 168 ±7	28 days after last dose ±3 days
Biomarker Urine Collection	Master			X			X		X		X	
DNA Collection[7] (optional)	Master			X								
CSF Collection (optional)	Master			X					X[15]			
Concomitant Medication Review	Master	X	X	X		X	X	X	X	X	X	X
Adverse Event Review[6]	Master	X	X	X	X	X	X	X	X	X	X	X
Columbia-Suicide Severity Rating Scale	Master			X		X	X		X		X	
Install Smartphone App	Regimen			X								
Voice Recording[9]	Regimen			X		X	X		X		X	
Uninstall Smartphone App	Regimen										X	
Randomization to Regimen	Master	X										
Randomization within Regimen	Master		X									
Administer/Dispense Investigational product	Master			X[8]			X		X		X[10]	
Drug Accountability/Compliance	Master				X	X	X	X	X	X	X	
Exit Questionnaire	Master										X	
Vital Status Determination[13]	Master										X	

1 Master Protocol Screening procedures must be completed within 42 days to 1 day prior to the Baseline Visit. The Regimen-Specific Screening Visit and Baseline Visit should be combined, if possible.
2 During the Master Protocol Screening Visit, participants will be consented via the Master Protocol informed consent form (ICF). After a participant is randomized to a regimen, participants will be consented a second time via the regimen-specific ICF.
3 At the Regimen Specific Screening Visit, participants will have regimen-specific eligibility criteria assessed.
4 Vital signs include weight, systolic and diastolic pressure, respiratory rate, heart rate and temperature. Height measured at Master Protocol Screening Visit only.

5 Clinical safety labs include hematology (CBC with differential), complete chemistry panel, thyroid function and urinalysis. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during the study. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.

6 Adverse events that occur after signing the consent form will be recorded.

7 The DNA sample can be collected after baseline if a baseline sample is not obtained or the sample is not usable.

8 Administer first dose of investigational product only after Baseline Visit procedures are completed.

9 In addition to study visits outlined in the SOA, participants will be asked to complete twice weekly voice recordings at home.

10 Drug will only be dispensed at this visit if the participant continues in the OLE.

11 Participants will only have a Follow-Up Safety Call at this time if they *do not* continue on in the OLE. Participants who continue into OLE will have a Follow-Up Safety Call after their last dose of study drug during the OLE phase.

12 Whole blood and plasma will be collected prior to administration of the daily dose at the Baseline and Week 24 visit. Plasma will be collected prior to administration of the daily dose at the Week 4 and Week 8 visits. Refer to the RSA laboratory manual for collection and processing parameters.

13 Vital status, defined as a determination of date of death or death equivalent or date last known alive, will be determined for each randomized participant at the end of the placebo-controlled portion of their follow-up (generally the Week 24 visit, as indicated). If at that time the participant is alive, his or her vital status should be determined again at the time of the last patient last visit (LPLV) of the placebo-controlled portion of a given regimen.

14 The maximum window between study visits during the placebo-controlled period may not exceed 64 days.

15 If the CSF collection is unable to be performed for logistical reasons, such as scheduling, at the Week 16 Visit, it may be performed at the Week 24 Visit.

16 Participants should be instructed to hold the study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

17 Visit may be conducted via phone or telemedicine with remote services instead of in-person if this is needed to protect the safety of the participant due to a pandemic.

18 If required due to pandemic-related restrictions, Forced Vital Capacity (FVC) performed in a Pulmonary Function Laboratory may be used for eligibility (Master Protocol Screening ONLY).

SCHEDULE OF ACTIVITIES – OPEN LABEL EXTENSION (OPTIONAL)[7]

Activity (page 1 of 1)	Master Protocol or Regimen-Specific	Open Label Extension (Optional)[5]									
		Week 2	Week 4[9]	Week 8[9]	Week 12	Week 16[8,9]	Week 20	Week 24	Week 28[8,9]	Week 40[9]	Week 52[8]
		Phone	Clinic	Clinic	Phone	Clinic	Phone	Phone	Clinic	Clinic	Clinic
		Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day168 ±3	Day 196 ± 14	Day 280 ± 14	Day 364 ± 14
Vital Signs[1]	Master		X	X		X			X	X	X
Slow Vital Capacity	Master		X	X		X			X	X	X
ALSFRS-R	Master		X	X	X	X	X	X	X	X	X
ALSAQ-40	Regimen								X		X
CNS Bulbar Function Scale	Regimen			X		X			X	X	X
Clinical Safety Labs[2]	Master		X	X		X			X	X	X
PK/PD Biomarker Blood Collection	Regimen					X			X		X
Biomarker PD Urine Collection	Regimen					X			X		X
Biomarker Blood Collection	Master					X			X		X
Biomarker Urine Collection	Master					X			X		X
Concomitant Medication Review	Master		X	X		X			X	X	X
Adverse Event Review[3]	Master	X	X	X	X	X	X	X	X	X	X
Columbia-Suicide Severity Rating Scale	Master		X	X		X			X	X	X
Administer/Dispense Investigational product	Master			X		X			X	X	X

Drug Accountability/Compliance	Master	X	X	X	X	X	X	X	X	X	X

1 Vital signs include weight, systolic and diastolic pressure, respiratory rate, heart rate and temperature. Height in cm measured at Master Protocol Screening Visit only.
2 Clinical safety labs include hematology (CBC with differential), complete chemistry panel, liver function tests, thyroid function and urinalysis. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during the study. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.
3 Adverse events that occur after signing the consent form will be recorded.
4 Participants who continue into OLE will have a Follow-Up Safety Call (as described in the body of this RSA) after their last dose of study drug during the OLE phase.
5 The duration of the OLE is 52 weeks.
6 Participants who continue into the OLE and then withdraw consent or early terminate will be asked to complete an Early Termination Visit and Follow-Up Safety Call as described in the body of this RSA.
7 The maximum window between study visits during the OLE may not exceed 64 days for the Week 8 and Week 16 visits; and 96 days for the Week 28, Week 40, and Week 52 visits.
8 Participants should be instructed to hold the study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.
9 Visit may be conducted via phone or telemedicine with remote services instead of in-person if this is needed to protect the safety of the participant due to a pandemic.

1. INTRODUCTION

Regimen C: CNM-Au8

1.1 CNM-Au8 Background Information

Amyotrophic lateral sclerosis (ALS) is a late-onset, fatal neurodegenerative disease affecting motor neurons with an incidence of approximately 1 in 100,000. Most ALS cases are sporadic, with 5–10% of cases being familial ALS (Zarei et al., 2015). Both sporadic and familial ALS are associated with degeneration and loss of cortical and spinal motor neurons, resulting in muscle wasting and weakness, eventually leading to respiratory failure and death. The etiology of ALS remains unknown. Emerging research has discovered interconnecting pathophysiological mechanisms amongst the seemingly disparate genetic mutations, implicated in both familial (fALS) and sporadic (sALS) ALS. Importantly, these emerging data suggest that impairments related to oxidative stress and mitochondrial function appear to be common across many underlying genetic abnormalities. Genetic mutations in superoxide dismutase (SOD1), regulation of RNA processing and metabolism (represented by mutations in FUS, TDP43 and several more genes) all appear to result in both oxidative stress and RNA dysregulation, indicative of an underlying bioenergetic failure in ALS. Bioenergetic dysfunction affects not only motor neurons themselves but also astrocytes and oligodendrocytes that provide crucial energetic support to motor neuron survival. Postmortem analyses of brain tissues from both fALS and sALS participants exhibit evidence of widespread accumulation of oxidative damage to proteins, lipids, and DNA. Transgenic mice expressing mutant SOD1 forms have evidence of nerve and spinal cord tissue damage that recapitulates key aspects of ALS and demonstrates clear evidence of excessive protein and lipid oxidation (D'Amico et al., 2013b). Oxidative stress also leads to the accumulation of protein aggregates such as TDP43 (Bozzo et al., 2017a), which are invariably found in ALS motor neurons, and which indicate failures of protein degradation systems such as autophagy (Blokhuis et al., 2013).

Elevated oxidative stress has been shown to impact the processing of gene transcripts important for motor neuron survival. This observation connecting oxidative stress to RNA processing was demonstrated when paraquat, a well-known inducer of oxidative stress, was shown to induce striking changes in RNA splicing in neuroblastoma cells (Maracchioni et al., 2007). Two of the pre-mRNAs whose splicing was shown to be altered in this study are transcripts of genes known to be important for motor neuron survival, Survival Motor Neuron (SMN) and Apoptotic Peptidase Activating Factor 1 (APAF1) (Maracchioni et al., 2007). The mis-localization of the RNA-binding protein TDP43 in ALS is further evidence of the connection between oxidative stress and RNA dysmetabolism. In general, TDP43 is involved in pre-RNA processing in the nucleus, where it functions in transcriptional control. However, oxidative stress causes the

TDP43 to translocate to the cytoplasm, where it accumulates in stress granules, as protein aggregates. Sequestration of an RNA regulator such as TDP43 in the cytoplasm and away from the nucleus is thought to prevent it from carrying out its normal functions and thereby can lead to dysregulation of RNA processing and transcription. The converse relationship has similarly been shown, namely that mis-regulation of RNA metabolism increases oxidative stress within the CNS. Mutations in RNA regulators FUS and TDP43 cause disruptions in the expression of key nuclear-encoded mitochondrial genes (Zhang et al., 2014) as well as in the expression of energy metabolism regulators such as PGC-1 alpha and oxidative stress reducers such as MnSOD and catalase (Sanchez-Ramos et al., 2011). Therefore, the pathophysiologic disease mechanism of ALS appears to involve a positive feedback cycle of oxidative stress and protein aggregation impacting RNA regulation which in turn leads to a sustained increased oxidative stress disease state, ending in motor neuron death (Bozzo et al., 2017b).

Unifying both themes of RNA dysregulation and oxidative stress is the concept that ALS is a disease of energetic dysmetabolism (Dupuis et al., 2011; Tefera and Borges, 2016; Vandoorne et al., 2018). Motor neurons consume high amounts of energy to function and are therefore exquisitely sensitive to apoptosis if their energetic demands cannot be met. The main energy-producing organelle of the cell, the mitochondria, exhibit dysfunction in ALS on multiple levels. For example, overall there are fewer mitochondria, as measured by mitochondrial DNA, in the spinal cords of both fALS and sALS participants (Wiedemann et al., 2002). Presynaptic mitochondrial swelling in motor neurons has been observed in both ALS participants (Siklos et al., 1996), as well as in a SOD1 transgenic mouse model prior to disease onset (Manfredi and Xu, 2005). Swollen and vacuolarized mitochondria are indicative of impaired mitochondrial function (Siklos et al., 1996), which is corroborated by studies showing that there is decreased activity of the electron transport chain in spinal cord mitochondria of ALS participants (Wiedemann et al., 2002). Because reduced respiration and reduced ATP synthesis preceded clinical symptom onset in SOD1G93A mice (Jung et al., 2002; Mattiazzi et al., 2002; Szelechowski et al., 2018), energetic dysmetabolism appears to play an important role in the etiology of this disease. Overexpression of PGC-1 alpha, which stimulates mitochondrial biogenesis, improved survival, motor neuron function and motor neuron survival in mutant SOD1G93A mice (Zhao et al., 2011), indicating that improving mitochondrial function and/or efficiency may be a successful therapeutic strategy for ALS (Vandoorne et al., 2018).

Energetic dysmetabolism in ALS is further underscored by studies uncovering carbohydrate metabolism abnormalities and ATP deficits in ALS motor neurons (Vandoorne et al., 2018). FDG-PET imaging has demonstrated a correlation between reduced ALS brain glucose uptake and cognitive impairment (Canosa et al., 2016) as well as overall disease severity (Dalakas et al., 1987). Some groups who conducted FDG-PET studies showing reduced glucose uptake in ALS participant brains suggested CNS hypometabolism can be used as an early diagnostic indication of ALS (Van Laere et al., 2014; Van Weehaeghe et al., 2016). While it is difficult to determine whether lowered glucose utilization detected in imaging studies is due to reduced neuronal catabolism of glucose or due to motor neuron loss, there are further lines of evidence to suggest energetic dysmetabolism in ALS cells. A proteomic study of ALS participant fibroblasts indicated a significant reduction in key enzymes involved in glycolysis (Szelechowski et al., 2018), and expression profiling of sALS participant motor cortexes showed downregulation of glycolytic genes (Lederer et al., 2007). A GWAS analysis involving 12577 cases and 23475 controls identified glycolysis/gluconeogenesis as among the top seven biological pathways represented by the genes identified as associated with ALS (Du et al., 2018). Because glia preferentially utilize glycolysis, these results may indicate a dysfunction of glia in providing energetic support to motor neurons in ALS participants. Energetic stress on motor neurons may increase their levels of oxidative stress, as observed in post-mortem brain samples of ALS participants (Ferrante et al., 1997), and lead to motor neuron death.

1.2 CNM-Au8 Therapeutic Rationale

CNM-Au8 is a concentrated aqueous suspension of clean-surfaced nanocrystals consisting solely of Au atoms organized into clean surfaced crystals of highly faceted, substantially uniform geometrical shapes that act as nanocatalysts, supporting cellular bioenergetic reactions.

CNM-Au8 is supplied for dosing orally in sodium bicarbonate buffered USP purified water. Each mL of CNM-Au8 suspension at 500 µg/mL is estimated to contain between 100 - 500 trillion highly faceted Au nanocrystals. IP will consist of 60 mL high density polyethylene (HDPE) bottles containing CNM-Au8 at concentrations of 250 µg/mL (15 mg) or 500 µg/mL (30 mg), or placebo, administered orally, once daily in the morning, at least 30 minutes prior to food intake. Study participants will consume two (2) bottles of IP each morning for total daily intake of 120 mL equivalent to CNM-Au8 30mg or 60mg, or placebo.

1.2.1 Investigational Product Characteristics

The median diameter of CNM-Au8 nanocrystals is approximately 13 nm, as determined by transmission electron microscopy (TEM). [redacted] of measured nanocrystal diameters and approximate geometrical shapes (e.g., low volume estimate: disc-like approximation, aspect ratio of 0.2; maximum volume estimate: spheroid, aspect ratio of 1.0), each Au nanocrystal has an approximate composition ranging from 13,000 – 66,000 Au atoms per nanocrystal at the 13 nm median diameter with a corresponding molar mass ranging between 2.7 x 10^3 kDA to 1.3 x 10^4 kDA. Summary estimates for CNM-Au8 mass, volume, and particle characteristics are described below in Table 1. Each mL of CNM-Au8 suspension at 500 µg/mL is estimated to contain between 100 - 500 trillion highly faceted Au nanocrystals.

Table 1. Estimated Volume, Mass, and Particle Characteristics of CNM-Au8

Metric (CNM-Au8 500 µg/mL, 60 mL Dose)	**Disc-Like Approximation Minimum (Aspect 0.2)**	**Spherical Approximation Maximum (Aspect 1.0)**
Median Au Nanocrystal Diameter (nm)	13	
Au Nanocrystal Volume (nm^3)	2.3 x 10^2	1.2 x 10^3
Au Nanocrystal Surface Area (nm^2)	3.2 x 10^2	5.3 x 10^2
Au Atoms per Nanocrystal (count)	1.4 x 10^4	6.8 x 10^4
Au Nanocrystal Molecular Weight (kDa)	2.7 x 10^3	1.3 x 10^4
Total Au Nanocrystal Surface Area per mL (cm^2)	3.6 x10^2	1.2 x10^2
Au Nanocrystals per mL CNM-Au8 (count)	1.1 x 10^14	2.3 x 10^13
Au Nanocrystals per 60 mL Dose (count)	3.4 x 10^15	6.8 x 10^14

1.2.2 Preclinical Data Supporting CNM-Au8 in the Treatment of ALS

CNM-Au8 passes into the systemic circulation via intestinal absorption. Cellular uptake of CNM-Au8, following oral administration, has been demonstrated in a variety of cells and tissue matrices including macrophages, oligodendrocytes, neurons, and brain tissue, utilizing Inductively Coupled Plasma Mass Spectrometry (ICP-MS) and high resolution TEM visualization of nanocrystals.

Preclinical characterization of the *in vitro* and *in vivo* properties of CNM-Au8 demonstrated that it acts to protect neuronal populations from chemical, inflammatory, and hypoxic insults through a series of unique catalytic mechanisms involving [redacted] the nicotine adenine dinucleotide redox couple (NAD+/NADH) to boost glycolytic energy production, [redacted] 3) [redacted] the SOD-like inactivation of reactive oxygen species [redacted] to protect cells from oxidative damage and mitochondrial dysfunction.
The nicotinamide adenine dinucleotide redox couple (NAD+, NADH) plays a central role in the energy metabolism of all living cells. NAD+, NADH and their relative intracellular ratio (NAD+/NADH) are linked to regulation of energy homeostasis, neuroprotection, immune function, chromosome stability, DNA repair mechanisms, sleep and circadian rhythms, and longevity (Canto et al., 2015; Imai, 2010a, b; Nikiforov et al., 2015; Ying, 2008).

Fundamentally, NAD+ and NADH are essential coenzymes in the adenosine triphosphate (ATP)-generating reactions driving both glycolysis and oxidative phosphorylation. More specifically, in aerobic glycolysis, NAD+ availability is integral to the enzymatic cascade from glyceraldehyde 3-phosphate via glyceraldehyde phosphate dehydrogenase to 1,3-bisphosphoglyceric acid. Similarly, [redacted] electrons are removed via NADH oxidation in Complex I of the electron transport chain during oxidative phosphorylation, resulting in NAD+, and ultimately these electrons are transferred to a lipid-soluble carrier, ubiquinone.
In addition, NAD+ and its metabolites act as binding substrates for a wide range of proteins including the metabolic and transcription-regulating sirtuins, the poly-ADP-ribose polymerases (PARPs) involved in DNA repair, and the cyclic ADP-ribose synthases (cADPRs) such as CD38 that serve to regulate Ca2+ signaling involved in cell cycle control and insulin signaling (Canto et al., 2015). In the Wallerian Degeneration Slow Dominant Mutant Mouse, heightened expression of NMNAT1, resulting in increased NAD+, has been attributed to protect against the axonal degeneration, otherwise observed in these mice (Sasaki et al., 2009).

Quantitative measurement of regional brain area NAD+ concentrations and redox states in humans has recently been demonstrated using a novel ^{31}P-Magnetic Resonance Spectroscopy (MRS) imaging technique (Chouinard et al., 2017; Lu et al., 2016). In people, brain NAD+/NADH redox potential is inversely correlated with age. ^{31}P-MRS of the visual cortex of young (21-26 year old), middle-aged (33-36 year old), and aged (59-68 year old) healthy individuals demonstrated a linear decline of NAD+ redox potential with increasing age (Zhu et al., 2015). This age-related decrease in cerebral energy metabolism is believed to be associated with the bioenergetic failure underlying many neurodegenerative diseases, because NAD+/NADH redox potential is essential for fundamental ATP-generating processes such as glycolysis and mitochondrial oxidative phosphorylation. These bioenergetic processes power all cellular processes including 'housekeeping' functions that are responsible for maintaining overall cellular health including processes of autophagy, apoptosis, and the unfolded protein response (UPR) (Canto et al., 2015; Chua and Tang, 2013a; Villanueva-Paz et al., 2016). Boosting brain NAD+ levels in a mouse model of AD reduced DNA damage, neuroinflammation, and apoptosis while improving cognitive function in multiple behavioral tests and restored hippocampal synaptic plasticity (Hou et al., 2018).

The first demonstration of catalyzed oxidation of NADH to NAD+ by gold nanoparticles was demonstrated by Huang et al. (2005) using a simple cell-free assay. In order to compare the catalytic oxidation rate of clean-surfaced CNM-Au8 against similarly-sized gold nanoparticles made using citrate reduction, the same cell-free assay utilized by Huang et al. was repeated comparing gold nanoparticles sourced from the U.S. National Institutes of Standards and Technology (NIST). The resulting change in the 339 nm NADH absorbance peak was assessed while investigating each of the gold nanoparticles at the same concentrations. In all cases, CNM-Au8 nanocrystals consistently showed significantly superior catalytic activities regardless of size and/or method of preparation of the comparator gold nanoparticles.

Figure 1. CNM-Au8 NADH Catalysis Rates

Figure 1A. CNM-Au8 Catalytic Effects vs. NIST Standard Citrated AuNP

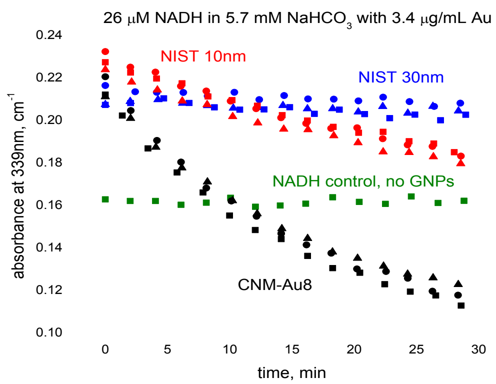
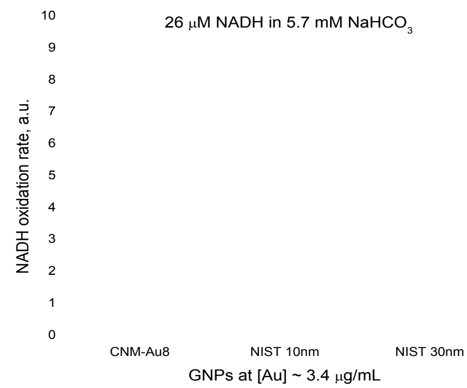


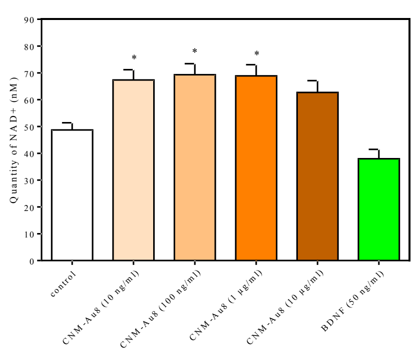
Figure 1B. Relative NADH Oxidation Rates vs. NIST Citrated AuNP



CNM-Au8 treatment of rodent central nervous system cells in vitro also significantly increases levels of both NAD+ and NADH (Figure 2).

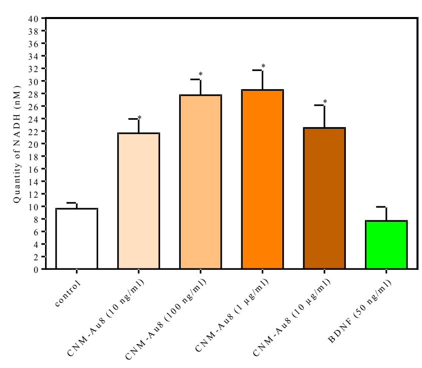
Figure 2. Effects of CNM-Au8 on NAD Levels in Primary Rodent Mesencephalic Cultures

Figure 2A. Effect of CNM-Au8 on NAD+ Levels



*p<0.05 vs control condition One-Way ANOVA followed by Dunnett's test

Figure 2B. Effect of CNM-Au8 on NADH Levels



*p<0.05 vs control condition One-Way ANOVA followed by Dunnett's test

[redacted]

[redacted]

[redacted]

[redacted]

Excessive reactive oxygen exposure disrupts cellular redox homeostasis, damages intracellular organelles as well as DNA and proteins, and may also underpin pathophysiologic mechanisms of ALS. Super oxide dismutases (SODs) evolved as a component of the cell's antioxidant defense system to regulate the levels of reactive oxygen species (ROS) and prevent damage from oxidative stress. SOD enzymes are found in the cytoplasm and mitochondria where they convert oxygen radicals to O_2 or H_2O_2, which can in turn be converted to water and O_2 by catalases. A dose-dependent reduction of ROS levels was observed in differentiating oligodendrocyte precursor cells in primary culture with CNM-Au8 treatment.

Figure 4 . Effects of CNM-Au8 on SOD and ROS Generation

Figure 4A. Effects on SOD Activity Assay

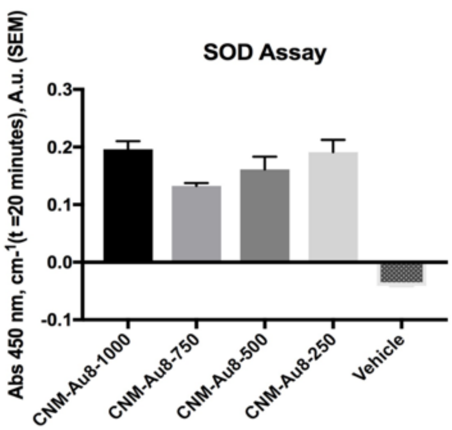


Figure 4B. Effects on ROS Generation in Purified Murine OPC cultures

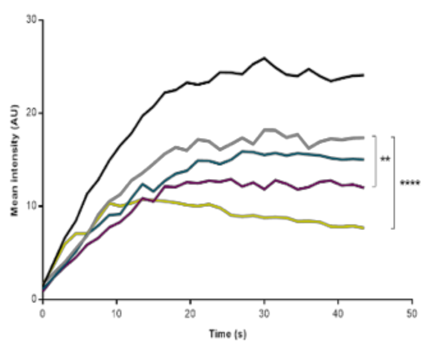
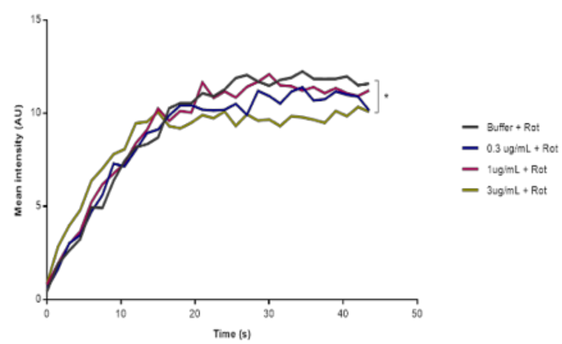
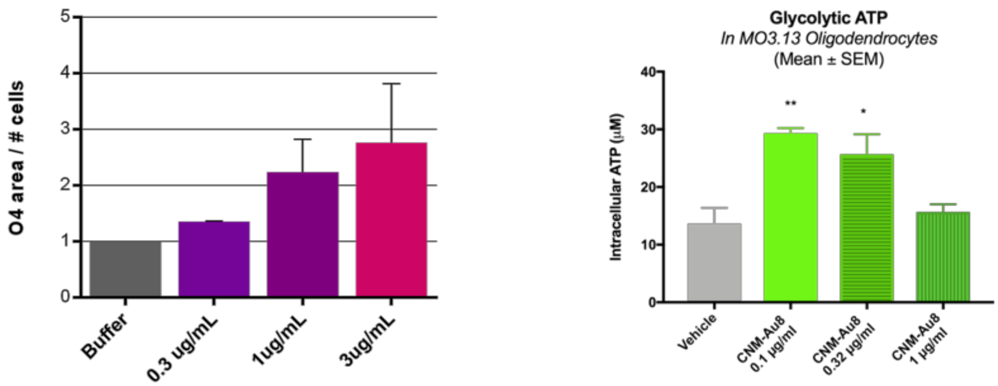


Figure 4C. Effects on ROS Generation in Murine OPC Cultures Plus Rotenone



Demonstration of SOD-like catalytic activity with CNM-Au8 treatment has several important implications. First, neurons and glial cells are energetically demanding and generate large amounts of ROS as byproducts of the energetic processes necessary for their function. Theories of aging and neurodegeneration suggest that as CNS cells age, they are less able to efficiently remove ROS and thereby succumb to the consequences of long-term ROS exposure. As CNS cells are exquisitely sensitive to ROS, they are among the first cell types to degenerate, leading to the cognitive, psychomotor, and movement impairments observed in diseases such as ALS, Alzheimer's disease, and Parkinson's disease (Angelova and Abramov, 2018). In addition, SOD1 may act not only as a regulator of ROS but also as a part of a glucose-oxygen sensing mechanism of a cell to determine whether the cell utilizes oxidative phosphorylation or aerobic glycolysis for ATP production (Reddi and Culotta, 2013). A study of oligodendrocyte (OL) energetics showed that human OLs preferentially use aerobic glycolysis during differentiation and myelination (Rone et al., 2016). Data with CNM-Au8 demonstrate that treatment switches OPCs from proliferation to differentiation, while simultaneously stimulating aerobic glycolysis production of ATP likely through NADH oxidation. Stimulation of SOD-like activity therefore is consistent with a role for CNM-Au8 in regulating cellular bioenergetics.

Figure 5. Effect of CNM-Au8 on OPC Differentiation and ATP Production



Effect of CNM-Au8 on a) OPC differentiation (O4+ cells) in isolated murine OPCs and b) ATP production in MO3.13 oligodendrocytes (72hr). Statistical analysis performed using one-way ANOVA ($p < 0.05$).

Nonclinical Summary of *In Vitro* and *In Vivo* ALS Neuroprotection Studies

1.2.2.1 *In vitro* ALS Neuroprotection Models

CNM-Au8 exhibits neuroprotective effects in a broad range of neural cell types in *in vitro* and *in vivo* preclinical models of ALS.

1.2.2.1.1 CNM-Au8 Protects Primary Rodent Motor Neurons from Excitotoxicity

Excitotoxicity is the pathological process by which neuronal death occurs as a result of excessive stimulation of receptors at excitatory synapses such as the NMDA receptor. Excitotoxicity has been implicated in acute neurological damage from ischemia and traumatic brain injury and in the chronic neurodegeneration in Huntington's disease. Glutamate excitoxicity is also believed to be a significant contributor to motor neuron death in ALS. Excitotoxic neuronal death via over-activation of NMDA receptors contributes to excessive flux of calcium (Ca2+) into the cell. This triggers a range of responses resulting in cell death, including increased oxidative stress, inappropriate activation of proteases such as calpain, dysregulation of Ca2+-related pathways, mitochondrial damage, and the apoptotic cascade.

Determination of the neuroprotective effects of CNM-Au8 on motor neurons in an in vitro glutamate challenge model was carried out in ventral spinal cord cultures from E14 rat embryos (Martinou et al., 1992). On Day 11 after seeding, cultures were pre-treated with CNM-Au8 or vehicle. The positive control riluzole was added for 1 hour of pre-treatment before glutamate addition. On Day 13, glutamate was added to the cultures for 20 minutes, followed by treatment of riluzole or CNM-Au8 for another 48 hours before cells were fixed and stained. Cultures were stained with anti-MAP-2 to assess motor neuron survival and neurite lengths, and also with TDP-43 and Hoechst stain to assess extranuclear TDP43. CNM-Au8 dose-dependently protects motor neurons from glutamate-induced cell death (Figure 6A) while also preserving neurite length (Figure 6B). CNM-Au8 also outperformed riluzole on neurite network preservation. In addition, CNM-Au8 treatment prevented the accumulation of cytoplasmic TDP-43, a hallmark of ALS cytotoxicity (Figure 6D).

Figure 6 . Neuroprotection of Motor Neurons from Glutamate Excitotoxicity by CNM-Au8

Figure 6A. Motor Neuron Survival

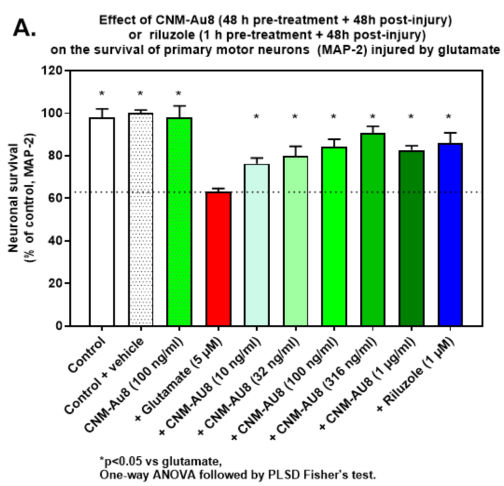


Figure 6B. Motor Neuron Neurtie Network Area

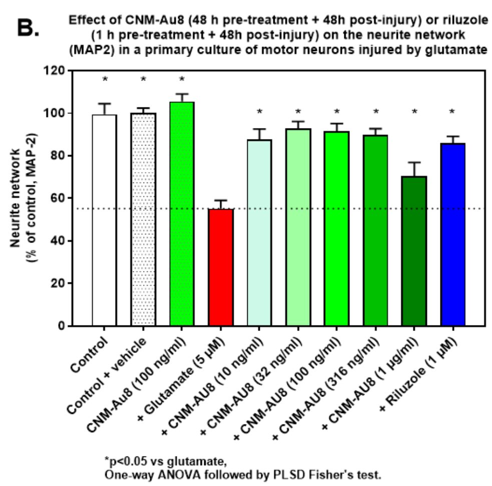


Figure 6B. Effect Of CNM-Au8 on MAP-2 Staining in Rat Hippocampal Neurons Following Glutamate Excitotoxicity Injury

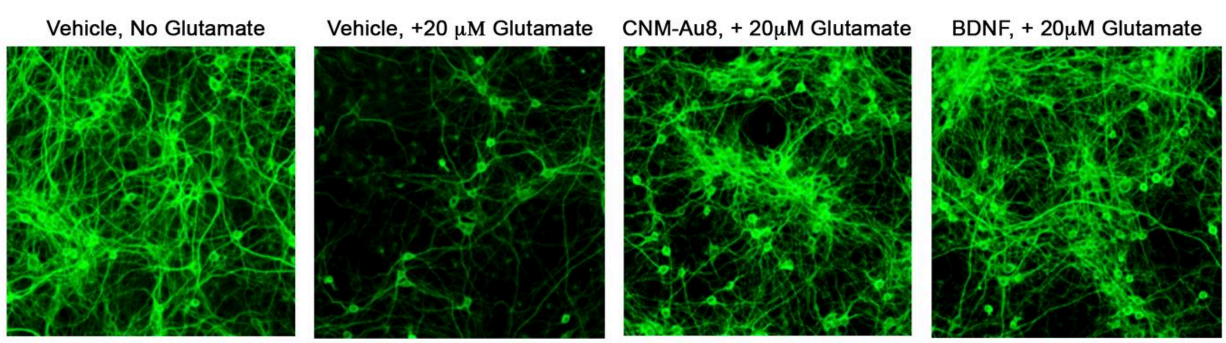
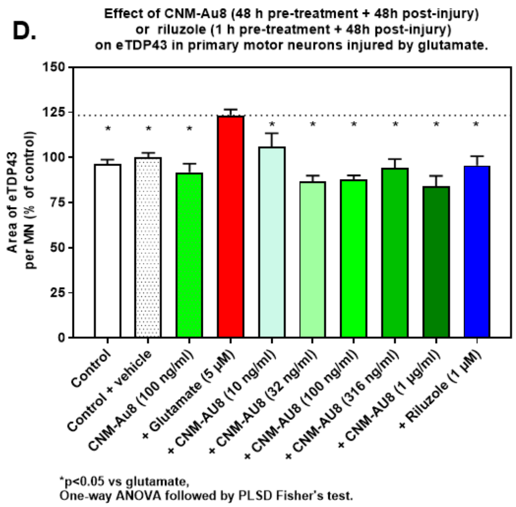


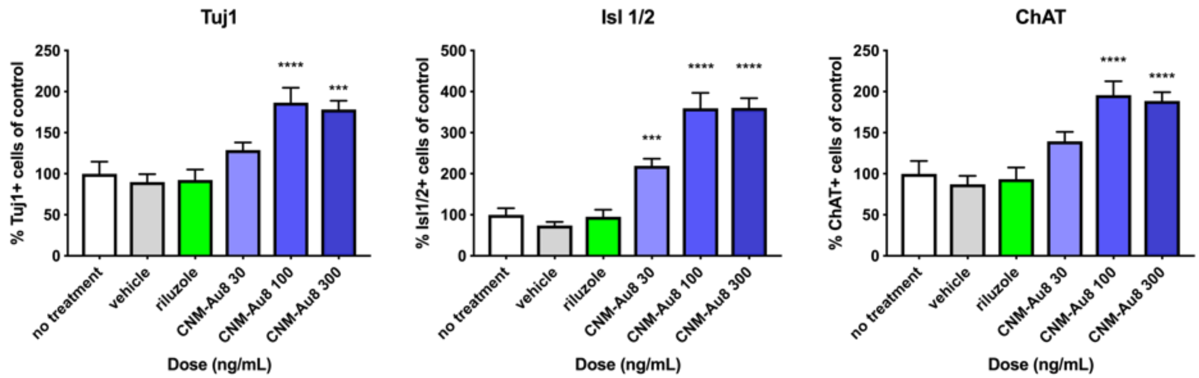
Figure 6D. Reduction in Cytoplasmic TDP43



1.2.2.1.2 CNM-Au8 Protects Normal iPSC Derived Human Motor Neurons From ALS-Participant-Derived Toxic Astrocytes

Astrocytes from both sporadic and familial ALS participants display toxicity against normal human motor neurons (MN) in co-culture (Meyer et al., 2014a). Toxic ALS-participant-derived astrocytes appear to secrete an unidentified toxic factor; and when co-cultured with human MNs, the MNs die (Meyer et al., 2014a). Similar studies have shown that astrocytes from sporadic participants also exhibit toxicity against healthy MNs, likely by similar mechanisms (Haidet-Phillips et al., 2011; Meyer et al., 2014b; Qian et al., 2017). To determine whether CNM-Au8 protects MNs from toxic ALS-participant-derived astrocytes, healthy human iPSC-derived MNs were plated with SOD1A4V astrocytes for 14 days in the presence of various doses of CNM-Au8 or vehicle. Neuroprotection was assessed by quantification of the expression of neuronal markers (Tuj1, Isl1/2, and ChAT) compared to vehicle treated control. Measurement of primary neurite lengths and a neurite network complexity Sholl analysis was also conducted (Sholl, 1953). As shown in Figure 7 below significant neuroprotective dose-dependent effects of CNM-Au8 treatment were consistently observed on survival in the presence of $SOD1^{A4V}$ participant-derived toxic astrocytes.

Figure 7. CNM-Au8 Protects Normal iPSC Derived Human Motor Neurons From ALS-Participant-Derived Toxic Astrocytes



1.2.2.2 *In vivo* ALS Neuroprotection Models

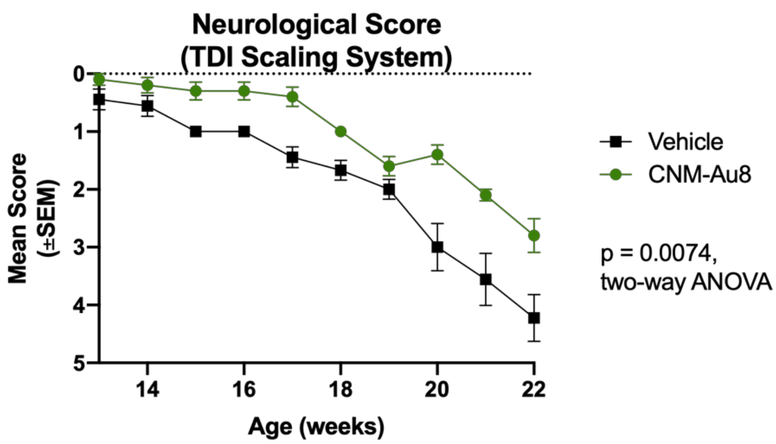
To demonstrate the efficacy of CNM-Au8 treatment in an in vivo model of ALS, two studies were conducted in separate $SOD1^{G93A}$ mouse model strains. These included the rapidly progressive $SOD1^{G93A}$ transgenic line on a mixed SJL/C57BL6 background (Heiman-Patterson et al., 2005) and the more slowly progressive $SOD1^{G93A}$ transgenic line on a congenic C57Bl/6 background (Lutz, 2018) with lifespans of ~157 days compared to ~129 days, respectively. The study using rapidly progressing $SOD1^{G93A}$ animals showed only minor improvement in clinical onset (p=0.13, Mantel-Cox test), as well as significant lack of brainstem atrophy ($p<0.05$, unpaired t-test) in the CNM-Au8-treated group (N=15 animals per group;); all other functional measures were not significant (data not shown). In the study of the slower progressing $SOD1^{G93A}$ (N=20 female mice, 10 per group), four-week-old female SOD1 mice were randomly assigned one of two groups with each group balanced for weight.

Beginning at four weeks of age, mice were provided either CNM-Au8 or vehicle ad libitum in their drinking water. Beginning at Week 8 of age each mouse was tested weekly for strength and motor coordination utilizing several measures including the triple horizontal bar hang time test, static rod orientation test, inverted screen hang time test, weight hold test, and home cage wheel activity (average speed) in addition to standard ALS clinical scoring (Deacon, 2013a, 2013b; Hatzipetros et al., 2015). The CNM-Au8-treated group significantly outperforms the vehicle treated group at virtually all timepoints across these tests (Figure 8). In addition, survival benefits have been observed through Week 22 of the study (Figure 9).

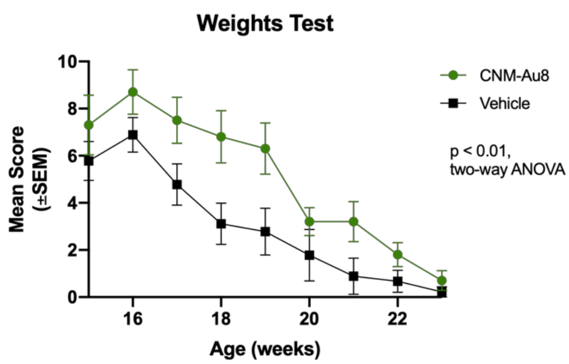
Taken together, these [redacted] data from an [redacted] study in a transgenic ALS mouse model demonstrate preservation of motor function following chronic treatment with CNM-Au8.

Figure 8. Locomotor Functional Efficacy of CNM-Au8 in an SOD1^{G93A} (C57Bl/6 congenic strain) Murine Study

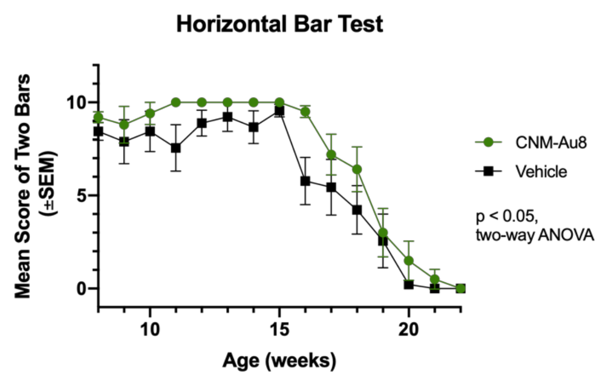
A. ALS TDI Neuro Score



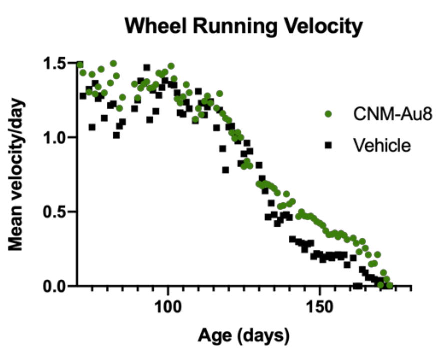
B. Weights Hold Test



C. Horizontal Bar Test



D. Home Wheel Velocity



E. Home Wheel Velocity By Period

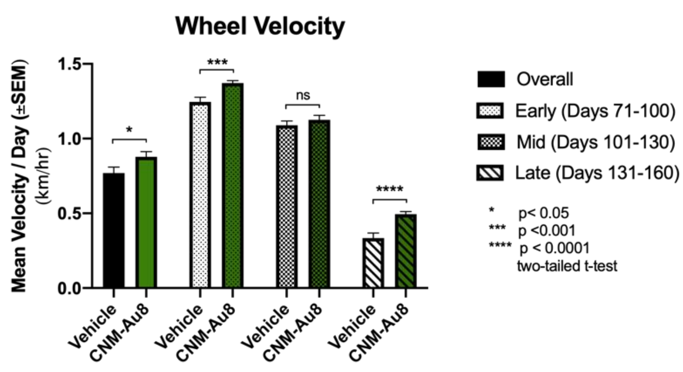
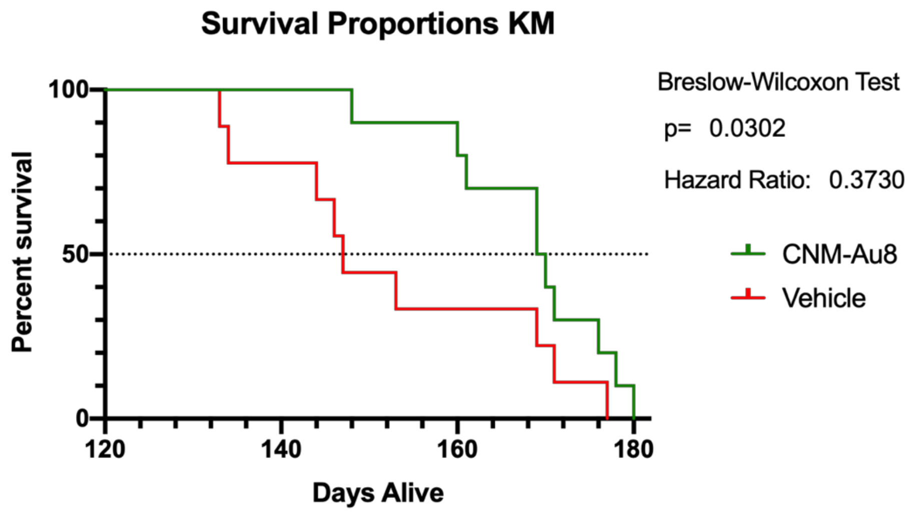


Figure 9. Effect of CNM-Au8 on Survival in a Murine SOD1^{G93A} Model



1.2.3 Summary CNM-Au8 Treatment Rationale for ALS

CNM-Au8-treated motor neurons are protected against cell death by glutamate excitotoxicity in a dose-dependent manner in a primary rat neural-glial co-culture where cytoplasmic levels of TDP-43 are significantly reduced. Aberrant cytoplasmic localization and aggregation of TDP-43 has been observed in over 90% of cases of ALS in humans (Neumann et al. 2006). Further, results from these multiple iPSC human motor neuron studies derived from [redacted] familial, and healthy lines indicate that CNM-Au8 is neuroprotective to MNs challenged with multiple different ALS disease related stressors, namely, excitotoxic and toxic astrocytic stress. Results from the SOD1^{G93A}-overexpression mouse study demonstrate significant improvements in locomotor performance of CNM-Au8-treated mice with a significant survival benefit.
In summary, CNM-Au8 has demonstrated neuroprotection across a range of neuronal cell types, antioxidant activity to counteract the accumulation of ROS in ALS, and preservation of motor function in an ALS mouse model. CNM-Au8 therefore represents an important therapeutic candidate for ALS.

2. OBJECTIVES

2.1 Study Objectives and Endpoints

Primary Efficacy Objective:
To evaluate the efficacy CNM-Au8 as compared to placebo on ALS disease progression.

Secondary Efficacy Objective:

- To evaluate the effect of CNM-Au8 on selected secondary measures of disease progression, including survival.

Safety Objective:

- To evaluate the safety of CNM-Au8 for ALS.

Exploratory Efficacy Objective:

- To evaluate the effect of CNM-Au8 on selected biomarkers and endpoints.

Primary Efficacy Endpoint:
Change in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) using a Bayesian repeated measures model that accounts for loss to follow-up due to mortality.

Secondary Efficacy Endpoints:

- Change in respiratory function as assessed by slow vital capacity (SVC).
- Change in muscle strength as measured isometrically using hand-held dynamometry (HHD) and grip strength.
- Survival.

Safety Endpoints:

- Treatment-emergent adverse and serious adverse events.
- Changes in laboratory values and treatment-emergent and clinically significant laboratory abnormalities.
- Changes in ECG parameters and treatment-emergent and clinically significant ECG abnormalities.
- Treatment-emergent suicidal ideation and suicidal behavior.

Exploratory Efficacy Endpoints:

- Changes in quantitative voice characteristics.
- Difference in the proportion of patients experiencing a ≥ 6-point decline in the ALSFRS-R between active treatment and placebo from Baseline to Week 24.
- Changes in biofluid biomarkers of neurodegeneration.

- Changes in patient reported outcomes.

3. RSA DESIGN

This study is a multi-center, randomized, placebo-controlled trial, testing two active doses of CNM-Au8 (30 mg, 60 mg), given orally daily versus color-matched placebo. Participants will be randomized 3:1 active (CNM-Au8): placebo. Participants randomized to active will be equally allocated between the CNM-Au8 30 mg and 60 mg doses.

3.1 Scientific Rationale for RSA Design

The comparator for this study is placebo. Placebo will consist of a sodium bicarbonate solution in USP water, color matched to active CNM-Au8. Placebo was chosen as the comparator because there are no highly effective, disease-modifying therapies currently available for people with ALS. Both riluzole and edaravone, the only two therapies currently approved for ALS, are available to participants as concomitant therapy during this study. In this manner, participants are not sacrificing any potential benefit from currently approved therapies to participate in this study.

3.2 End of Participation Definition

If a participant chooses *not* to continue in the optional Open Label Extension phase, he or she is considered to have ended participation the Regimen if he or she has completed the placebo-controlled phase of the RSA (24 weeks of treatment), including the last visit or the last scheduled procedure shown in the Schedule of Activities (SoA).

If a participant *chooses* to continue in the Open Label Extension phase, he or she is considered to have ended participation the Regimen if he or she has completed the full placebo-controlled phase as well as all planned Open Label Extension visits, including the last visit or the last scheduled procedure shown in the Schedule of Activities (SoA).

The end of the Regimen is defined as completion of the last visit or procedure shown in the SoA in the trial globally, including the Open Label Extension phase.

4. RSA ENROLLMENT

4.1 Number of Study Participants

Approximately one hundred-sixty (160) participants will be randomized for this Regimen.

4.2 Inclusion and Exclusion Criteria

In order to be randomized to an RSA, participants must meet the Master Protocol eligibility criteria. In addition, participants meeting all of the following inclusion and exclusion criteria will be allowed to enroll in this Regimen:

4.2.1 RSA Inclusion Criteria

Per the Master Protocol.

4.2.2 RSA Exclusion Criteria

1. History of allergy to gold, gold salts, or colloidal gold preparations.

4.3 Treatment Assignment Procedures

Each participant who meets all eligibility criteria for the RSA will be randomized to receive either CNM-Au8 or placebo for approximately 24-weeks of double-blind treatment. An additional 52-weeks of open-label treatment may be offered to participants following completion of the placebo-controlled portion of the study.

5. INVESTIGATIONAL PRODUCT

5.1 Investigational Product Manufacturer

Manufacturing, testing, product characterization, release of CNM-Au8 will be carried out at the Sponsor's facility at 500 Principio Parkway West, Suite 400, North East, MD 21901, USA.

Table 2. CNM-Au8 Investigational Product Dosing Administration

Description	CNM-Au8 30 mg	CNM-Au8 60 mg	Matched Placebo
Total Daily Dosage	30 mg	60mg	NA
Concentration	250 µg/mL	500 µg/mL	NA
Volume per Bottle	60 mL	60 mL	60 mL
Bottles per Day	2	2	2
Daily Volume	120 mL	120 mL	120 mL
Route of Administration	Oral	Oral	Oral
Time and Frequency	Once Daily; Same Time Each Day (± 1 hour)	Once Daily; Same Time Each Day (± 1 hour)	Once Daily; Same Time Each Day (± 1 hour)

The investigational product components and quality standards are described in Table 6 below.

Table 3. CNM-Au8 Components and Quality Standards

Description	Quality Standard	CNM-Au8 30 mg	CNM-Au8 60 mg	Matched Placebo
$NaHCO_3$ (mg) per bottle	ACS, USP identity	32.8 mg	32.8 mg	32.8 mg
Au (mg) per Bottle	Conforms with ASTM B562-95 and USP <233>	15 mg	30 mg	NA
USP Purified Water	USP for total organic carbon, and conductivity	60 mL	60 mL	60 mL

5.2 Labeling, Packaging, and Resupply

All investigational drug products will be labeled according to applicable local and legislative requirements. Label text will be approved according to the Sponsor's agreed procedures, and a copy of the labels will be made available to the study site upon request. For all study drugs, a system of numbering in accordance with all requirements of Good Manufacturing Practice (GMP) will be used, ensuring that each dose of study drug can be traced back to the respective bulk ware of the ingredients. The Sponsor's Quality Assurance group will maintain lists linking all numbering levels. A complete record of batch numbers and expiry dates of all study treatment as well as the labels will be maintained in the Sponsor study file.

Study drug label may include, but is not limited to, the following information:

- Batch number
- Storage information
- Unique number/code

5.3 Acquisition, Storage, and Preparation

Investigational product will be stored at the investigational site in accordance with Good Clinical Practice (GCP) and GMP requirements and will be inaccessible to unauthorized personnel. A complete record of batch numbers and expiry dates can be found in the regimen industry partner study file; the site-relevant elements of this information will be available in the Investigator site file. The responsible site personnel will confirm receipt of IP via the interactive web response system (IWRS) and will use the IP only within the framework of this clinical study and in accordance with this protocol. Receipt, distribution, return and destruction (if any) of the IP must be properly documented according to the Sponsor's agreed and specified procedures.

All IP must be kept in a locked area with limited access and stored at 15°- 25°C (59°- 77°F). Mean kinetic temperature should not to exceed 25°C. Excursions between 15°C and 30°C (59° and 86° F) that may be experienced in pharmacies, hospitals, or warehouses, and during shipping are allowed.

5.4 Study Medication/Intervention, Administration, Escalation, and Duration

Participants will either receive CNM-Au8 30 mg, 60 mg, or color-matched placebo in the study. The Investigational Product (IP) will be self-administered by participants who will be directed to take the IP each day in the morning at approximately the same time, at home at least thirty (30) minutes prior to food intake. The drug formulations will be identical in appearance (size, shape, volume, color) and smell. The packaging and labeling will be designed to maintain blinding to the Site Investigator's team and to participants. There are no visible differences between CNM-Au8 30 mg, 60 mg, and placebo dosing units.

Participants may take the study drug by mouth or through a gastric tube. Participants taking study drug by gastric tube should be directed to first flush the tube with approximately 20-30 mL of distilled water, administer study drug, and then flush the tube post-study drug administration with another 20-30 mL of distilled water.

Based on the supply of the IP provided, the maximum window between in-clinic study visits during the placebo-controlled period may not exceed 64 days. The maximum window between in-clinic study visits during the Open Label Extension period [redacted] may not exceed 64 days for the first 16 weeks, and may not exceed 96 days for the duration of the OLE study period.

5.5 Dosage Adjustment

No dosage adjustments are anticipated during the placebo-controlled period of the study. If participants have difficulty tolerating IP, the Investigator may permit the participants to reduce the daily dose by one (1) bottle. Potential tolerability issues may include gastrointestinal disturbances, headache, somnolence, or other treatment emergent adverse events are not otherwise explained.

Down titration of IP to one (1) bottle daily will only be allowed to address tolerability issues and only with Medical Monitor approval. After a participant has been down titrated to, he/she may be allowed to retry two (2) bottle dose (re-challenge), after a discussion with and approval by the Site Investigator and Medical Monitor. Any dosing changes must be appropriately documented including start date(s) and re-challenge date(s). If the participant cannot tolerate the re-challenge, he/she may remain at the lower dose of one (1) bottle daily. Drug holidays (e.g., treatment-free periods) are not permitted.

During the Open Label Extension, participants will maintain their current blinded dose. Participants previously randomized to placebo during the placebo-controlled period of the study will be re-randomized to receive CNM-Au8 30 mg or 60 mg for the Open Label Extension. IP tolerability issues encountered during the Open Label Extension will be managed on a case-by-case basis by the Site Investigator in coordination with the Medical Monitor.

5.6 Justification for Dosage

Dosage selection has been made based on human safety margins from nonclinical toxicology studies and the exposure data from pharmacology studies that demonstrated neuroprotection benefits.

5.6.1. Human Dosing Safety Margins

Based upon the maximum doses (mg/kg/day) evaluated in the nonclinical GLP repeat-dose 21-day toxicokinetic studies, the completed first-in-human Phase 1 study (AU8.1000-14-01) doses of 15, 30, 60, 90 mg provided a safety margin to the NOAEL based on dose ratios (mg/m^2) ranging from 26x – 4x in rodents, and 195x – 32x in canines, as described in the tables below. Therefore, at the top dose of 90 mg tested in humans, there was a minimum 4x safety margin to the NOAEL in rats.

Table 4. Summary of CNM-Au8 Conversion of Animal 21-Day Repeat Doses To Human Equivalent Dose (HED) Based On Body Surface Area (mg/m2) for 60 kg Human

Species	21-Day NOAEL Dose (mg/kg/day)	21-Day NOAEL Dose (mg/m^2)	Safety Margin Based on Dosing Ratios (mg/m^2) (For 21-Day dosing Studies)			
			15 mg (9.3 mg/m^2)	30 mg (18.5 mg/m^2)	60 mg (37.0 mg/m^2)	90 mg (55.5 mg/m^2)
Rat	40	240	25.9	13.0	6.5	4.3
Canine	90	1800	194.6	97.3	48.6	32.4

Further, when evaluating Au exposure based on the end of study Day 21 AUC(0-24) (ng*hr/mL) in the canine, and rodent studies in comparison with the human 21-day MAD study, the human doses provided an exposure safety margin ranging from 3.3x – 1.6x compared with rodents, and 18.5x – 9.0x compared with canines, as described in the Table 3 below.

Table 5. Summary of CNM-Au8 Exposure Safety Margin Based on End of Study AUC(0-24) ng*hr/mL. Ratio of Animal Toxicokinetic to Human Pharmacokinetic AUC Results From 21-Day Repeat Dose Studies.

Species	21-Day NOAEL Dose (mg/kg/day)	Animal End of Study AUC(0-24) (ng*hr/mL)[a]	Safety Margin Based on Animal/Human AUC(0-24) Ratio (For 21-Day Dosing Studies)			
			Human 15 mg AUC (32.3 ng*hr/mL)	Human 30 mg AUC (41.4 ng*hr/mL)	Human 60 mg AUC (50.3 ng*hr/mL)	Human 90 mg AUC (66.0 ng*hr/mL)
Rat	40	106	3.3	2.6	2.1	1.6
Canine	90	596	18.5	14.4	11.8	9.0

Notes: [a] Average of Male and Female AUC(0-24) ng*hr/mL values at End of Study

CNM-Au8 exposure at the NOAEL at the end of the chronic toxicokinetic studies in the 9-Month canine and 6-Month rodent chronic dosing studies, provided an exposure safety margin ranging from 6.5x – 3.2x, and 13.6x – 6.7x in rodents and canines, respectively in comparison with the human exposures in the 21-day MAD study, as described below in Table 4.

Table 6. Summary of CNM-Au8 Exposure Safety Margin Based on End of Study AUC(0-24) ng*hr/mL Ratio of Chronic Animal Toxicokinetic 6 and 9-Month Rodent and Canine Repeat Dose Studies to Human 21-Day Pharmacokinetic Results

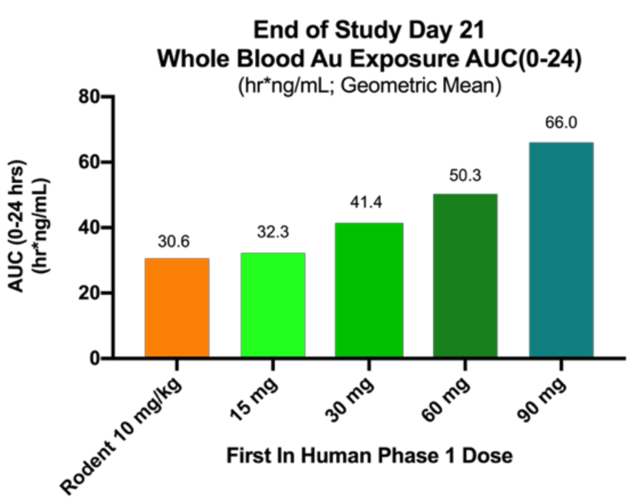
Species (Study)	NOAEL Chronic Dosing (mg/kg/day)	Animal End of Study AUC(0-24) (ng*hr/mL)[a]	Safety Margin Based on Chronic Animal End-of-Study/ Human 21-Day AUC(0-24) Ratio			
			Human 15 mg AUC (32.3 ng*hr/mL)	Human 30 mg AUC (41.4 ng*hr/mL)	Human 60 mg AUC (50.3 ng*hr/mL)	Human 90 mg AUC (66.0 ng*hr/mL)
Rat (6-Month)	40	209	6.5	5.0	4.2	3.2
Canine (9-Month)	10	440	13.6	10.6	8.7	6.7

[a] Average of Male and Female AUC(0-24) ng*hr/mL values at End of Study

5.6.2 Neuroprotection Pharmacologic Exposure Data

Based on the blood gold exposure observed in rodent toxicokinetic (TK) studies where significant remyelination and neuroprotection benefits were observed in preclinical neuroprotection models at mean AUC value of 30.6 (hr*ng/mL), which suggests a minimum human equivalent exposure to achieve a positive bioenergetic response to CNM-Au8 of 15 – 30 mg. Increased AUC exposure of 50.3 (hr*ng/mL) was observed at the CNM-Au8 60 mg dose.

Figure 10. 21-Day AUC Exposure Ranges Between Rodent Toxicokinetic Studies, Preclinical Efficacy Models, and First-In-Human Dosing



In conclusion, based on CNM-Au8 exposures at the NOAELs in the chronic rodent and canine studies along with observed CNM-Au8 exposures in 21-day multiple dosing in humans while also considering safety and tolerability, the dosage chosen for this study in participants with ALS will be CNM-Au8 30 mg/day and 60 mg/day.

5.7 Participant Compliance

Participants will be requested to return any unused IP including empty packaging and used bottles at each study visit. Treatment compliance will be assessed at study visits through bottle counts and will be documented and summarized by a drug-dispensing log for each participant. Overall treatment compliance with IP intake for the per protocol analysis set should be between 80% and 120% of the planned dose and will be assessed at specified study visits. In the event participants are not compliant within this range, discontinuation may be considered by the Site Investigator in consultation with the Medical Monitor or designee.

The date of dispensing the IP to the participant will be documented in the eCRF. Changes in IP dosing for tolerability issues will also be documented in the eCRF.

If a dose of IP is missed, the participant should take the dose that day and continue with the planned dosing interval for the following day. The dose should not be doubled to make up for a missed dose within the same day.

5.8 Overdose

In the event of overdose, study staff should monitor the study participant and provide supportive care as needed. The SI should also contact the Medical Monitor within twenty-four (24) hours. No prior experience has been documented regarding overdosage in humans.

5.9 CNM-Au8 Known Potential Risks and Benefits

5.9.1 Known Potential Risks

5.9.1.1 Immediate Risks

- There has been limited human exposure to clean-surfaced Au nanocrystals. There could be unexpected adverse effects from 24 weeks of exposure to 30 mg or 60 mg of CNM-Au8. These risks have been mitigated by an extensive preclinical toxicology program in multiple species demonstrating No-Adverse-Effect-Levels (NOAEL) at the highest doses studied (up to 10 mg/kg/day in canines; 40 mg/kg/day in rats). Therefore, no Maximum Tolerated Dose (MTD) could be established in these nonclinical studies. In addition, single and multiple ascending dose studies have been performed in healthy human volunteers, which showed no significant adverse events (SAEs) or safety issues. CNM-Au8 was well tolerated at the doses studied up to 90 mg per day over twenty-one days of consecutive dosing.

- There have been no clinical trials of CNM-Au8 in humans with ALS. There is a risk that CNM-Au8 may not be effective and a very small risk that CNM-Au8 could make participants worse. These risks have been mitigated by a robust preclinical program, demonstrating that CNM-Au8 is effective in multiple models of protection from neurodegeneration.

5.9.1.2 Long Range Risks

There has been limited human exposure to CNM-Au8. There could be unexpected adverse effects from chronic exposure to 30 mg or 60 mg of CNM-Au8. Based on rodent and canine toxicokinetic data, when taken orally with daily administration, CNM-Au8 can be expected to be absorbed, and eliminated, relatively slowly with increasing blood and tissue concentrations accumulating over time.

The risks of long-term exposure to CNM-Au8 are mitigated by toxicology studies with chronic dosing, demonstrating no significant toxicities or maximum-tolerated dose.

5.9.1.3 Summary of Participant Risk

Given the potential benefits of CNM-Au8 demonstrated in pre-clinical models of ALS, the safety of CNM-Au8 demonstrated in preclinical toxicology studies and Phase 1 clinical trials in healthy controls, overall, the benefit-risk of CNM-Au8 is positive for participants with ALS.

5.9.2 Known Potential Benefits

As described previously, in preclinical models of ALS, CNM-Au8 has demonstrated efficacy in several relevant models of neurodegeneration. CNM-Au8 has been shown to protect neuronal populations from chemical, inflammatory, and hypoxic insults through a series of unique catalytic mechanisms involving 1) the nicotine adenine dinucleotide redox couple (NAD+/NADH) to boost glycolytic energy production, [redacted] an[redacted] the SOD-like inactivation of reactive oxygen species and nitric oxide to protect cells from oxidative damage and mitochondrial dysfunction. These data suggest that CNM-Au8 could improve participant functioning and/or prolong life in participants with ALS. The safety of CNM-Au8, at the doses to be tested in this study have been shown to be safe and well tolerated in preclinical toxicology and Phase 1 clinical trials in healthy controls.

5.10 Regimen-Specific Lab Alerts

Site Investigators and the Medical Monitor will receive an alert from the Central Lab for certain lab changes. (See below.) These alerts are informative only, and they do not mean a participant must reduce drug dosage or discontinue study drug. These alerts indicate that the Site Investigator and Medical Monitor should follow up with additional testing and management as clinically indicated. Discontinuation of any participant on the basis of abnormal laboratory findings will be at the discretion of the Site Investigator.

Chemistry Panel Test Name	**Outlier Criteria Warranting Further Investigation**
ALT and AST	>3x ULN
Creatinine	>1.5 x Baseline
Hematology Panel Test Name	**Outlier Criteria Warranting Further Investigation**
Platelet count	$<75{,}000/mm^3$

6. REGIMEN SCHEDULE

In addition to procedures in the Master Protocol, the following regimen-specific procedures will be conducted during the study:

- ALSAQ-40
- CNS Bulbar Function Scale
- Voice recording
- Smartphone installation and removal
- Whole blood collection for PK and PD analyses
- Plasma collection for PK and PD analyses
- Urine collection for PD analyses

Modifications to Regimen Schedule Due to a Pandemic

Designated visits in the Schedule of Activities (i.e., Week 4, Week 8, and Week 16) may be conducted via telemedicine (or phone if telemedicine is not available) with remote services instead of in-person if needed to protect the safety of the participant due to a pandemic. If a planned in-clinic visit is conducted via telemedicine (or phone if telemedicine is not available) with remote services, only selected procedures will be performed. Instructions on how to document missed procedures are included in the MOP.

In addition to the procedures in the Master Protocol that should be conducted during the phone or telemedicine and remote visits, the following regimen-specific procedures should be completed:

- Voice Recording
- CNS Bulbar Function Scale (Week 8 and 16 only).

Details on collection of the CNS Bulbar Function Scale, dispensing IP during remote visits, and documenting participants' willingness to participate in OLE are described in the MOP.

Blood samples for PK and PD analyses are **not** collected during the remote visits by the home health agency and this should be recorded as such in the applicable source documentation and EDC.

6.1 Regimen-Specific Screening Visit

This visit will take place in-person after the Master Protocol randomization to a regimen. There are no regimen-specific procedures for this visit. For this regimen, the Regimen-Specific Screening Visit and Baseline Visit should be combined, if possible.

6.2 Baseline Visit

This visit will take place on Day 0. The following procedures will be performed for the regimen schedule:

- ALSAQ-40
- CNS Bulbar Function Scale
- Install smartphone app
- Voice recording
- Whole blood collection for PK and PD analyses
- Plasma collection for PK and PD analyses
- Urine collection for PD analyses
- Dispense IP
- Administer first dose of IP in clinic *after* all Baseline procedures have been completed
- Remind participant to bring in investigational product to the next visit

6.3 Week 2 Telephone Visit

This visit will take place 14 ± 3 days after the Baseline Visit via telephone. No regimen-specific procedures will be performed at this visit.

6.4 Week 4 and 8 Visits

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

These visits will take place on Days 28 ± 7 and 56 ± 7 days, respectively. The following procedures will be performed for the regimen schedule:

- CNS Bulbar Function Scale [Week 8 only]
- Voice recording
- Plasma collection for PK analyses
- Dispense IP (Week 8 only)Remind participant to bring in investigational product to the next visit

6.5 Week 12 Telephone Visit

This visit will take place 84 ± 3 days after the Baseline Visit via telephone. No regimen-specific procedures will be performed at this visit.

6.6 Week 16 Visit

This visit will take place on Day 112 ± 7 days. The following procedures will be performed for the regimen schedule:

- ALSAQ-40
- CNS Bulbar Function Scale
- Voice Recording
- Dispense IP
- Document participant's willingness to participate in the OLE
- Remind participant to bring in investigational product to the next visit

6.7 Week 20 Telephone Visit

This visit will take place 140 ± 3 days after the Baseline Visit via telephone. No regimen-specific procedures will be performed at this visit.

6.8 Week 24 Visit or Early Termination Visit

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

This visit will take place on Day 168 ± 7 days. The following procedures will be performed for the regimen schedule:

- ALSAQ-40
- CNS Bulbar Function Scale
- Voice recording
- Uninstall smartphone app
- Whole blood collection for PK and PD analyses
- Plasma collection for PD analyses
- Urine collection for PD analyses
- Dispense IP [See note below.]

- Remind participant to bring investigational product to the next visit (only if continuing in OLE)

Note: Drug is only dispensed at this visit if the participant is continuing in the Open Label Extension.

6.9 Follow-Up Safety Call

Participants will have a Follow-Up Safety Call 28±3 days after their last dose of study drug. Only those participants NOT continuing on in the Open Label Extension will have the Follow-Up Safety Call at the end of their participation in the placebo-controlled portion of the trial. The following procedures will be performed:

- Assess and document AEs
- Review and document concomitant medications

6.10 Process for Early Terminations

Participants who withdraw consent or early terminate from the study and do not complete the protocol will be asked to be seen for an in-person Early Termination Visit and complete a Follow-Up Safety Call.

The in-person Early Termination Visit should be scheduled as soon as possible after a participant early terminates. If the participant early terminates or withdraws consent during the placebo-controlled portion of the Regimen, all assessments that are collected at the Week 24 in-clinic visit should be conducted. If the participant early terminates or withdraws consent during the OLE phase, all assessments that are collected at the OLE Week 52 in-clinic visit should be conducted. The Follow-Up Safety Call should be completed approximately 28 days after the last dose of study drug.

If the Early Termination Visit occurs approximately 28±3 days after the last dose of study drug, the information for the Follow-Up Safety Call can be collected during the Early Termination Visit, and a separate Follow-Up Safety Call does not need to be completed. If the in-person Early Termination Visit does not occur within 28±3 days of the last dose of study drug, the Follow-Up Safety Call should occur approximately 28 days after the last dose of study drug and the Early Termination Visit will be completed after the Follow-Up Safety Call.

If a participant decides to discontinue study drug, but will complete the protocol, an in-person Early Termination Visit and Follow-Up Safety Call is not necessary.

6.11 Open Label Extension

Participants who have completed the placebo-controlled portion of the trial will be eligible to continue in the Open Label Extension (OLE) as outlined in the SOA. Participants will first be asked about their desire to continue in the OLE at the Regimen-Specific Screening Visit. They will also be asked to *re-confirm* whether they want to continue in the OLE at the Week 16 Visit of the placebo-controlled period.

Modifications to OLE Schedule Due to a Pandemic
Designated visits in the Schedule of Activities for the OLE (i.e. Week 4, Week 8, Week 16, Week 28, and Week 40) may be conducted via telemedicine (or phone if telemedicine is not available) with remote services instead of in-person if needed to protect the safety of the participant due to a pandemic. If a planned in-clinic visit is conducted via telemedicine (or phone if telemedicine is not available) with remote services, only selected procedures will be performed. Instructions on how to document missed procedures are included in the MOP.

In addition to the procedures in the Master Protocol that should be conducted during the phone or telemedicine and remote visits, the following regimen-specific procedures should be completed:

- CNS Bulbar Function Scale (Not done at OLE Week 4).

Blood and urine samples for PK and PD analysis are **not** collected during the remote visits by the home health agency, and this should be recorded as such in the applicable source documentation and EDC.

6.11.1 Week 2 OLE Telephone Visit

This visit will take place via telephone 14 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.2 Week 4 OLE Visit

This visit will take place in-person 28 ± 10 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Administer ALSFRS-R questionnaire
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.3 Week 8 OLE Visit

This visit will take place in-person at 56 ± 7 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Administer ALSFRS-R questionnaire
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Dispense investigational product to participant
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.4 Week 12 OLE Telephone Visit

This visit will take place via telephone at 84 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Administer ALSFRS-R questionnaire
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.5 Week 16 OLE Visit

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

This visit will take place in-person at 112 ± 7 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Administer ALSFRS-R questionnaire
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Collect urine sample biomarker analyses
- Collect blood sample for biomarker analyses
- Whole blood PK and PD collection
- Plasma PD collection
- PD urine collection
- Dispense investigational product to participant
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.6 Week 20 OLE Telephone Visit

This visit will take place via telephone at 140 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Administer ALSFRS-R questionnaire
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.7 Week 24 OLE Telephone Visit

This visit will take place via telephone at 168 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Administer ALSFRS-R questionnaire
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.8 Week 28 OLE Visit

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

The Week 28 OLE Visit will take place in-person 196 ± 14 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Administer ALSFRS-R questionnaire
- ALSAQ-40
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Collect urine sample biomarker analyses
- Collect blood sample for biomarker analyses
- Whole blood PK and PD collection
- Plasma PD collection
- PD urine collection
- Dispense investigational product to participant
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.9 Week 40 OLE Visit

The Week 40 OLE Visit will take place in-person 280 ± 14 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC

- Administer ALSFRS-R questionnaire
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Dispense investigational product to participant
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.10 Week 52 OLE Visit

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

The Week 52 OLE Visit will take place in-person 364 ± 14 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Administer ALSFRS-R questionnaire
- ALSAQ-40
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Collect urine sample biomarker analyses
- Collect blood sample for biomarker analyses
- Whole blood PK and PD collection
- Plasma PD collection
- PD urine collection
- Perform investigational product compliance

7 OUTCOME MEASURES AND ASSESSMENTS

7.1 Voice Analysis

In addition to the scheduled in clinic voice recordings, voice samples will be collected twice per week, using an app installed on either an android or iOS based smartphone. The app characterizes ambient noise, then asks patients to perform a set of speaking tasks: reading sentences -- 5 fixed and 5 chosen at random from a large sentence bank-- repeating a consonant-vowel sequence, producing a sustained phonation, and counting on a single breath. Voice signals are uploaded to a HIPAA-compliant web server, where an AI-based analysis identifies relevant vocal attributes. Quality control (QC) of individual samples will occur by evaluation of voice records by trained personnel.

7.2 ALSAQ-40

The Amyotrophic Lateral Sclerosis Assessment Questionnaire-40 (ALSAQ-40) is a patient self-report health status patient-reported outcome. The ALSAQ-40 consists of forty questions that are specifically used to measure the subjective well-being of patients with ALS and motor neuron disease.

Participants will be handed the questionnaire and asked to write their answers themselves. Caregivers can also help, if needed.

7.3 Center for Neurologic Study Bulbar Function Scale

The Center for Neurologic Study Bulbar Function Scale (CNS-BFS) is a patient self-report scale that has been developed for use as an endpoint in clinical trials and as a clinical measure for evaluating and following ALS patients. The CNS-BFS consists of three domains (swallowing, speech, and salivation), which are assessed with a 21-question, self-report questionnaire.

Participants will be handed the questionnaire and asked to write their answers themselves. Caregivers can also help, if needed.

Instructions on administering the questionnaire during a phone or telemedicine visit will be included in the MOP.

8 BIOFLUID COLLECTION

8.1 Pharmacokinetic (PK) Assessments

8.1.1 Whole Blood PK Assessments of CNM-Au8

Samples for the measurement of whole blood concentrations of Au will be collected before (pre-dose) administration of investigational product per the Schedule of Assessments. PK collection and processing parameters are specified in the RSA laboratory manual.

PK analyses of CNM-Au8 will be specified in a separate PK analysis plan.

PK blood samples will be shipped to and analyzed under the responsibility of the Clene Nanomedicine, Inc.'s bioanalytical laboratory at:

Clene Nanomedicine, Inc.
Bioanalytics Laboratory
500 Principio Parkway West, Suite 400 North East, MD 21901-2912
USA

8.1.2 Plasma PK Assessments of Riluzole

Samples for the measurement of plasma concentrations of riluzole will be collected before (pre-dose) administration of both riluzole and the IP per the Schedule of Assessments. Riluzole plasma PK collection and processing parameters are specified in the laboratory manual.

Population PK analyses of riluzole will be specified in the separate PK analysis plan. The first forty (40) participants taking riluzole and randomized to Regimen C (CNM-Au8) to reach the Week 8 visit will be evaluated by the DSMB, or unblinded DSMB PK designee, to determine whether there are any changes in riluzole population PK parameters following treatment with CNM-Au8 versus placebo at the Week 4 and Week 8 timepoints. The unblinded riluzole population PK analyses and designee will be implemented as per the DSMB charter.

Riluzole PK plasma samples will be shipped to and analyzed under the responsibility of Covance Central Laboratory Services LP.

8.2 Pharmacodynamic (PD) Assessments

Plasma, whole blood, and urine samples for pharmacodynamic measurements will be collected before (pre-dose) administration of the investigational product per the Schedule of

Assessments. PD collection and processing parameters are specified in the RSA laboratory manual.

PD plasma, whole blood, and urine samples will be shipped to the Clene Nanomedicine, Inc.'s bioanalytical laboratory at the address noted above for long-term storage. PD analyses will be conducted with external vendors and specified in a separate PD analysis plan.

PD analyses may include metabolomic markers such as the redox coenzymes: NAD+, NADH, NADP+, NADPH; energetic coenzymes: ATP, ADP, AMP and antioxidants: GSSG and GSH). Disease progression PD analyses may include urinary neurotrophin receptor $P75^{ECD}$ levels, and serum neurofilament light chain (NfL).

9 REGIMEN-SPECIFIC STATISTICAL CONSIDERATIONS

9.1 Deviations from the Default Master Protocol Trial Design

The statistical design for this regimen will be in accordance with the default statistical design described in Appendix I of the master protocol with only one deviation. This regimen will not include interim analyses for early success.

9.2 Regimen Specific Operating Characteristics

Clinical trial simulation is used to quantify operating characteristics for this regimen (refer to the regimen SAP for further details).

9.3 Sharing of Controls from other Regimens

The primary analysis of this regimen will include sharing of all controls from the other regimens. This is justified by the minor differences in inclusion/exclusion criteria of the RSA, such that there are no expected systematic differences in the primary endpoint between the controls across regimens.

APPENDIX I: THE ALSAQ-40

ALSAQ-40

Please complete this questionnaire as soon as possible. If you have any difficulties filling in this questionnaire by yourself, please have someone help you. However it is **your** responses that we are interested in.

The questionnaire consists of a number of statements about difficulties that you may have experienced **during the last 2 weeks.** There are no right or wrong answers: your first response is likely to be the most accurate for you. **Please check the box that best describes your own experiences or feelings.**

Please answer every question even though some may seem very similar to others, or may not seem relevant to you.

All the information you provide is **confidential.**

The following statements all refer to difficulties that you may have had **during the last 2 weeks**. Please indicate, by checking the appropriate box, how often the following statements have been true for you.

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot walk at all
please check **Always/cannot walk at all.**

How often <u>during the last 2 weeks</u> have the following been true?

*Please check **one box** for each question.*

	Never	Rarely	Some-times	Often	Always or cannot walk at all
1. I have found it difficult to walk short distances, e.g. around the house.	☐	☐	☐	☐	☐
2. I have fallen over while walking.	☐	☐	☐	☐	☐
3. I have stumbled or tripped while walking.	☐	☐	☐	☐	☐
4. I have lost my balance while walking.	☐	☐	☐	☐	☐
5. I have had to concentrate while walking.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you are not able to perform the activity at all please check **Always/cannot at all**

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
6. Walking had worn me out.	☐	☐	☐	☐	☐
7. I have had pains in my legs while walking.	☐	☐	☐	☐	☐
8. I have found it difficult to go up and down the stairs.	☐	☐	☐	☐	☐
9. I have found it difficult to stand up.	☐	☐	☐	☐	☐
10. I have found it difficult to move from sitting in a chair to standing upright.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all please check **Always/cannot do at all.**

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
11. I have had difficulty using my arms and hands.	☐	☐	☐	☐	☐
12. I have found turning and moving in bed difficult.	☐	☐	☐	☐	☐
13. I have had difficulty picking things up.	☐	☐	☐	☐	☐
14. I have had difficulty holding books or newspapers, or turning pages.	☐	☐	☐	☐	☐
15. I have had difficulty writing clearly.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had during the last 2 weeks. Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all please check **Always/cannot do at all.**

How often during the last 2 weeks have the following been true?

*Please check **one box** for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
16. I have found it difficult to do jobs around the house.	☐	☐	☐	☐	☐
17. I have found it difficult to feed myself.	☐	☐	☐	☐	☐
18. I have had difficulty combing my hair or brushing and/or flossing my teeth.	☐	☐	☐	☐	☐
19. I have had difficulty getting dressed.	☐	☐	☐	☐	☐
20. I have had difficulty washing at the bathroom sink.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had during the last 2 weeks. Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all please check **Always/cannot do at all.**

How often during the last 2 weeks have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
21. I have had difficulty swallowing.	☐	☐	☐	☐	☐
22. I have had difficulty eating solid food.	☐	☐	☐	☐	☐
23. I have had difficulty drinking liquids.	☐	☐	☐	☐	☐
24. I have had difficulty participating in conversations.	☐	☐	☐	☐	☐
25. I have felt that my speech has not been easy to understand.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all please check **Always/cannot do at all.**

How often <u>during the last 2 weeks</u> have the following been true?

*Please check **one box** for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
26. I have stuttered or slurred my speech.	☐	☐	☐	☐	☐
27. I have had to talk very slowly.	☐	☐	☐	☐	☐
28. I have talked less than I used to do.	☐	☐	☐	☐	☐
29. I have been frustrated with my speech.	☐	☐	☐	☐	☐
30. I have felt self-conscious about my speech.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

How often <u>during the last 2 weeks</u> have the following been true?

*Please check **one box** for each question*

	Never	Rarely	Some-times	Often	Always
31. I have felt lonely.	☐	☐	☐	☐	☐
32. I have been bored.	☐	☐	☐	☐	☐
33. I have felt embarrassed in social situations.	☐	☐	☐	☐	☐
34. I have felt hopeless about the future.	☐	☐	☐	☐	☐
35. I have worried that I am a burden to other people.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always
36. I have wondered why I keep going.	☐	☐	☐	☐	☐
37. I have felt angry because of the disease.	☐	☐	☐	☐	☐
38. I have felt depressed.	☐	☐	☐	☐	☐
39. I have worried about how the disease will affect me in the future.	☐	☐	☐	☐	☐
40. I have felt as if I have lost my independence	☐	☐	☐	☐	☐

Please make sure that you have checked one ***box for each question.***

Thank you for completing this questionnaire.

APPENDIX II: THE BULBAR FUNCTION SCALE (CNS-BFS)

BULBAR FUNCTION SCALE (CNS-BFS)						
SIALORRHEA	**Does Not Apply (1)**	**Applies Rarely (2)**	**Applies Occasionally (3)**	**Applies Frequently (4)**	**Applies Most of the Time (5)**	
1. Excessive saliva is a concern to me.	❍	❍	❍	❍	❍	
2. I take medication to control drooling.	❍	❍	❍	❍	❍	
3. Saliva causes me to gag or choke.	❍	❍	❍	❍	❍	
4. Drooling causes me to be frustrated or embarrassed.	❍	❍	❍	❍	❍	
5. In the morning I notice saliva on my pillow.	❍	❍	❍	❍	❍	
6. My mouth needs to be dabbed to prevent drooling.	❍	❍	❍	❍	❍	
7. My secretions are not manageable.	❍	❍	❍	❍	❍	
				TOTAL Sialorrhea Score: _____		
SPEECH	**Does Not Apply (1)**	**Applies Rarely (2)**	**Applies Occasionally (3)**	**Applies Frequently (4)**	**Applies Most of the Time (5)**	**Unable to Communicate by Speaking (6)**
1. My speech is difficult to understand.	❍	❍	❍	❍	❍	❍
2. To be understood I repeat myself.	❍	❍	❍	❍	❍	❍
3. People who understand me tell other people what I said.	❍	❍	❍	❍	❍	❍
4. To communicate I write things down or use devices such as a computer.	❍	❍	❍	❍	❍	❍
5. I am talking less because it takes so much effort to speak.	❍	❍	❍	❍	❍	❍

6. My speech is slower than usual.	❍	❍	❍	❍	❍	❍
7. It is hard for people to hear me.	❍	❍	❍	❍	❍	❍
				TOTAL Speech Score: _____		
SWALLOWING	**Does Not Apply (1)**	**Applies Rarely (2)**	**Applies Occasionally (3)**	**Applies Frequently (4)**	**Applies Most of the Time (5)**	
☐ Feeding tube is in place						
1. Swallowing is a problem.	❍	❍	❍	❍	❍	
2. Cutting my food makes it easier to chew and swallow.	❍	❍	❍	❍	❍	
3. To get food down I have switched to a soft diet.	❍	❍	❍	❍	❍	
4. After swallowing I gag or choke.	❍	❍	❍	❍	❍	
5. It takes longer to eat.	❍	❍	❍	❍	❍	
6. My weight is dropping because I can't eat normally.	❍	❍	❍	❍	❍	
7. Food gets stuck in my throat.	❍	❍	❍	❍	❍	
				TOTAL Swallowing Score: _____		
				OVERALL SCORE: _____		

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REGIMEN-SPECIFIC APPENDIX C

FOR CNM-Au8

Regimen-Specific Appendix Date: 07 November 2022

Version Number: 6.0

TABLE OF CONTENTS

SIGNATURE PAGE

I have read the attached Regimen-Specific Appendix (RSA) entitled, "REGIMEN C: CNM-Au8" dated November 07, 2022 (Version 6.0) and agree to abide by all described RSA procedures. I agree to comply with the International Conference on Harmonisation Tripartite Guideline on Good Clinical Practice, applicable FDA regulations and guidelines identified in 21 CFR Parts 11, 50, 54, and 312, central Institutional Review Board (IRB) guidelines and policies, and the Health Insurance Portability and Accountability Act (HIPAA).

By signing the RSA, I agree to keep all information provided in strict confidence and to request the same from my staff. Study documents will be stored appropriately to ensure their confidentiality. I will not disclose such information to others without authorization, except to the extent necessary to conduct the study.

Site Name: ______________________________

Site Investigator: ______________________________

Signed: ____________________ Date: ____________

LIST OF ABBREVIATIONS

Abbreviation	Definition
ALSAQ-40	Amyotrophic Lateral Sclerosis Assessment Questionnaire-40
ASTM	American Society for Testing and Materials
ATP	Adenosine Trinucleotide Phosphate
Au	Gold
AUC	Area under the curve
AUC(0-24)	Area under the curve for 24 hours
CNM-Au8	Aqueous suspension of clean surfaced nanocrystals consisting of gold atoms self-organized into crystals of various faceted, geometrical shapes
CNS	Central nervous system
CNS-BFS	Center for Neurologic Study Bulbar Function Scale
CSF	Cerebrospinal fluid or cerebral spinal fluid
fALS	Familial ALS
FVC	Forced Vital Capacity
HED	Human equivalent dose
HDPE	High density polyethylene
hr	Hour
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
IP	Investigational Product
K2EDTA	Dipotassium ethylenediaminetetraacetic acid
kg	Kilogram
m	meter
MAD	Multiple ascending dose
mg	Milligram
mL	Milliliter
MRS	Magnetic Resonance Spectroscopy
mRNA	Messenger Ribonucleic acid (RNA)
MTD	Maximum tolerated dose
NAD+	Oxidized form of nicotinamide adenine dinucleotide
NADH	Reduced form of nicotinamide adenine dinucleotide
NADPH	Reduced form of nicotinamide adenine dinucleotide phosphate
NaHCO3	Sodium bicarbonate (baking soda)

Abbreviation	Definition
ng	nanogram
nm	nanometer
NOAEL	No observed adverse effect level
OLE	Open Label Extension
PD	Pharmacodynamic
PPP	Pentose phosphate pathway
RSA	Regimen-Specific Appendix
SAE	Serious adverse event
SOD1	Superoxide dismutase
SVC	Slow Vital Capacity
TEM	Transmission electron microscopy
TK	Toxicokinetic
USP	United States Pharmacopeia
VC	Vital Capacity

REGIMEN-SPECIFIC APPENDIX SUMMARY

<u>Regimen-Specific Appendix C</u>
For CNM-Au8

<u>Rationale and RSA Design</u>

CNM-Au8 is a suspension of clean-surfaced, faceted gold nanocrystals whose catalytic activity independently impacts energetic, metabolic, and redox pathways, resulting in significant neuroprotection from oxidative and excitotoxic insults.

CNM-Au8 is a nanocatalyst that is administered orally, penetrates the blood brain barrier, and has a unique mechanism of action involving: (1) the catalytic production of nicotinamide adenine dinucleotide (NAD), a key co-factor in the ATP-generating pathways of all cells, (2) catalytic anti-oxidative activity resulting in cellular protection from oxidative stress, [redacted] Importantly, CNM-Au8 demonstrated significant neuroprotection of healthy human motor neurons from cell death induced by toxic ALS-participant derived astrocytes, and of primary rat motor neurons from glutamatergic excitotoxicity. Oral delivery of CNM-Au8 to a mouse model of ALS resulted in the recovery of functional behaviors and significant extension of lifespan.

Given the involvement of cellular bioenergetic failure in the pathogenesis of ALS, the development of a disease modifying therapeutic that addresses the energetic dysregulation underlying the progressive accumulation of oxidative stress [redacted] is a rational therapeutic strategy.

This clinical protocol will test CNM-Au8 at doses of 30 mg and 60 mg versus placebo in a randomized, double-blind manner, to determine if CNM-Au8 can demonstrate sufficient efficacy and safety on accepted clinical and paraclinical outcomes for the treatment of ALS. The investigation of two doses of the study drug is based on counsel from the FDA recommending including a dose finding strategy in the study. The 30 mg and 60 mg doses of CNM-Au8 selected for the study achieve at least comparable AUC exposure in humans as those shown to demonstrate neuroprotection efficacy in nonclinical animals studies, and demonstrated minimal treatment emergent adverse events in the Phase 1 First-In-Human study in healthy volunteers.

<u>Allocation to Treatment Regimens</u>
Participants must first be screened under the Master Protocol before they are randomized to an RSA.

As soon as pre-defined criteria for futility for the RSA are met, or the target number of randomized participants has been reached, enrollment will stop in the RSA.

Number of Planned Participants and Treatment Groups

The number of planned participants for this regimen is approximately160.

There are 2 treatment groups for this regimen, active and placebo. Participants will be randomized in a 3:1 ratio to active treatment or placebo (i.e., 120 active: 40 placebo). Half of the active treatment group will receive CNM-Au8 30mg/day and the other half will receive 60mg/day (i.e., 60 at 30mg/day: 60 at 60mg/day).

Planned Number of Sites

Research participants will be enrolled from approximately 60 centers in the US.

Treatment Duration

The maximum duration of the placebo-controlled treatment portion is 24 weeks.

Follow-up Duration

At the conclusion of the 24-week placebo-controlled Treatment Period of the study, all participants will either schedule a 28-day follow up phone call and end their participation in the regimen or have the option to receive CNM-Au8 in the Open Label Extension (OLE) phase of the study.

In the OLE, CNM-Au8 will be provided by Clene Nanomedicine, until the primary results of the 24 week double-blind portion of the study are available, or Clene or sponsor terminate support or development of CNM-Au8 for ALS.

Total Planned Trial Duration

For participants completing the placebo-controlled Treatment Period of the Study, the planned amount of time for a participant in the trial is up to 34 weeks, or about 8 months. This duration assumes a 6-week screening window, a 24-week placebo-controlled treatment period, and a 4-week safety follow-up period for those participants who do not enter the Open Label Extension. Participants will complete approximately 10 study visits during the placebo-controlled treatment period of the study.

SCHEDULE OF ACTIVITIES – PLACEBO-CONTROLLED PERIOD[14]

As per the Schedule of Activities (SOA) below, visits must occur every 4 weeks and will be alternatively clinic-, phone-, or telemedicine-based, as applicable. There is a maximum 24-week duration of placebo-controlled treatment for a Regimen.

Activity (page 1 of 2)	Master Protocol or Regimen-Specific	Master Protocol Screening[1]	Regimen Specific Screening[1]	Baseline	Week 2	Week 4[16, 17]	Week 8[16, 17]	Week 12	Week 16[17]	Week 20	Week 24 or Early Term. Visit[14, 16]	Follow-Up Safety Call[11, 14]
		Clinic	Clinic	Clinic	Phone	Clinic	Clinic	Phone	Clinic	Phone	Clinic	Phone
		-42 to -1 Days	-41 to 0 Day	Day 0	Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day 168 ±7	28 days after last dose ±3 days
Written Informed Consent[2]	Master	X	X									
Written Informed Consent - OLE	Master								X			
Inclusion/Exclusion Review	Master	X	X[3]									
ALS & Medical History	Master	X										
Demographics	Master	X										
Physical Examination	Master	X										
Neurological Exam	Master	X										
Vital Signs[4]	Master	X		X		X	X		X		X	
Slow Vital Capacity	Master	X[18]		X			X		X		X	
Home Spirometry	Regimen	X[18]		X			X		X		X	
Muscle Strength Assessment	Master			X			X		X		X	
ALSFRS-R	Master	X		X		X	X	X	X	X	X	
ALSAQ-40	Regimen			X							X	
CNS Bulbar Function Scale	Regimen			X			X		X		X	
12-Lead ECG	Master	X									X	
Clinical Safety Labs[5]	Master	X		X		X	X		X		X	
PK/PD Biomarker Blood Collection[12]	Regimen			X		X	X				X	
Biomarker PD Urine Collection	Regimen			X							X	

Biomarker Blood Collection	Master			X			X		X		X	

Activity (page 2 of 2)	Master Protocol or Regimen-Specific	Master Protocol Screening[1]	Regimen Specific Screening[1]	Baseline	Week 2	Week 4[16, 17]	Week 8[16, 17]	Week 12	Week 16[17]	Week 20	Week 24 or Early Term. Visit[14, 16]	Follow-Up Safety Call[11, 14]
		Clinic	Clinic	Clinic	Phone	Clinic	Clinic	Phone	Clinic	Phone	Clinic	Phone
		-42 to -1 Days	-41 to 0 Day	Day 0	Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day 168 ±7	28 days after last dose ±3 days
Biomarker Urine Collection	Master			X			X		X		X	
DNA Collection[7] (optional)	Master			X								
CSF Collection (optional)	Master			X					X[15]			
Concomitant Medication Review	Master	X	X	X	X	X	X	X	X	X	X	
Adverse Event Review[6]	Master	X	X	X	X	X	X	X	X	X	X	X
Columbia-Suicide Severity Rating Scale	Master			X		X	X		X		X	
Install Smartphone App[19]	Regimen			X								
Voice Recording[9]	Regimen			X		X	X		X		X	
Uninstall Smartphone App	Regimen										X	
Assignment to the Regimen	Master	X										
Randomization within the Regimen	Master		X									
Administer/Dispense Investigational product	Master			X[8]			X		X		X[10]	
Study Drug Accountability/Compliance	Master				X[20]	X	X	X[20]	X	X[20]	X	
Exit Questionnaire	Master										X	
Vital Status Determination[13]	Master										X	

[1] Master Protocol Screening procedures must be completed within 42 days to 1 day prior to the Baseline Visit. The Regimen-Specific Screening Visit and Baseline Visit should be combined, if possible.

[2] During the Master Protocol Screening Visit, participants will be consented via the Master Protocol informed consent form (ICF). After a participant is randomized to a regimen, participants will be consented a second time via the regimen-specific ICF.
[3] At the Regimen Specific Screening Visit, participants will have regimen-specific eligibility criteria assessed.
[4] Vital signs include weight, systolic and diastolic pressure, respiratory rate, heart rate and temperature. Height measured at Master Protocol Screening Visit only.
[5] Clinical safety labs include hematology (CBC with differential), complete chemistry panel, thyroid function and urinalysis. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during the study. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.
[6] Adverse events that occur after signing the consent form will be recorded.
[7] The DNA sample can be collected after baseline if a baseline sample is not obtained or the sample is not usable.
[8] Administer first dose of investigational product only after Baseline Visit procedures are completed.
[9] In addition to study visits outlined in the SOA, participants will be asked to complete twice weekly voice recordings at home.
[10] Drug will only be dispensed at this visit if the participant continues in the OLE.
[11] Participants will only have a Follow-Up Safety Call at this time if they *do not* continue on in the OLE. Participants who continue into OLE will have a Follow-Up Safety Call after their last dose of study drug during the OLE phase.
[12] Whole blood and plasma will be collected prior to administration of the daily dose at the Baseline and Week 24 visit. Plasma will be collected prior to administration of the daily dose at the Week 4 and Week 8 visits. Refer to the RSA laboratory manual for collection and processing parameters.
[13] Vital status, defined as a determination of date of death or death equivalent or date last known alive, will be determined for each randomized participant at the end of the placebo-controlled portion of their follow-up (generally the Week 24 visit, as indicated). If at that time the participant is alive, his or her vital status should be determined again at the time of the last patient last visit (LPLV) of the placebo-controlled portion of a given regimen. We may also ascertain vital status at later time points by using publicly available data sources as described in section 8.15 of the Master Protocol.
[14] The maximum window between study visits during the placebo-controlled period may not exceed 64 days.
[15] If the CSF collection is unable to be performed for logistical reasons, such as scheduling, at the Week 16 Visit, it may be performed at the Week 24 Visit.
[16] Participants should be instructed to hold the study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.
[17] Visit may be conducted via phone or telemedicine with remote services instead of in-person if this is needed to protect the safety of the participant due to a pandemic, or other reason.
[18] If required due to pandemic-related restrictions, Forced Vital Capacity (FVC) performed by a Pulmonary Function Laboratory evaluator or with a study-approved home spirometer, or sustained phonation using a study approved method may be used for eligibility (Master Protocol Screening ONLY).
[19] Two smartphone apps should be installed on the participant's phone, one to collect the voice recordings and one to collect home spirometry.
[20] Drug accountability will not be done at phone visits. A drug compliance check-in must be held during phone visits to ensure participant is taking drug per dose regimen and to note any report of missed doses.

SCHEDULE OF ACTIVITIES – REGIMEN SPECIFIC OPEN LABEL EXTENSION PHASE (OPTIONAL)

Activity (page 1 of 1)	Master Protocol or Regimen-Specific	Open Label Extension (Optional)[5]								
		Week 2	Week 4[9]	Week 8[9]	Week 12	Week 16[8,9]	Week 20	Week 24	Week 28 and Q12 Weeks[8,9,11,12]	Follow-Up Safety Call[6]
		Phone	Clinic	Clinic	Phone	Clinic	Phone	Phone	Clinic	Phone
		Day 14 ±3	Day 28 ±7	Day 56 ±7	Day 84 ±3	Day 112 ±7	Day 140 ±3	Day168 ±3	Day 196 ± 14	28±7 days after last dose
Vital Signs[1]	Master		X	X		X			X	
Slow Vital Capacity	Master		X	X		X			X	
Home Spirometry	Regimen		X	X		X			X	
ALSFRS-R	Master		X	X	X	X	X	X	X	
ALSAQ-40[12]	Regimen								X[12]	
CNS Bulbar Function Scale	Regimen			X		X			X	
Clinical Safety Labs[2]	Master		X	X		X			X	
PK/PD Biomarker Blood Collection[11]	Regimen					X			X[11]	
Biomarker PD Urine Collection[11]	Regimen					X			X[11]	
Biomarker Blood Collection[11]	Master					X			X[11]	
Biomarker Urine Collection[11]	Master					X			X[11]	
Concomitant Medication Review	Master	X	X	X	X	X	X	X	X	
Adverse Event Review[3]	Master	X	X	X	X	X	X	X	X	X[6]
Columbia-Suicide Severity Rating Scale	Master		X	X		X			X	

Administer/Dispense Investigational product	Master			X		X			X	
Drug Accountability/Compliance	Master	X[10]	X	X	X[10]	X	X[10]	X[10]	X	

[1] Vital signs include weight, systolic and diastolic pressure, respiratory rate, heart rate and temperature. Height in cm measured at Master Protocol Screening Visit only.

[2] Clinical safety labs include hematology (CBC with differential), complete chemistry panel, liver function tests, thyroid function and urinalysis. Serum pregnancy testing will occur in women of child-bearing potential at the Master Protocol Screening Visit and as necessary during the study. Pregnancy testing is only repeated as applicable if there is a concern for pregnancy.

[3] Adverse events that occur after signing the consent form will be recorded.

[4] Participants who continue into OLE will have a Follow-Up Safety Call (as described in the body of this RSA) after their last dose of study drug during the OLE phase.

[5] The duration of the OLE is until the primary results of the 24 week double-blind portion of the study are available, or Clene or sponsor terminates support or development of CNM-Au8 for ALS.

[6] Participants who continue into the OLE and then withdraw consent or early terminate will be asked to complete an Early Termination Visit and Follow-Up Safety Call as described in the body of this RSA.

[7] The maximum window between study visits during the OLE may not exceed 64 days for the Week 8 and Week 16 visits; and 96 days for the Week 28 and subsequent visits on a 12-week interval.

[8] Participants should be instructed to hold the study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

[9] Visit may be conducted via phone or telemedicine with remote services instead of in-person if this is needed to protect the safety of the participant due to a pandemic or other reasons.

[10] Drug accountability will not be done at phone visits. A drug compliance check-in must be held during phone visits to ensure participant is taking drug per dose regimen and to note any report of missed doses

[11] Blood and urine biomarker collection in OLE phase will only occur at week 16, 28, and 52. For Regimen specific collection only, samples will continue to be collected every 24-weeks thereafter.

[12] ALSAQ-40 in OLE phase will only occur at week 28 and every 24 weeks (every other visit) thereafter.

1. INTRODUCTION

Regimen C: CNM-Au8

1.1 CNM-Au8 Background Information

Amyotrophic lateral sclerosis (ALS) is a late-onset, fatal neurodegenerative disease affecting motor neurons with an incidence of approximately 1 in 100,000. Most ALS cases are sporadic, with 5–10% of cases being familial ALS (Zarei et al., 2015). Both sporadic and familial ALS are associated with degeneration and loss of cortical and spinal motor neurons, resulting in muscle wasting and weakness, eventually leading to respiratory failure and death. The etiology of ALS remains unknown. Emerging research has discovered interconnecting pathophysiological mechanisms amongst the seemingly disparate genetic mutations, implicated in both familial (fALS) and sporadic (sALS) ALS. Importantly, these emerging data suggest that impairments related to oxidative stress and mitochondrial function appear to be common across many underlying genetic abnormalities. Genetic mutations in superoxide dismutase (SOD1), regulation of RNA processing and metabolism (represented by mutations in FUS, TDP43 and several more genes) all appear to result in both oxidative stress and RNA dysregulation, indicative of an underlying bioenergetic failure in ALS. Bioenergetic dysfunction affects not only motor neurons themselves but also astrocytes and oligodendrocytes that provide crucial energetic support to motor neuron survival. Postmortem analyses of brain tissues from both fALS and sALS participants exhibit evidence of widespread accumulation of oxidative damage to proteins, lipids, and DNA. Transgenic mice expressing mutant SOD1 forms have evidence of nerve and spinal cord tissue damage that recapitulates key aspects of ALS and demonstrates clear evidence of excessive protein and lipid oxidation (D'Amico et al., 2013b). Oxidative stress also leads to the accumulation of protein aggregates such as TDP43 (Bozzo et al., 2017a), which are invariably found in ALS motor neurons, and which indicate failures of protein degradation systems such as autophagy (Blokhuis et al., 2013).

Elevated oxidative stress has been shown to impact the processing of gene transcripts important for motor neuron survival. This observation connecting oxidative stress to RNA processing was demonstrated when paraquat, a well-known inducer of oxidative stress, was shown to induce striking changes in RNA splicing in neuroblastoma cells (Maracchioni et al., 2007). Two of the pre-mRNAs whose splicing was shown to be altered in this study are transcripts of genes known to be important for motor neuron survival, Survival Motor Neuron (SMN) and Apoptotic Peptidase Activating Factor 1 (APAF1) (Maracchioni et al., 2007). The mis-localization of the RNA-binding protein TDP43 in ALS is further evidence of the connection between oxidative stress and RNA dysmetabolism. In general, TDP43 is involved in pre-RNA processing in the nucleus, where it functions in transcriptional control. However, oxidative stress causes the

TDP43 to translocate to the cytoplasm, where it accumulates in stress granules, as protein aggregates. Sequestration of an RNA regulator such as TDP43 in the cytoplasm and away from the nucleus is thought to prevent it from carrying out its normal functions and thereby can lead to dysregulation of RNA processing and transcription. The converse relationship has similarly been shown, namely that mis-regulation of RNA metabolism increases oxidative stress within the CNS. Mutations in RNA regulators FUS and TDP43 cause disruptions in the expression of key nuclear-encoded mitochondrial genes (Zhang et al., 2014) as well as in the expression of energy metabolism regulators such as PGC-1 alpha and oxidative stress reducers such as MnSOD and catalase (Sanchez-Ramos et al., 2011). Therefore, the pathophysiologic disease mechanism of ALS appears to involve a positive feedback cycle of oxidative stress and protein aggregation impacting RNA regulation which in turn leads to a sustained increased oxidative stress disease state, ending in motor neuron death (Bozzo et al., 2017b).

Unifying both themes of RNA dysregulation and oxidative stress is the concept that ALS is a disease of energetic dysmetabolism (Dupuis et al., 2011; Tefera and Borges, 2016; Vandoorne et al., 2018). Motor neurons consume high amounts of energy to function and are therefore exquisitely sensitive to apoptosis if their energetic demands cannot be met. The main energy-producing organelle of the cell, the mitochondria, exhibit dysfunction in ALS on multiple levels. For example, overall there are fewer mitochondria, as measured by mitochondrial DNA, in the spinal cords of both fALS and sALS participants (Wiedemann et al., 2002). Presynaptic mitochondrial swelling in motor neurons has been observed in both ALS participants (Siklos et al., 1996), as well as in a SOD1 transgenic mouse model prior to disease onset (Manfredi and Xu, 2005). Swollen and vacuolarized mitochondria are indicative of impaired mitochondrial function (Siklos et al., 1996), which is corroborated by studies showing that there is decreased activity of the electron transport chain in spinal cord mitochondria of ALS participants (Wiedemann et al., 2002). Because reduced respiration and reduced ATP synthesis preceded clinical symptom onset in SOD1G93A mice (Jung et al., 2002; Mattiazzi et al., 2002; Szelechowski et al., 2018), energetic dysmetabolism appears to play an important role in the etiology of this disease. Overexpression of PGC-1 alpha, which stimulates mitochondrial biogenesis, improved survival, motor neuron function and motor neuron survival in mutant SOD1G93A mice (Zhao et al., 2011), indicating that improving mitochondrial function and/or efficiency may be a successful therapeutic strategy for ALS (Vandoorne et al., 2018).

Energetic dysmetabolism in ALS is further underscored by studies uncovering carbohydrate metabolism abnormalities and ATP deficits in ALS motor neurons (Vandoorne et al., 2018). FDG-PET imaging has demonstrated a correlation between reduced ALS brain glucose uptake and cognitive impairment (Canosa et al., 2016) as well as overall disease severity (Dalakas et al., 1987). Some groups who conducted FDG-PET studies showing reduced glucose uptake in ALS participant brains suggested CNS hypometabolism can be used as an early diagnostic indication of ALS (Van Laere et al., 2014; Van Weehaeghe et al., 2016). While it is difficult to determine whether lowered glucose utilization detected in imaging studies is due to reduced neuronal catabolism of glucose or due to motor neuron loss, there are further lines of evidence to suggest energetic dysmetabolism in ALS cells. A proteomic study of ALS participant fibroblasts indicated a significant reduction in key enzymes involved in glycolysis (Szelechowski et al., 2018), and expression profiling of sALS participant motor cortexes showed downregulation of glycolytic genes (Lederer et al., 2007). A GWAS analysis involving 12577 cases and 23475 controls identified glycolysis/gluconeogenesis as among the top seven biological pathways represented by the genes identified as associated with ALS (Du et al., 2018). Because glia preferentially utilize glycolysis, these results may indicate a dysfunction of glia in providing energetic support to motor neurons in ALS participants. Energetic stress on motor neurons may increase their levels of oxidative stress, as observed in post-mortem brain samples of ALS participants (Ferrante et al., 1997), and lead to motor neuron death.

1.2 CNM-Au8 Therapeutic Rationale

CNM-Au8 is a concentrated aqueous suspension of clean-surfaced nanocrystals consisting solely of Au atoms organized into clean surfaced crystals of highly faceted, substantially uniform geometrical shapes that act as nanocatalysts, supporting cellular bioenergetic reactions.

CNM-Au8 is supplied for dosing orally in sodium bicarbonate buffered USP purified water. Each mL of CNM-Au8 suspension at 500 µg/mL is estimated to contain between 100 - 500 trillion highly faceted Au nanocrystals. IP will consist of 60 mL high density polyethylene (HDPE) bottles containing CNM-Au8 at concentrations of 250 µg/mL (15 mg) or 500 µg/mL (30 mg), or placebo, administered orally, once daily in the morning, at least 30 minutes prior to food intake. Study participants will consume two (2) bottles of IP each morning for total daily intake of 120 mL equivalent to CNM-Au8 30mg or 60mg, or placebo.

For the OLE phase, CNM-Au8 will either be supplied the same as described above, or in a single 120 mL bottle at concentrations of 250 µg/mL (30 mg). In this revised volume, only a single bottle of IP will be consumed each morning.

1.2.1 Investigational Product Characteristics

The median diameter of CNM-Au8 nanocrystals is approximately 13 nm, as determined by transmission electron microscopy (TEM). [redacted] of measured nanocrystal diameters and approximate geometrical shapes (e.g., low volume estimate: disc-like approximation, aspect ratio of 0.2; maximum volume estimate: spheroid, aspect ratio of 1.0), each Au nanocrystal has an approximate composition ranging from 13,000 – 66,000 Au atoms per nanocrystal at the 13 nm median diameter with a corresponding molar mass ranging between 2.7 x 10^3 kDA to 1.3 x 10^4 kDA. Summary estimates for CNM-Au8 mass, volume, and particle characteristics are described below in Table 1. Each mL of CNM-Au8 suspension at 500 µg/mL is estimated to contain between 100 - 500 trillion highly faceted Au nanocrystals.

Table 1. Estimated Volume, Mass, and Particle Characteristics of CNM-Au8

Metric (CNM-Au8 500 µg/mL, 60 mL Dose)	**Disc-Like Approximation Minimum (Aspect 0.2)**	**Spherical Approximation Maximum (Aspect 1.0)**
Median Au Nanocrystal Diameter (nm)	13	
Au Nanocrystal Volume (nm^3)	2.3 x 10^2	1.2 x 10^3
Au Nanocrystal Surface Area (nm^2)	3.2 x 10^2	5.3 x 10^2
Au Atoms per Nanocrystal (count)	1.4 x 10^4	6.8 x 10^4
Au Nanocrystal Molecular Weight (kDa)	2.7 x 10^3	1.3 x 10^4
Total Au Nanocrystal Surface Area per mL (cm^2)	3.6 x10^2	1.2 x10^2
Au Nanocrystals per mL CNM-Au8 (count)	1.1 x 10^14	2.3 x 10^13
Au Nanocrystals per 60 mL Dose (count)	3.4 x 10^15	6.8 x 10^14

1.2.2 Preclinical Data Supporting CNM-Au8 in the Treatment of ALS

CNM-Au8 passes into the systemic circulation via intestinal absorption. Cellular uptake of CNM-Au8, following oral administration, has been demonstrated in a variety of cells and tissue matrices including macrophages, oligodendrocytes, neurons, and brain tissue, utilizing Inductively Coupled Plasma Mass Spectrometry (ICP-MS) and high resolution TEM visualization of nanocrystals.

Preclinical characterization of the *in vitro* and *in vivo* properties of CNM-Au8 demonstrated that it acts to protect neuronal populations from chemical, inflammatory, and hypoxic insults through a series of unique catalytic mechanisms involving [redacted] the nicotine adenine dinucleotide redox couple (NAD+/NADH) to boost glycolytic energy production, [redacted] 3) [redacted] the SOD-like inactivation of reactive oxygen species [redacted] to protect cells from oxidative damage and mitochondrial dysfunction.

The nicotinamide adenine dinucleotide redox couple (NAD+, NADH) plays a central role in the energy metabolism of all living cells. NAD+, NADH and their relative intracellular ratio (NAD+/NADH) are linked to regulation of energy homeostasis, neuroprotection, immune function, chromosome stability, DNA repair mechanisms, sleep and circadian rhythms, and longevity (Canto et al., 2015; Imai, 2010a, b; Nikiforov et al., 2015; Ying, 2008).

Fundamentally, NAD+ and NADH are essential coenzymes in the adenosine triphosphate (ATP)-generating reactions driving both glycolysis and oxidative phosphorylation. More specifically, in aerobic glycolysis, NAD+ availability is integral to the enzymatic cascade from glyceraldehyde 3-phosphate via glyceraldehyde phosphate dehydrogenase to 1,3-bisphosphoglyceric acid. Similarly, [redacted] electrons are removed via NADH oxidation in Complex I of the electron transport chain during oxidative phosphorylation, resulting in NAD+, and ultimately these electrons are transferred to a lipid-soluble carrier, ubiquinone.
In addition, NAD+ and its metabolites act as binding substrates for a wide range of proteins including the metabolic and transcription-regulating sirtuins, the poly-ADP-ribose polymerases (PARPs) involved in DNA repair, and the cyclic ADP-ribose synthases (cADPRs) such as CD38 that serve to regulate Ca2+ signaling involved in cell cycle control and insulin signaling (Canto et al., 2015). In the Wallerian Degeneration Slow Dominant Mutant Mouse, heightened expression of NMNAT1, resulting in increased NAD+, has been attributed to protect against the axonal degeneration, otherwise observed in these mice (Sasaki et al., 2009).

Quantitative measurement of regional brain area NAD+ concentrations and redox states in humans has recently been demonstrated using a novel ^{31}P-Magnetic Resonance Spectroscopy (MRS) imaging technique (Chouinard et al., 2017; Lu et al., 2016). In people, brain NAD+/NADH redox potential is inversely correlated with age. ^{31}P-MRS of the visual cortex of young (21-26 year old), middle-aged (33-36 year old), and aged (59-68 year old) healthy individuals demonstrated a linear decline of NAD+ redox potential with increasing age (Zhu et al., 2015). This age-related decrease in cerebral energy metabolism is believed to be associated with the bioenergetic failure underlying many neurodegenerative diseases, because NAD+/NADH redox potential is essential for fundamental ATP-generating processes such as glycolysis and mitochondrial oxidative phosphorylation. These bioenergetic processes power all cellular processes including ‘housekeeping’ functions that are responsible for maintaining overall cellular health including processes of autophagy, apoptosis, and the unfolded protein response (UPR) (Canto et al., 2015; Chua and Tang, 2013a; Villanueva-Paz et al., 2016). Boosting brain NAD+ levels in a mouse model of AD reduced DNA damage, neuroinflammation, and apoptosis while improving cognitive function in multiple behavioral tests and restored hippocampal synaptic plasticity (Hou et al., 2018).

The first demonstration of catalyzed oxidation of NADH to NAD+ by gold nanoparticles was demonstrated by Huang et al. (2005) using a simple cell-free assay. In order to compare the catalytic oxidation rate of clean-surfaced CNM-Au8 against similarly-sized gold nanoparticles made using citrate reduction, the same cell-free assay utilized by Huang et al. was repeated comparing gold nanoparticles sourced from the U.S. National Institutes of Standards and Technology (NIST). The resulting change in the 339 nm NADH absorbance peak was assessed while investigating each of the gold nanoparticles at the same concentrations. In all cases, CNM-Au8 nanocrystals consistently showed significantly superior catalytic activities regardless of size and/or method of preparation of the comparator gold nanoparticles.

Figure 1. CNM-Au8 NADH Catalysis Rates

Figure 1A. CNM-Au8 Catalytic Effects vs. NIST Standard Citrated AuNP

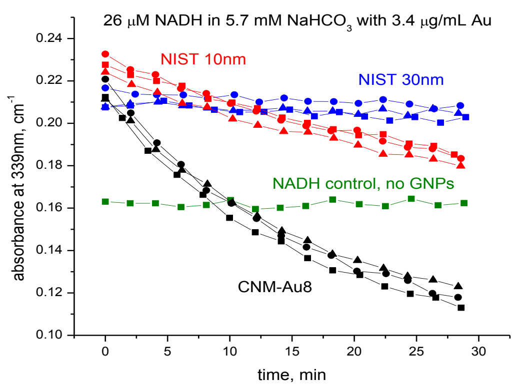
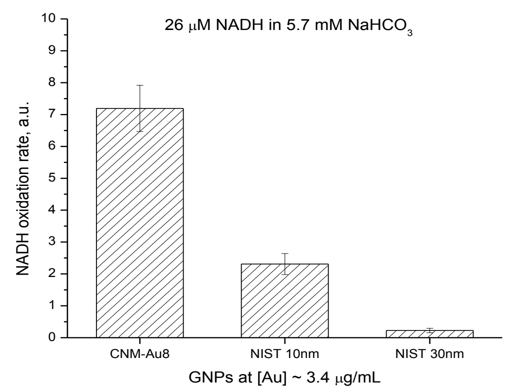


Figure 1B. Relative NADH Oxidation Rates vs. NIST Citrated AuNP

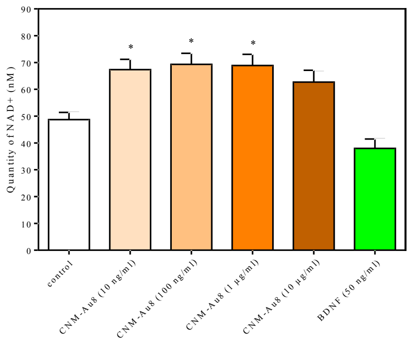


CNM-Au8 treatment of rodent central nervous system cells in vitro also significantly increases levels of both NAD+ and NADH (Figure 2).

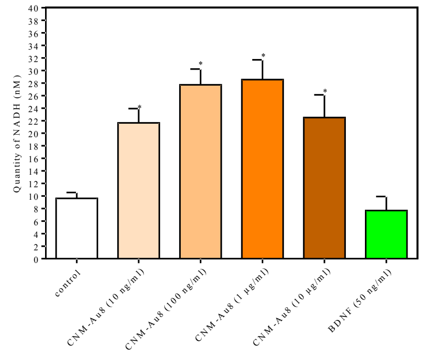
Figure 2. Effects of CNM-Au8 on NAD Levels in Primary Rodent Mesencephalic Cultures

Figure 2A. Effect of CNM-Au8 on NAD+ Levels

Figure 2B. Effect of CNM-Au8 on NADH Levels



*p<0.05 vs control condition One-Way ANOVA followed by Dunnett's test



*p<0.05 vs control condition One-Way ANOVA followed by Dunnett's test

[redacted]

Excessive reactive oxygen exposure disrupts cellular redox homeostasis, damages intracellular organelles as well as DNA and proteins, and may also underpin pathophysiologic mechanisms of

ALS. Super oxide dismutases (SODs) evolved as a component of the cell's antioxidant defense system to regulate the levels of reactive oxygen species (ROS) and prevent damage from oxidative stress. SOD enzymes are found in the cytoplasm and mitochondria where they convert oxygen radicals to O_2 or H_2O_2, which can in turn be converted to water and O_2 by catalases. A dose-dependent reduction of ROS levels was observed in differentiating oligodendrocyte precursor cells in primary culture with CNM-Au8 treatment.

Figure 4 . Effects of CNM-Au8 on SOD and ROS Generation

Figure 4A. Effects on SOD Activity Assay

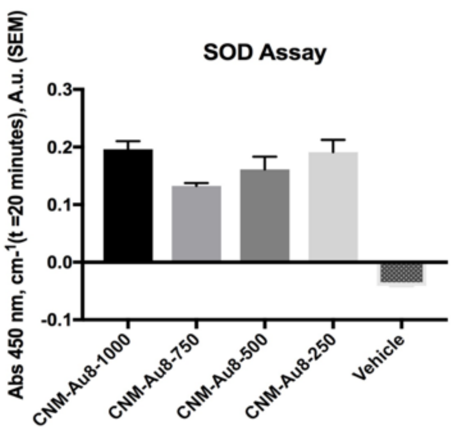


Figure 4B. Effects on ROS Generation in Purified Murine OPC cultures

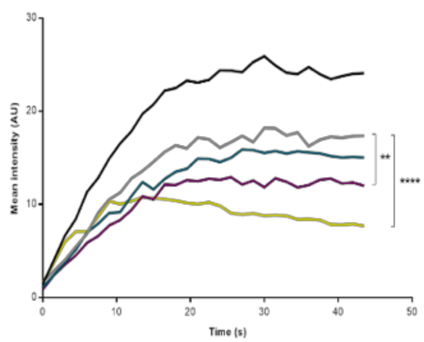
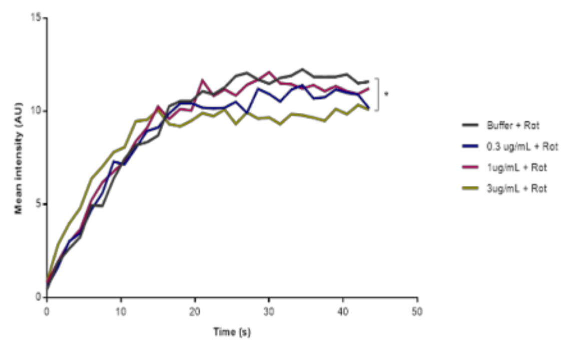
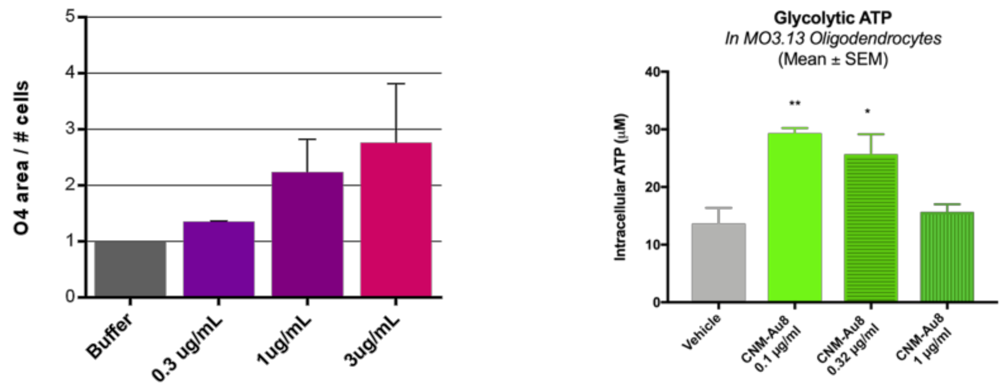


Figure 4C. Effects on ROS Generation in Murine OPC Cultures Plus Rotenone



Demonstration of SOD-like catalytic activity with CNM-Au8 treatment has several important implications. First, neurons and glial cells are energetically demanding and generate large amounts of ROS as byproducts of the energetic processes necessary for their function. Theories of aging and neurodegeneration suggest that as CNS cells age, they are less able to efficiently remove ROS and thereby succumb to the consequences of long-term ROS exposure. As CNS cells are exquisitely sensitive to ROS, they are among the first cell types to degenerate, leading to the cognitive, psychomotor, and movement impairments observed in diseases such as ALS, Alzheimer's disease, and Parkinson's disease (Angelova and Abramov, 2018). In addition, SOD1 may act not only as a regulator of ROS but also as a part of a glucose-oxygen sensing mechanism of a cell to determine whether the cell utilizes oxidative phosphorylation or aerobic glycolysis for ATP production (Reddi and Culotta, 2013). A study of oligodendrocyte (OL) energetics showed that human OLs preferentially use aerobic glycolysis during differentiation and myelination (Rone et al., 2016). Data with CNM-Au8 demonstrate that treatment switches OPCs from proliferation to differentiation, while simultaneously stimulating aerobic glycolysis production of ATP likely through NADH oxidation. Stimulation of SOD-like activity therefore is consistent with a role for CNM-Au8 in regulating cellular bioenergetics.

Figure 5. Effect of CNM-Au8 on OPC Differentiation and ATP Production



Effect of CNM-Au8 on a) OPC differentiation (O4+ cells) in isolated murine OPCs and b) ATP production in MO3.13 oligodendrocytes (72hr). Statistical analysis performed using one-way ANOVA ($p < 0.05$).

Nonclinical Summary of *In Vitro* and *In Vivo* ALS Neuroprotection Studies

1.2.2.1 *In vitro* ALS Neuroprotection Models

CNM-Au8 exhibits neuroprotective effects in a broad range of neural cell types in *in vitro* and *in vivo* preclinical models of ALS.

1.2.2.1.1 CNM-Au8 Protects Primary Rodent Motor Neurons from Excitotoxicity

Excitotoxicity is the pathological process by which neuronal death occurs as a result of excessive stimulation of receptors at excitatory synapses such as the NMDA receptor. Excitotoxicity has been implicated in acute neurological damage from ischemia and traumatic brain injury and in the chronic neurodegeneration in Huntington's disease. Glutamate excitoxicity is also believed to be a significant contributor to motor neuron death in ALS. Excitotoxic neuronal death via over-activation of NMDA receptors contributes to excessive flux of calcium (Ca2+) into the cell. This triggers a range of responses resulting in cell death, including increased oxidative stress, inappropriate activation of proteases such as calpain, dysregulation of Ca2+-related pathways, mitochondrial damage, and the apoptotic cascade.

Determination of the neuroprotective effects of CNM-Au8 on motor neurons in an in vitro glutamate challenge model was carried out in ventral spinal cord cultures from E14 rat embryos (Martinou et al., 1992). On Day 11 after seeding, cultures were pre-treated with CNM-Au8 or vehicle. The positive control riluzole was added for 1 hour of pre-treatment before glutamate addition. On Day 13, glutamate was added to the cultures for 20 minutes, followed by treatment of riluzole or CNM-Au8 for another 48 hours before cells were fixed and stained. Cultures were stained with anti-MAP-2 to assess motor neuron survival and neurite lengths, and also with TDP-43 and Hoechst stain to assess extranuclear TDP43. CNM-Au8 dose-dependently protects motor neurons from glutamate-induced cell death (Figure 6A) while also preserving neurite length (Figure 6B). CNM-Au8 also outperformed riluzole on neurite network preservation. In addition, CNM-Au8 treatment prevented the accumulation of cytoplasmic TDP-43, a hallmark of ALS cytotoxicity (Figure 6D).

Figure 6 . Neuroprotection of Motor Neurons from Glutamate Excitotoxicity by CNM-Au8

Figure 6A. Motor Neuron Survival

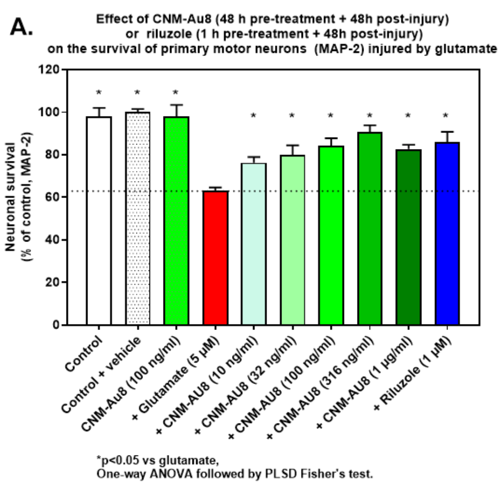


Figure 6B. Motor Neuron Neurtie Network Area

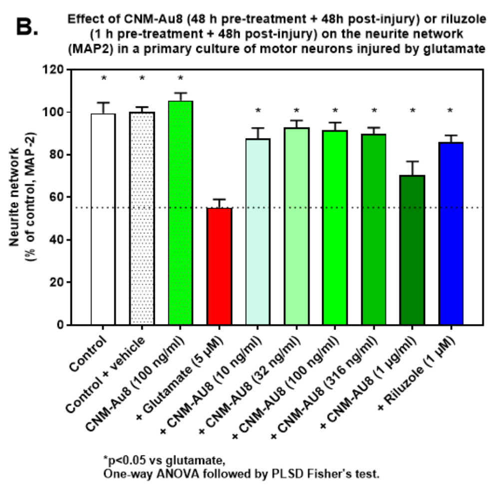


Figure 6B. Effect Of CNM-Au8 on MAP-2 Staining in Rat Hippocampal Neurons Following Glutamate Excitotoxicity Injury

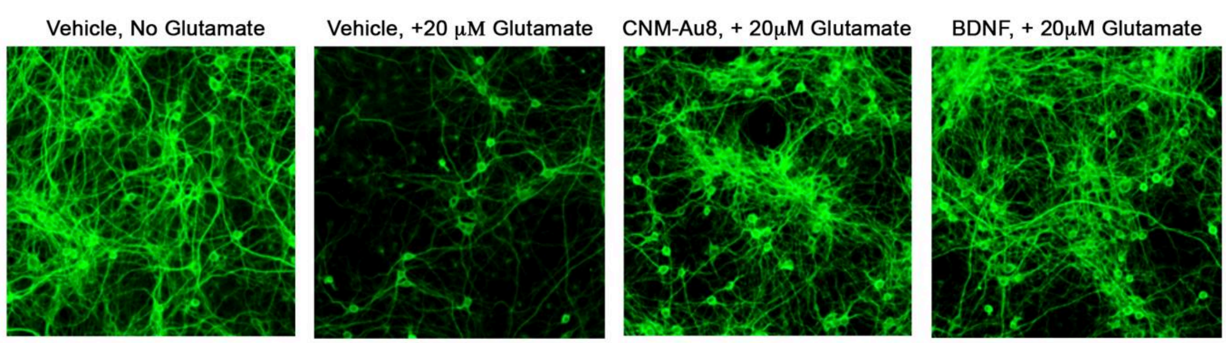
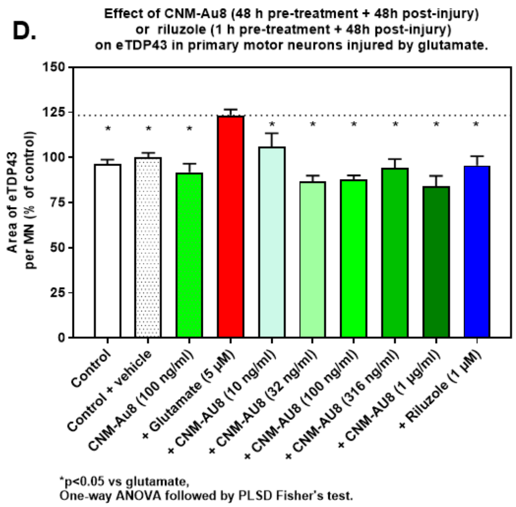


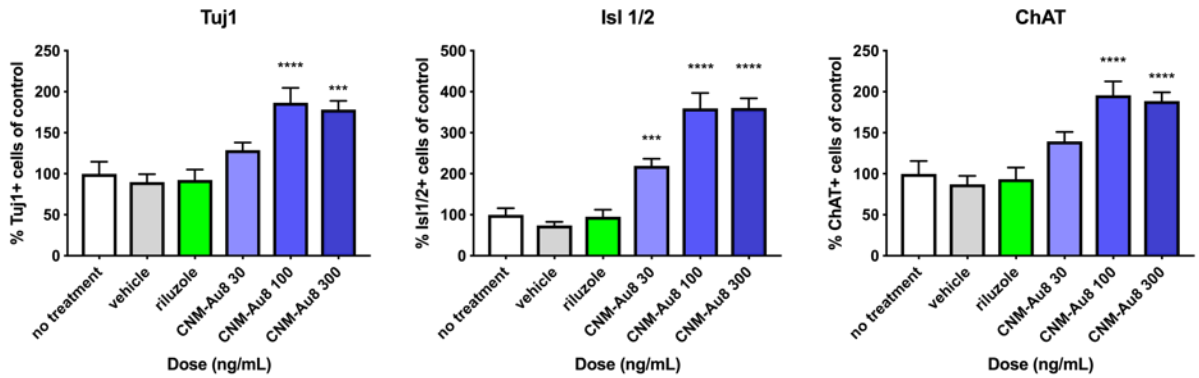
Figure 6D. Reduction in Cytoplasmic TDP43



1.2.2.1.2 CNM-Au8 Protects Normal iPSC Derived Human Motor Neurons From ALS-Participant-Derived Toxic Astrocytes

Astrocytes from both sporadic and familial ALS participants display toxicity against normal human motor neurons (MN) in co-culture (Meyer et al., 2014a). Toxic ALS-participant-derived astrocytes appear to secrete an unidentified toxic factor; and when co-cultured with human MNs, the MNs die (Meyer et al., 2014a). Similar studies have shown that astrocytes from sporadic participants also exhibit toxicity against healthy MNs, likely by similar mechanisms (Haidet-Phillips et al., 2011; Meyer et al., 2014b; Qian et al., 2017). To determine whether CNM-Au8 protects MNs from toxic ALS-participant-derived astrocytes, healthy human iPSC-derived MNs were plated with SOD1A4V astrocytes for 14 days in the presence of various doses of CNM-Au8 or vehicle. Neuroprotection was assessed by quantification of the expression of neuronal markers (Tuj1, Isl1/2, and ChAT) compared to vehicle treated control. Measurement of primary neurite lengths and a neurite network complexity Sholl analysis was also conducted (Sholl, 1953). As shown in Figure 7 below significant neuroprotective dose-dependent effects of CNM-Au8 treatment were consistently observed on survival in the presence of $SOD1^{A4V}$ participant-derived toxic astrocytes.

Figure 7. CNM-Au8 Protects Normal iPSC Derived Human Motor Neurons From ALS-Participant-Derived Toxic Astrocytes



1.2.2.2 *In vivo* ALS Neuroprotection Models

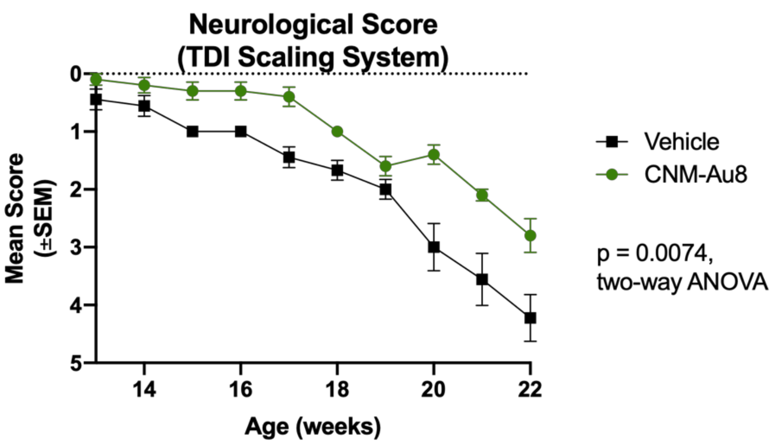
To demonstrate the efficacy of CNM-Au8 treatment in an in vivo model of ALS, two studies were conducted in separate $SOD1^{G93A}$ mouse model strains. These included the rapidly progressive $SOD1^{G93A}$ transgenic line on a mixed SJL/C57BL6 background (Heiman-Patterson et al., 2005) and the more slowly progressive $SOD1^{G93A}$ transgenic line on a congenic C57Bl/6 background (Lutz, 2018) with lifespans of ~157 days compared to ~129 days, respectively. The study using rapidly progressing $SOD1^{G93A}$ animals showed only minor improvement in clinical onset ($p=0.13$, Mantel-Cox test), as well as significant lack of brainstem atrophy ($p<0.05$, unpaired t-test) in the CNM-Au8-treated group (N=15 animals per group;); all other functional measures were not significant (data not shown). In the study of the slower progressing $SOD1^{G93A}$ (N=20 female mice, 10 per group), four-week-old female SOD1 mice were randomly assigned one of two groups with each group balanced for weight.

Beginning at four weeks of age, mice were provided either CNM-Au8 or vehicle ad libitum in their drinking water. Beginning at Week 8 of age each mouse was tested weekly for strength and motor coordination utilizing several measures including the triple horizontal bar hang time test, static rod orientation test, inverted screen hang time test, weight hold test, and home cage wheel activity (average speed) in addition to standard ALS clinical scoring (Deacon, 2013a, 2013b; Hatzipetros et al., 2015). The CNM-Au8-treated group significantly outperforms the vehicle treated group at virtually all timepoints across these tests (Figure 8). In addition, survival benefits have been observed through Week 22 of the study (Figure 9).

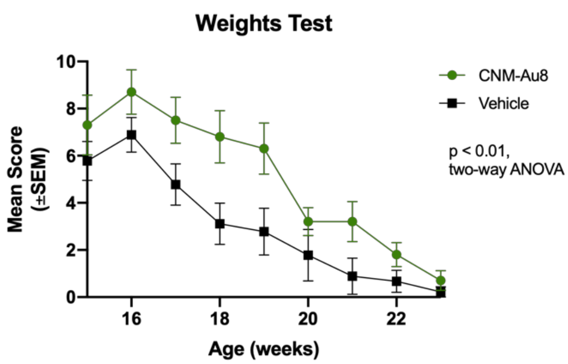
Taken together, these [redacted] data from an [redacted] study in a transgenic ALS mouse model demonstrate preservation of motor function following chronic treatment with CNM-Au8.

Figure 8. Locomotor Functional Efficacy of CNM-Au8 in an SOD1^{G93A} (C57Bl/6 congenic strain) Murine Study

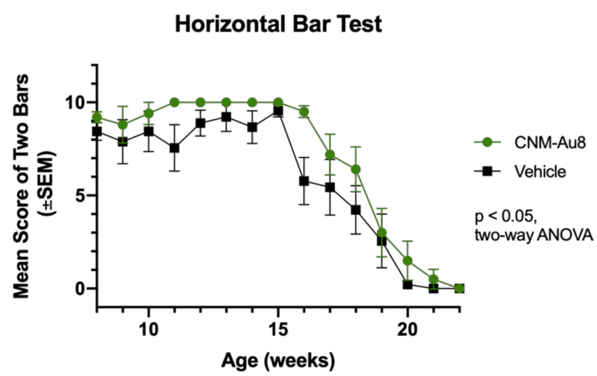
A. ALS TDI Neuro Score



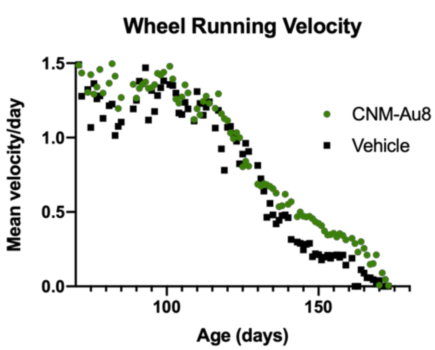
B. Weights Hold Test



C. Horizontal Bar Test



D. Home Wheel Velocity



E. Home Wheel Velocity By Period

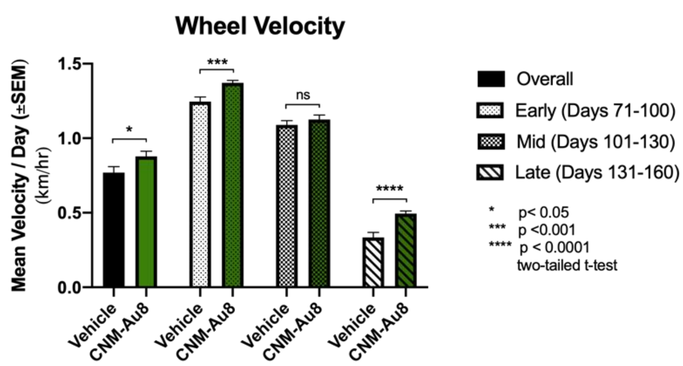
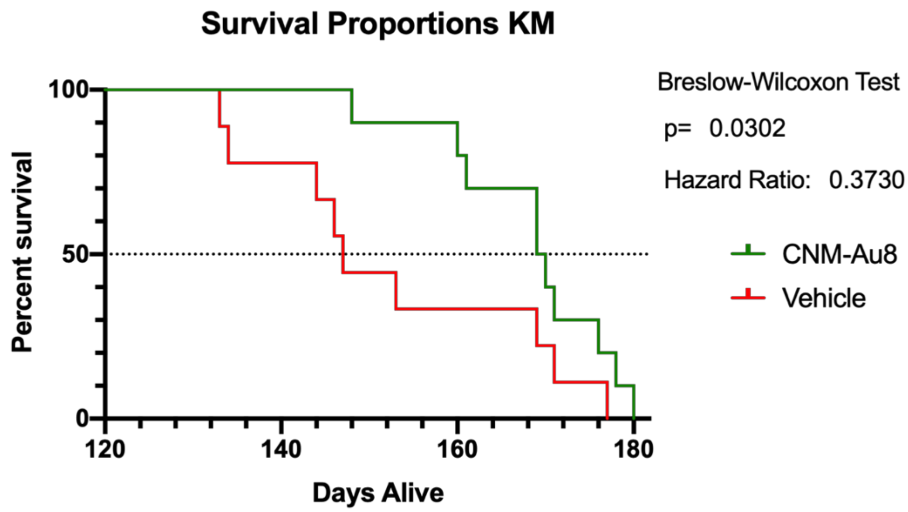


Figure 9. Effect of CNM-Au8 on Survival in a Murine SOD1^{G93A} Model



1.2.3 Summary CNM-Au8 Treatment Rationale for ALS

CNM-Au8-treated motor neurons are protected against cell death by glutamate excitotoxicity in a dose-dependent manner in a primary rat neural-glial co-culture where cytoplasmic levels of TDP-43 are significantly reduced. Aberrant cytoplasmic localization and aggregation of TDP-43 has been observed in over 90% of cases of ALS in humans (Neumann et al. 2006). Further, results from these multiple iPSC human motor neuron studies derived from [redacted], familial, and healthy lines indicate that CNM-Au8 is neuroprotective to MNs challenged with multiple different ALS disease related stressors, namely, excitotoxic and toxic astrocytic stress. Results from the SOD1^{G93A}-overexpression mouse study demonstrate significant improvements in locomotor performance of CNM-Au8-treated mice with a significant survival benefit.

In summary, CNM-Au8 has demonstrated neuroprotection across a range of neuronal cell types, antioxidant activity to counteract the accumulation of ROS in ALS, and preservation of motor function in an ALS mouse model. CNM-Au8 therefore represents an important therapeutic candidate for ALS.

2. OBJECTIVES

2.1 Study Objectives and Endpoints

Primary Efficacy Objective:
To evaluate the efficacy CNM-Au8 as compared to placebo on ALS disease progression.

Secondary Efficacy Objective:

- To evaluate the effect of CNM-Au8 on selected secondary measures of disease progression, including survival.

Safety Objective:

- To evaluate the safety of CNM-Au8 for ALS.

Exploratory Efficacy Objective:

- To evaluate the effect of CNM-Au8 on selected biomarkers and endpoints.

Primary Efficacy Endpoint:
Change in disease severity as measured by the ALS Functional Rating Scale-Revised (ALSFRS-R) total score using a Bayesian repeated measures model that accounts for loss to follow-up due to mortality.

Secondary Efficacy Endpoints:

- Change in respiratory function as assessed by slow vital capacity (SVC).
- Change in muscle strength as measured isometrically using hand-held dynamometry (HHD) and grip strength.
- Survival.

Safety Endpoints:

- Treatment-emergent adverse and serious adverse events.
- Changes in laboratory values and treatment-emergent and clinically significant laboratory abnormalities.
- Changes in ECG parameters and treatment-emergent and clinically significant ECG abnormalities.
- Treatment-emergent suicidal ideation and suicidal behavior.

Exploratory Efficacy Endpoints:

- Changes in quantitative voice characteristics.
- Difference in the proportion of patients experiencing a $\geq$ 6-point decline in the ALSFRS-R between active treatment and placebo from Baseline to Week 24.
- Changes in biofluid biomarkers of neurodegeneration.

- Changes in patient reported outcomes.
- Change in respiratory function as assessed by home spirometry.

3. RSA DESIGN

This study is a multi-center, randomized, placebo-controlled trial, testing two active doses of CNM-Au8 (30 mg, 60 mg), given orally daily versus color-matched placebo. Participants will be randomized 3:1 active (CNM-Au8): placebo. Participants randomized to active will be equally allocated between the CNM-Au8 30 mg and 60 mg doses.

3.1 Scientific Rationale for RSA Design

The comparator for this study is placebo. Placebo will consist of a sodium bicarbonate solution in USP water, color matched to active CNM-Au8. Placebo was chosen as the comparator because there are no highly effective, disease-modifying therapies currently available for people with ALS. Both riluzole and edaravone, the only two therapies currently approved for ALS, are available to participants as concomitant therapy during this study. In this manner, participants are not sacrificing any potential benefit from currently approved therapies to participate in this study.

3.2 End of Participation Definition

A participant is considered to have ended his or her participation in the placebo-controlled period of the Regimen if they:

- Complete planned placebo-controlled period visits, as described in the SOA, including participants on or off study drug
- Early terminate from the study and complete the Early Termination Visit and Follow-Up Phone call as described in Section 6.1.11
- Withdraw consent to continue participation in the study, or are lost to follow-up

If a participant initiates open-label study drug in the OLE period, he or she is considered to have completed his or her participation in the OLE period of the Regimen if they choose to discontinue participation or if all planned OLE period visits, including the last visit or the last scheduled procedure shown in the SOA, have been completed.

3.3 End of Regimen Definition

The end of the placebo-controlled period in a Regimen occurs when all randomized participants have completed their participation in the placebo-controlled period as defined in section 3.2.

The end of the OLE period in a Regimen occurs when all participants who initiated open-label study drug in the OLE period have completed their participation in the OLE period as defined in section 3.2.

4. RSA ENROLLMENT

4.1 Number of Study Participants

Approximately one hundred-sixty (160) participants will be randomized for this Regimen.

4.2 Inclusion and Exclusion Criteria

In order to be randomized to an RSA, participants must meet the Master Protocol eligibility criteria. In addition, participants meeting all of the following inclusion and exclusion criteria will be allowed to enroll in this Regimen:

4.2.1 RSA Inclusion Criteria

Per the Master Protocol.

4.2.2 RSA Exclusion Criteria

1. History of allergy to gold, gold salts, or colloidal gold preparations.

4.3 Treatment Assignment Procedures

Each participant who meets all eligibility criteria for the RSA will be randomized to receive either CNM-Au8 or placebo for approximately 24-weeks of double-blind treatment. An additional period of open-label treatment may be offered to participants following completion of the placebo-controlled portion of the study.

5. INVESTIGATIONAL PRODUCT

5.1 Investigational Product Manufacturer

Manufacturing, testing, product characterization, release of CNM-Au8 will be carried out at the Sponsor's facility at 500 Principio Parkway West, Suite 400, North East, MD 21901, USA.

Table 2. CNM-Au8 Investigational Product Dosing Administration for the Double-Blind Period

Description	CNM-Au8 30 mg	CNM-Au8 60 mg	Matched Placebo
Total Daily Dosage	30 mg	60mg	NA
Concentration	250 µg/mL	500 µg/mL	NA
Volume per Bottle	60 mL	60 mL	60 mL
Bottles per Day	2	2	2
Daily Volume	120 mL	120 mL	120 mL
Route of Administration	Oral	Oral	Oral
Time and Frequency	Once Daily; Same Time Each Day (± 1 hour)	Once Daily; Same Time Each Day (± 1 hour)	Once Daily; Same Time Each Day (± 1 hour)

For the OLE phase of the trial, daily doses may be provided in the format listed in Table 2 above, or, in a single 120 mL bottle, maintaining the same total daily volume of administration, but increasing the volume per bottle to 120 mL and reducing the number of bottles administered per day from 2 to 1.

The investigational product components and quality standards are described in Table 6 below.

Table 3. CNM-Au8 Components and Quality Standards

Description	Quality Standard	CNM-Au8 30 mg	CNM-Au8 60 mg	Matched Placebo	CNM-Au8 30mg (OLE)
$NaHCO_3$ (mg) per bottle	ACS, USP identity	32.8 mg	32.8 mg	32.8 mg	65.5 mg
Au (mg) per Bottle	Conforms with ASTM B562-95 and USP <233>	15 mg	30 mg	NA	30 mg

USP Purified Water	USP for total organic carbon, and conductivity	60 mL	60 mL	60 mL	120 mL

For the OLE phase of the trial, daily doses of CNM-Au8 may be provided in either two 60 mL bottles, or in a single 120 mL bottle.

5.2 Labeling, Packaging, and Resupply

All investigational drug products will be labeled according to applicable local and legislative requirements. Label text will be approved according to the Sponsor's agreed procedures, and a copy of the labels will be made available to the study site upon request. For all study drugs, a system of numbering in accordance with all requirements of Good Manufacturing Practice (GMP) will be used, ensuring that each dose of study drug can be traced back to the respective bulk ware of the ingredients. The Sponsor's Quality Assurance group will maintain lists linking all numbering levels. A complete record of batch numbers and expiry dates of all study treatment as well as the labels will be maintained in the Sponsor study file.

Study drug label may include, but is not limited to, the following information:

- Batch number
- Storage information
- Unique number/code

5.3 Acquisition, Storage, and Preparation

Investigational product will be stored at the investigational site in accordance with Good Clinical Practice (GCP) and GMP requirements and will be inaccessible to unauthorized personnel. A complete record of batch numbers and expiry dates can be found in the regimen industry partner study file; the site-relevant elements of this information will be available in the Investigator site file. The responsible site personnel will confirm receipt of IP via the interactive web response system (IWRS) and will use the IP only within the framework of this clinical study and in accordance with this protocol. Receipt, distribution, return and destruction (if any) of the IP must be properly documented according to the Sponsor's agreed and specified procedures.

All IP must be kept in a locked area with limited access and stored at 15°- 25°C (59°- 77°F). Mean kinetic temperature should not exceed 25°C. Excursions between 2°C and 60°C (36° and 140° F) that may be experienced in pharmacies, hospitals, or warehouses, and during shipping are allowed.

5.4 Study Medication/Intervention, Administration, Escalation, and Duration

Participants will either receive CNM-Au8 30 mg, 60 mg, or color-matched placebo in the study. The Investigational Product (IP) will be self-administered by participants who will be directed to take the IP each day in the morning at approximately the same time, at home at least thirty (30) minutes prior to food intake. The drug formulations will be identical in appearance (size, shape, volume, color) and smell. The packaging and labeling will be designed to maintain blinding to the Site Investigator's team and to participants. There are no visible differences between CNM-Au8 30 mg, 60 mg, and placebo dosing units.

Participants may take the study drug by mouth or through a gastric tube. Participants taking study drug by gastric tube should be directed to first flush the tube with approximately 20-30 mL of distilled water, administer study drug, and then flush the tube post-study drug administration with another 20-30 mL of distilled water.

Based on the supply of the IP provided, the maximum window between in-clinic study visits during the placebo-controlled period may not exceed 64 days. The maximum window between in-clinic study visits during the Open Label Extension period [redacted] may not exceed 64 days for the first 16 weeks, and may not exceed 96 days for the duration of the OLE study period.

5.5 Drug Returns and Destruction

At each in person visit the steps outlined in Manual of Procedures must be followed for study drug accountability and compliance, as well as study drug return and destruction.

Prior to study drug destruction, all used and unused IP requires a second accountability verification to be completed by a different study team member, and both verifications should be documented on the study destruction logs. No study drug may be destroyed on-site until written approval is provided by the study monitoring team. Sites should follow their local drug destruction policies.

5.5 Dosage Adjustment

No dosage adjustments are anticipated during the placebo-controlled period of the study. If participants have difficulty tolerating IP, the Investigator may permit the participants to reduce the daily dose by one (1) bottle. Potential tolerability issues may include gastrointestinal disturbances, headache, somnolence, or other treatment emergent adverse events are not otherwise explained.

Down titration of IP to one (1) bottle daily will only be allowed to address tolerability issues and only with Medical Monitor approval. After a participant has been down titrated to, he/she may be allowed to retry two (2) bottle dose (re-challenge), after a discussion with and approval by the Site Investigator and Medical Monitor. Any dosing changes must be appropriately documented including start date(s) and re-challenge date(s). If the participant cannot tolerate the re-challenge, he/she may remain at the lower dose of one (1) bottle daily. Drug holidays (e.g., treatment-free periods) are not permitted.

During the Open Label Extension, participants will maintain their current blinded dose. Participants previously randomized to placebo during the placebo-controlled period of the study will be re-randomized to receive CNM-Au8 30 mg or 60 mg for the Open Label Extension. All dosing assignments will remain blinded throughout the Open Label Extension. IP tolerability issues encountered during the Open Label Extension will be managed on a case-by-case basis by the Site Investigator in coordination with the Medical Monitor. Following release of the topline results of the blinded duration phase of the trial, the dose of the OLE will be adjusted to 30mg after consultation between Clene and the sponsor. Participants and Investigators will remain blinded to the original randomized dose assignment.

5.6 Justification for Dosage

Dosage selection has been made based on human safety margins from nonclinical toxicology studies and the exposure data from pharmacology studies that demonstrated neuroprotection benefits.

5.6.1. Human Dosing Safety Margins

Based upon the maximum doses (mg/kg/day) evaluated in the nonclinical GLP repeat-dose 21-day toxicokinetic studies, the completed first-in-human Phase 1 study (AU8.1000-14-01) doses of 15, 30, 60, 90 mg provided a safety margin to the NOAEL based on dose ratios (mg/m^2) ranging from 26x – 4x in rodents, and 195x – 32x in canines, as described in the tables below.

Therefore, at the top dose of 90 mg tested in humans, there was a minimum 4x safety margin to the NOAEL in rats.

Table 4. Summary of CNM-Au8 Conversion of Animal 21-Day Repeat Doses To Human Equivalent Dose (HED) Based On Body Surface Area (mg/m2) for 60 kg Human

Species	21-Day NOAEL Dose (mg/kg/day)	21-Day NOAEL Dose (mg/m^2)	Safety Margin Based on Dosing Ratios (mg/m^2) (For 21-Day dosing Studies)			
			15 mg (9.3 mg/m^2)	30 mg (18.5 mg/m^2)	60 mg (37.0 mg/m^2)	90 mg (55.5 mg/m^2)
Rat	40	240	25.9	13.0	6.5	4.3
Canine	90	1800	194.6	97.3	48.6	32.4

Further, when evaluating Au exposure based on the end of study Day 21 AUC(0-24) (ng*hr/mL) in the canine, and rodent studies in comparison with the human 21-day MAD study, the human doses provided an exposure safety margin ranging from 3.3x – 1.6x compared with rodents, and 18.5x – 9.0x compared with canines, as described in the Table 3 below.

Table 5. Summary of CNM-Au8 Exposure Safety Margin Based on End of Study AUC(0-24) ng*hr/mL. Ratio of Animal Toxicokinetic to Human Pharmacokinetic AUC Results From 21-Day Repeat Dose Studies.

Species	21-Day NOAEL Dose (mg/kg/day)	Animal End of Study AUC(0-24) (ng*hr/mL)[a]	Safety Margin Based on Animal/Human AUC(0-24) Ratio (For 21-Day Dosing Studies)			
			Human 15 mg AUC (32.3 ng*hr/mL)	Human 30 mg AUC (41.4 ng*hr/mL)	Human 60 mg AUC (50.3 ng*hr/mL)	Human 90 mg AUC (66.0 ng*hr/mL)
Rat	40	106	3.3	2.6	2.1	1.6
Canine	90	596	18.5	14.4	11.8	9.0

Notes: [a] Average of Male and Female AUC(0-24) ng*hr/mL values at End of Study

CNM-Au8 exposure at the NOAEL at the end of the chronic toxicokinetic studies in the 9-Month canine and 6-Month rodent chronic dosing studies, provided an exposure safety margin ranging from 6.5x – 3.2x, and 13.6x – 6.7x in rodents and canines, respectively in comparison with the human exposures in the 21-day MAD study, as described below in Table 4.

Table 6. Summary of CNM-Au8 Exposure Safety Margin Based on End of Study AUC(0-24) ng*hr/mL Ratio of Chronic Animal Toxicokinetic 6 and 9-Month Rodent and Canine Repeat Dose Studies to Human 21-Day Pharmacokinetic Results

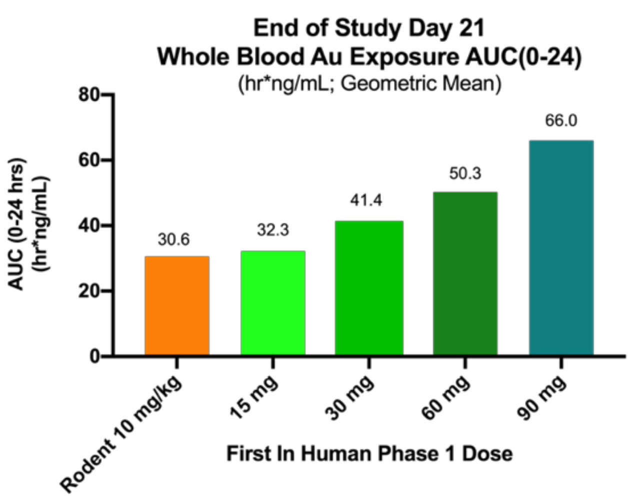
Species (Study)	NOAEL Chronic Dosing (mg/kg/day)	Animal End of Study AUC(0-24) (ng*hr/mL)[a]	Safety Margin Based on Chronic Animal End-of-Study/ Human 21-Day AUC(0-24) Ratio			
			Human 15 mg AUC (32.3 ng*hr/mL)	Human 30 mg AUC (41.4 ng*hr/mL)	Human 60 mg AUC (50.3 ng*hr/mL)	Human 90 mg AUC (66.0 ng*hr/mL)
Rat (6-Month)	40	209	6.5	5.0	4.2	3.2
Canine (9-Month)	10	440	13.6	10.6	8.7	6.7

[a] Average of Male and Female AUC(0-24) ng*hr/mL values at End of Study

5.6.2 Neuroprotection Pharmacologic Exposure Data

Based on the blood gold exposure observed in rodent toxicokinetic (TK) studies where significant remyelination and neuroprotection benefits were observed in preclinical neuroprotection models at mean AUC value of 30.6 (hr*ng/mL), which suggests a minimum human equivalent exposure to achieve a positive bioenergetic response to CNM-Au8 of 15 – 30 mg. Increased AUC exposure of 50.3 (hr*ng/mL) was observed at the CNM-Au8 60 mg dose.

Figure 10. 21-Day AUC Exposure Ranges Between Rodent Toxicokinetic Studies, Preclinical Efficacy Models, and First-In-Human Dosing



In conclusion, based on CNM-Au8 exposures at the NOAELs in the chronic rodent and canine studies along with observed CNM-Au8 exposures in 21-day multiple dosing in humans while also considering safety and tolerability, the dosage chosen for this study in participants with ALS will be CNM-Au8 30 mg/day and 60 mg/day.

5.7 Participant Compliance

Participants will be requested to return any unused IP including empty packaging and used bottles at each study visit. Treatment compliance will be assessed at in clinic study visits through bottle counts and will be documented and summarized by a drug-dispensing log for each participant. During phone visits, drug compliance check-in will be held to ensure participant is taking drug per dose regimen and to note any report of missed doses. Overall treatment compliance with IP intake for the per protocol analysis set should be between 80% and 120% of the planned dose and will be assessed at specified study visits. In the event participants are not compliant within this range, discontinuation may be considered by the Site Investigator in consultation with the Medical Monitor or designee.

The date of dispensing the IP to the participant will be documented in the eCRF. Changes in IP dosing for tolerability issues will also be documented in the eCRF.

If a dose of IP is missed, the participant should take the dose that day and continue with the planned dosing interval for the following day. The dose should not be doubled to make up for a missed dose within the same day.

5.8 Overdose

In the event of overdose, study staff should monitor the study participant and provide supportive care as needed. The SI should also contact the Medical Monitor within twenty-four (24) hours. No prior experience has been documented regarding overdosage in humans.

5.9 Prohibited Medications

There are no prohibited medications for this regimen.

5.10 CNM-Au8 Known Potential Risks and Benefits

5.10.1 Known Potential Risks

5.10.1.1 Immediate Risks

- There has been limited human exposure to clean-surfaced Au nanocrystals. There could be unexpected adverse effects from 24 weeks of exposure to 30 mg or 60 mg of CNM-Au8. These risks have been mitigated by an extensive preclinical toxicology program in multiple species demonstrating No-Adverse-Effect-Levels (NOAEL) at the highest doses studied (up to 10 mg/kg/day in canines; 40 mg/kg/day in rats). Therefore, no Maximum Tolerated Dose (MTD) could be established in these nonclinical studies. In addition, single and multiple ascending dose studies have been performed in healthy human volunteers, which showed no significant adverse events (SAEs) or safety issues. CNM-Au8 was well tolerated at the doses studied up to 90 mg per day over twenty-one days of consecutive dosing.

- There have been no clinical trials of CNM-Au8 in humans with ALS. There is a risk that CNM-Au8 may not be effective and a very small risk that CNM-Au8 could make participants worse. These risks have been mitigated by a robust preclinical program, demonstrating that CNM-Au8 is effective in multiple models of protection from neurodegeneration.

5.10.1.2 Long Range Risks

There has been limited human exposure to CNM-Au8. There could be unexpected adverse effects from chronic exposure to 30 mg or 60 mg of CNM-Au8. Based on rodent and canine toxicokinetic data, when taken orally with daily administration, CNM-Au8 can be expected to be absorbed, and eliminated, relatively slowly with increasing blood and tissue concentrations accumulating over time.

The risks of long-term exposure to CNM-Au8 are mitigated by toxicology studies with chronic dosing, demonstrating no significant toxicities or maximum-tolerated dose.

5.10.1.3 Summary of Participant Risk

Given the potential benefits of CNM-Au8 demonstrated in pre-clinical models of ALS, the safety of CNM-Au8 demonstrated in preclinical toxicology studies and Phase 1 clinical trials in healthy controls, overall, the benefit-risk of CNM-Au8 is positive for participants with ALS.

5.10.2 Known Potential Benefits

As described previously, in preclinical models of ALS, CNM-Au8 has demonstrated efficacy in several relevant models of neurodegeneration. CNM-Au8 has been shown to protect neuronal populations from chemical, inflammatory, and hypoxic insults through a series of unique catalytic mechanisms involving 1) the nicotine adenine dinucleotide redox couple (NAD+/NADH) to boost glycolytic energy production, [redacted] an [redacted] the SOD-like inactivation of reactive oxygen species and nitric oxide to protect cells from oxidative damage and mitochondrial dysfunction. These data suggest that

CNM-Au8 could improve participant functioning and/or prolong life in participants with ALS. The safety of CNM-Au8, at the doses to be tested in this study have been shown to be safe and well tolerated in preclinical toxicology and Phase 1 clinical trials in healthy controls.

5.11 Regimen-Specific Lab Alerts

Site Investigators and the Medical Monitor will receive an alert from the Central Lab for certain lab changes. (See below.) These alerts are informative only, and they do <u>not</u> mean a participant must reduce drug dosage or discontinue study drug. These alerts indicate that the Site Investigator and Medical Monitor should follow up with additional testing and management as clinically indicated. Discontinuation of any participant on the basis of abnormal laboratory findings will be at the discretion of the Site Investigator.

Chemistry Panel Test Name	**Outlier Criteria Warranting Further Investigation**
ALT and AST	>3x ULN
Creatinine	>1.5 x <u>Baseline</u>
Hematology Panel Test Name	**Outlier Criteria Warranting Further Investigation**
Platelet count	<75,000/mm^3

6. REGIMEN SCHEDULE

In addition to procedures in the Master Protocol, the following regimen-specific procedures will be conducted during the study:

- Home Spirometry
    - Note: Home spirometry should be collected within the visit window but will occur while the participant is not in the clinic (at home or other remote location)e.
- ALSAQ-40
- CNS Bulbar Function Scale
- Voice recording
- Smartphone installation and removal
- Whole blood collection for PK and PD analyses
- Plasma collection for PK and PD analyses
- Urine collection for PD analyses

Modifications to Regimen Schedule

Designated visits in the Schedule of Activities (i.e., Week 4, Week 8, and Week 16) may be conducted via telemedicine (or phone if telemedicine is not available) with remote services instead of in-person if needed to protect the safety of the participant due to a pandemic or other reason. If a planned in-clinic visit is conducted via telemedicine (or phone if telemedicine is not available) with remote services, only selected procedures will be performed. Instructions on how to document missed procedures are included in the MOP.

In addition to the procedures in the Master Protocol that should be conducted during the phone or telemedicine and remote visits, the following regimen-specific procedures should be completed:

- Home Spirometry (Week 8 and 16 only)
- Voice Recording
- CNS Bulbar Function Scale (Week 8 and 16 only)

Details on collection of the CNS Bulbar Function Scale, dispensing IP during remote visits, and documenting participants' willingness to participate in OLE are described in the MOP.

Blood samples for PK and PD analyses are **<u>not</u>** collected during the remote visits by the home health agency and this should be recorded as such in the applicable source documentation and EDC.

6.1 Regimen-Specific Screening Visit

This visit will take place in-person after the Master Protocol randomization to a regimen. There are no regimen-specific procedures for this visit. For this regimen, the Regimen-Specific Screening Visit and Baseline Visit should be combined, if possible.

Participants may be required to reconsent to the regimen if new procedures or information is added in the future. Should a participant need to reconsent, this should occur during the participant's next in-person visit. If the participant's next in-clinic visit is conducted remotely, reconsent may also be completed remotely using the following procedures:

1. The site staff sends copy of the informed consent form to the participant.
2. The participant reads through the consent form but does not sign.
3. The Site Investigator, or other study staff member approved and delegated to obtain informed consent, contacts the participant and reviews the informed consent form with the participant.
4. The participant signs the informed consent form and returns the original signed consent form back to the site.
5. Once received at the site, the individual who consented the participant signs the informed consent form.

6.2 Baseline Visit

This visit will take place on Day 0. The following procedures will be performed for the regimen schedule:

- ALSAQ-40
- CNS Bulbar Function Scale
- Install smartphone app
- Home Spirometry
- Voice recording
- Whole blood collection for PK and PD analyses
- Plasma collection for PK and PD analyses
- Urine collection for PD analyses
- Dispense IP
- Administer first dose of IP in clinic *after* all Baseline procedures have been completed
- Remind participant to bring in investigational product to the next visit

6.3 Week 2 Telephone Visit

This visit will take place 14 ± 3 days after the Baseline Visit via telephone. No regimen-specific procedures will be performed at this visit.

6.4 Week 4 and 8 Visits

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

These visits will take place on Days 28 ± 7 and 56 ± 7 days, respectively. The following procedures will be performed for the regimen schedule:

- Home Spirometry [Week 8 Only]
- CNS Bulbar Function Scale [Week 8 only]
- Voice recording
- Plasma collection for PK analyses
- Dispense IP (Week 8 only)
- Remind participant to bring in investigational product to the next visit

6.5 Week 12 Telephone Visit

This visit will take place 84 ± 3 days after the Baseline Visit via telephone. No regimen-specific procedures will be performed at this visit.

6.6 Week 16 Visit

This visit will take place on Day 112 ± 7 days. The following procedures will be performed for the regimen schedule:

- Home Spirometry
- ALSAQ-40
- CNS Bulbar Function Scale
- Voice Recording
- Dispense IP
- Document participant's willingness to participate in the OLE
    - If OLE consent is not obtained at Week 16, it may be obtained at Week 24.

- Remind participant to bring in investigational product to the next visit

6.7 Week 20 Telephone Visit

This visit will take place 140 ± 3 days after the Baseline Visit via telephone. No regimen-specific procedures will be performed at this visit.

6.8 Week 24 Visit or Early Termination Visit

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

This visit will take place on Day 168 ± 7 days. The following procedures will be performed for the regimen schedule:

- Home Spirometry
- ALSAQ-40
- CNS Bulbar Function Scale
- Voice recording
- Uninstall smartphone app
- Whole blood collection for PK and PD analyses
- Plasma collection for PD analyses
- Urine collection for PD analyses
- Dispense IP [See note below.]
- Remind participant to bring investigational product to the next visit (only if continuing in OLE)

Note: Drug is only dispensed at this visit if the participant is continuing in the Open Label Extension.

6.9 Follow-Up Safety Call

Participants will have a Follow-Up Safety Call 28±3 days after their last dose of study drug. Only those participants NOT continuing on in the Open Label Extension will have the Follow-Up Safety Call at the end of their participation in the placebo-controlled portion of the trial. The following procedures will be performed:

- Assess and document AEs

6.10 Process for Early Terminations

Participants who early terminate from the study and do not complete the protocol per ITT will be asked to be seen for an in-person Early Termination Visit and complete a Follow-Up Safety Call. If a participant is not able to be seen in-person, safety assessments and others that can be conducted remotely should be performed.

The in-person Early Termination Visit should be scheduled as soon as possible after a participant early terminates. If the participant early terminates or withdraws consent during the placebo-controlled portion of the Regimen, all assessments that are collected at the Week 24 in-clinic visit should be conducted. If the participant early terminates or withdraws consent during the OLE phase, all assessments that are collected at the OLE Week 52 in-clinic visit should be conducted. The Follow-Up Safety Call should be completed approximately 28 days after the last dose of study drug.

If the Early Termination Visit occurs approximately 28±3 days after the last dose of study drug, the information for the Follow-Up Safety Call can be collected during the Early Termination Visit, and a separate Follow-Up Safety Call does not need to be completed. If the in-person Early Termination Visit does not occur within 28±3 days of the last dose of study drug, the Follow-Up Safety Call should occur approximately 28 days after the last dose of study drug and the Early Termination Visit will be completed after the Follow-Up Safety Call.

If a participant decides to discontinue study drug, but will complete the protocol, an in-person Early Termination Visit and Follow-Up Safety Call is not necessary.

6.11 Open Label Extension

Participants who have completed the placebo-controlled portion of the trial on drug. will be eligible to continue in the Open Label Extension (OLE) as outlined in the SOA. Participants will first be asked about their desire to continue in the OLE at the Regimen-Specific Screening Visit. They will also be asked to *re-confirm* whether they want to continue in the OLE at the Week 16 Visit of the placebo-controlled period. The OLE, CNM-Au8 will be provided by Clene Nanomedicine, until the primary results of the 24 week double-blind portion of the study are available, or Clene or sponsor terminate support or development of CNM-Au8 for ALS.

Modifications to OLE Schedule

Designated visits in the Schedule of Activities for the OLE (i.e. Week 4, Week 8, Week 16, Week 28, and Week 40) may be conducted via telemedicine (or phone if telemedicine is not available) with remote services instead of in-person if needed to protect the safety of the participant due to a pandemic or other reason. If a planned in-clinic visit is conducted via telemedicine (or phone if telemedicine is not available) with remote services, only selected procedures will be performed. Instructions on how to document missed procedures are included in the Manual of Procedures.

In addition to the procedures in the Master Protocol that should be conducted during the phone or telemedicine and remote visits, the following regimen-specific procedures should be completed:

- Home Spirometry
- CNS Bulbar Function Scale (Not done at OLE Week 4).

Blood and urine samples for PK and PD analysis are **not** collected during the remote visits by the home health agency, and this should be recorded as such in the applicable source documentation and EDC.

6.11.1 Week 2 OLE Telephone Visit

This visit will take place via telephone 14 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Drug Compliance Check-In
- Remind participant to bring investigational product to the next visit

6.11.2 Week 4 OLE Visit

This visit will take place in-person 28 ± 10 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Collect Home Spirometry
- Administer ALSFRS-R questionnaire
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire

- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.3 Week 8 OLE Visit

This visit will take place in-person at 56 ± 7 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Collect Home Spirometry
- Administer ALSFRS-R questionnaire
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Dispense investigational product to participant
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.4 Week 12 OLE Telephone Visit

This visit will take place via telephone at 84 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Administer ALSFRS-R questionnaire
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Drug Compliance Check-In
- Remind participant to bring investigational product to the next visit

6.11.5 Week 16 OLE Visit

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

This visit will take place in-person at 112 ± 7 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Collect vital signs including weight
- Perform SVC
- Collect Home Spirometry
- Administer ALSFRS-R questionnaire
- CNS Bulbar Function Scale
- Collect blood samples for Clinical Safety Labs and, for WOCBP, for pregnancy test if applicable
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Administer the C-SSRS Since Last Visit questionnaire
- Collect urine sample biomarker analyses
- Collect blood sample for biomarker analyses
- Whole blood PK and PD collection
- Plasma PD collection
- PD urine collection
- Dispense investigational product to participant
- Perform investigational product compliance
- Remind participant to bring investigational product to the next visit

6.11.6 Week 20 OLE Telephone Visit

This visit will take place via telephone at 140 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Administer ALSFRS-R questionnaire
- Review concomitant medications
- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Drug Compliance Check-In
- Remind participant to bring investigational product to the next visit

6.11.7 Week 24 OLE Telephone Visit

This visit will take place via telephone at 168 ± 3 days after the Week 24 Visit of the placebo-controlled portion of the trial. The following procedures will be performed:

- Administer ALSFRS-R questionnaire
- Review concomitant medications

- Assess and document AEs, including Key Study Events (*see section 10.3 of Master Protocol*)
- Drug Compliance Check-In
- Remind participant to bring investigational product to the next visit

6.11.8 Week 28 and Q12 Week OLE Visits

Participants should be instructed to hold study drug on the day of the study visit. Study drug should not be taken until after study visit procedures are complete.

The Week 28 OLE Visit will take place in-person 196 ± 14 days after the Week 24 Visit of the placebo-controlled portion of the trial. The procedures listed in the SoA should be performed and participants should be provided with 12 weeks of drug and instructions for dosing. Following the Week 28 OLE Visit, visit will occur every 12 weeks ± 14 days.

7 OUTCOME MEASURES AND ASSESSMENTS

For all assessments listed below, please refer to the Manual of Procedures for detailed instructions.

7.1 Voice Analysis

In addition to the scheduled in clinic voice recordings, voice samples will be collected twice per week, using an app installed on either an android or iOS based smartphone. The app characterizes ambient noise, then asks patients to perform a set of speaking tasks: reading sentences -- 5 fixed and 5 chosen at random from a large sentence bank-- repeating a consonant-vowel sequence, producing a sustained phonation, and counting on a single breath. Voice signals are uploaded to a HIPAA-compliant web server, where an AI-based analysis identifies relevant vocal attributes. Quality control (QC) of individual samples will occur by evaluation of voice records by trained personnel.

The voice analysis app is only available in English, therefore participants who do not speak English should not complete the voice recording. Caregivers cannot provide language assistance when the participant is completing the voice recording.

7.2 ALSAQ-40

The Amyotrophic Lateral Sclerosis Assessment Questionnaire-40 (ALSAQ-40) is a patient self-report health status patient-reported outcome. The ALSAQ-40 consists of forty questions that are specifically used to measure the subjective well-being of patients with ALS and motor neuron disease.

Participants will be handed the questionnaire and asked to write their answers themselves. Caregivers can also help, if needed.

7.3 Center for Neurologic Study Bulbar Function Scale

The Center for Neurologic Study Bulbar Function Scale (CNS-BFS) is a patient self-report scale that has been developed for use as an endpoint in clinical trials and as a clinical measure for evaluating and following ALS patients. The CNS-BFS consists of three domains (swallowing, speech, and salivation), which are assessed with a 21-question, self-report questionnaire.

Participants will be handed the questionnaire and asked to write their answers themselves. Caregivers can also help, if needed.

Instructions on administering the questionnaire during a phone or telemedicine visit will be included in the MOP.

7.4 Home Spirometry

Remote/home-based forced vital capacity will be measured with the MIR Spirobank Smart spirometer. Instructions for use will be provided to the participant. The participant will perform the vital capacity maneuver at home (or other remote location) with real time video coaching (or phone coaching, if video is not available) by the evaluator. Three to five vital capacity maneuvers will be performed, consistent with the manner vital capacity is obtained in clinic.

8 BIOFLUID COLLECTION

8.1 Pharmacokinetic (PK) Assessments

8.1.1 Whole Blood PK Assessments of CNM-Au8

Samples for the measurement of whole blood concentrations of Au will be collected before (pre-dose) administration of investigational product per the Schedule of Assessments. PK collection and processing parameters are specified in the RSA laboratory manual.

PK analyses of CNM-Au8 will be specified in a separate PK analysis plan.

PK blood samples will be shipped to and analyzed under the responsibility of the Clene Nanomedicine, Inc.'s bioanalytical laboratory at:

Clene Nanomedicine, Inc.
Bioanalytics Laboratory
500 Principio Parkway West, Suite 400 North East, MD 21901-2912
USA

8.1.2 Plasma PK Assessments of Riluzole

Samples for the measurement of plasma concentrations of riluzole will be collected before (pre-dose) administration of both riluzole and the IP per the Schedule of Assessments. Riluzole plasma PK collection and processing parameters are specified in the laboratory manual.

Population PK analyses of riluzole will be specified in the separate PK analysis plan. The first forty (40) participants taking riluzole and randomized to Regimen C (CNM-Au8) to reach the Week 8 visit will be evaluated by the DSMB, or unblinded DSMB PK designee, to determine whether there are any changes in riluzole population PK parameters following treatment with CNM-Au8 versus placebo at the Week 4 and Week 8 timepoints. The unblinded riluzole population PK analyses and designee will be implemented as per the DSMB charter.

Riluzole PK plasma samples will be shipped to and analyzed under the responsibility of Covance Central Laboratory Services LP.

8.2 Pharmacodynamic (PD) Assessments

Plasma, whole blood, and urine samples for pharmacodynamic measurements will be collected before (pre-dose) administration of the investigational product per the Schedule of Assessments. PD collection and processing parameters are specified in the laboratory manual.

PD plasma, whole blood, and urine samples will be shipped to the Clene Nanomedicine, Inc.'s bioanalytical laboratory at the address noted above for long-term storage. PD analyses will be conducted with external vendors and specified in a separate PD analysis plan.

9 REGIMEN-SPECIFIC STATISTICAL CONSIDERATIONS

9.1 Deviations from the Default Master Protocol Trial Design

The statistical design for this regimen will be in accordance with the default statistical design described in Appendix I of the master protocol with only one deviation. This regimen will not include interim analyses for early success.

9.2 Regimen Specific Operating Characteristics

Clinical trial simulation is used to quantify operating characteristics for this regimen (refer to the regimen SAP for further details).

9.3 Sharing of Controls from other Regimens

The primary analysis of this regimen will include sharing of all controls from the other regimens. This is justified by the minor differences in inclusion/exclusion criteria of the RSA, such that there are no expected systematic differences in the primary endpoint between the controls across regimens.

APPENDIX I: THE ALSAQ-40

ALSAQ-40

Please complete this questionnaire as soon as possible. If you have any difficulties filling in this questionnaire by yourself, please have someone help you. However it is **your** responses that we are interested in.

The questionnaire consists of a number of statements about difficulties that you may have experienced **during the last 2 weeks.** There are no right or wrong answers: your first response is likely to be the most accurate for you. **Please check the box that best describes your own experiences or feelings.**

Please answer every question even though some may seem very similar to others, or may not seem relevant to you.

All the information you provide is **confidential.**

The following statements all refer to difficulties that you may have had **during the last 2 weeks**. Please indicate, by checking the appropriate box, how often the following statements have been true for you.

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot walk at all
please check **Always/cannot walk at all.**

How often <u>during the last 2 weeks</u> have the following been true?

*Please check **one box** for each question.*

	Never	Rarely	Some-times	Often	Always or cannot walk at all
1. I have found it difficult to walk short distances, e.g. around the house.	☐	☐	☐	☐	☐
2. I have fallen over while walking.	☐	☐	☐	☐	☐
3. I have stumbled or tripped while walking.	☐	☐	☐	☐	☐
4. I have lost my balance while walking.	☐	☐	☐	☐	☐
5. I have had to concentrate while walking.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you are not able to perform the activity at all please check **Always/cannot at all**

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
6. Walking had worn me out.	☐	☐	☐	☐	☐
7. I have had pains in my legs while walking.	☐	☐	☐	☐	☐
8. I have found it difficult to go up and down the stairs.	☐	☐	☐	☐	☐
9. I have found it difficult to stand up.	☐	☐	☐	☐	☐
10. I have found it difficult to move from sitting in a chair to standing upright.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all *please check* **Always/cannot do at all.**

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
11. I have had difficulty using my arms and hands.	☐	☐	☐	☐	☐
12. I have found turning and moving in bed difficult.	☐	☐	☐	☐	☐
13. I have had difficulty picking things up.	☐	☐	☐	☐	☐
14. I have had difficulty holding books or newspapers, or turning pages.	☐	☐	☐	☐	☐
15. I have had difficulty writing clearly.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all
please check **Always/cannot do at all.**

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
16. I have found it difficult to do jobs around the house.	☐	☐	☐	☐	☐
17. I have found it difficult to feed myself.	☐	☐	☐	☐	☐
18. I have had difficulty combing my hair or brushing and/or flossing my teeth.	☐	☐	☐	☐	☐
19. I have had difficulty getting dressed.	☐	☐	☐	☐	☐
20. I have had difficulty washing at the bathroom sink.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all please check **Always/cannot do at all.**

How often <u>during the last 2 weeks</u> have the following been true?

*Please check **one box** for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
21. I have had difficulty swallowing.	☐	☐	☐	☐	☐
22. I have had difficulty eating solid food.	☐	☐	☐	☐	☐
23. I have had difficulty drinking liquids.	☐	☐	☐	☐	☐
24. I have had difficulty participating in conversations.	☐	☐	☐	☐	☐
25. I have felt that my speech has not been easy to understand.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

If you cannot do the activity at all please check **Always/cannot do at all.**

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always or cannot do at all
26. I have stuttered or slurred my speech.	☐	☐	☐	☐	☐
27. I have had to talk very slowly.	☐	☐	☐	☐	☐
28. I have talked less than I used to do.	☐	☐	☐	☐	☐
29. I have been frustrated with my speech.	☐	☐	☐	☐	☐
30. I have felt self-conscious about my speech.	☐	☐	☐	☐	☐

Please make sure that you have checked ***one box for each question*** *before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

How often <u>during the last 2 weeks</u> have the following been true?

*Please check **one box** for each question*

	Never	Rarely	Some-times	Often	Always
31. I have felt lonely.	☐	☐	☐	☐	☐
32. I have been bored.	☐	☐	☐	☐	☐
33. I have felt embarrassed in social situations.	☐	☐	☐	☐	☐
34. I have felt hopeless about the future.	☐	☐	☐	☐	☐
35. I have worried that I am a burden to other people.	☐	☐	☐	☐	☐

*Please make sure that you have checked **one box for each question** before going on to the next page.*

The following statements all refer to certain difficulties that you may have had <u>during the last 2 weeks.</u> Please indicate, by checking the appropriate box, how often the following statements have been true for you.

How often <u>during the last 2 weeks</u> have the following been true?

Please check ***one box*** *for each question*

	Never	Rarely	Some-times	Often	Always
36. I have wondered why I keep going.	☐	☐	☐	☐	☐
37. I have felt angry because of the disease.	☐	☐	☐	☐	☐
38. I have felt depressed.	☐	☐	☐	☐	☐
39. I have worried about how the disease will affect me in the future.	☐	☐	☐	☐	☐
40. I have felt as if I have lost my independence	☐	☐	☐	☐	☐

Please make sure that you have checked one ***box for each question.***

Thank you for completing this questionnaire.

APPENDIX II: THE BULBAR FUNCTION SCALE (CNS-BFS)

BULBAR FUNCTION SCALE (CNS-BFS)						
SIALORRHEA	**Does Not Apply (1)**	**Applies Rarely (2)**	**Applies Occasionally (3)**	**Applies Frequently (4)**	**Applies Most of the Time (5)**	
1. Excessive saliva is a concern to me.	❍	❍	❍	❍	❍	
2. I take medication to control drooling.	❍	❍	❍	❍	❍	
3. Saliva causes me to gag or choke.	❍	❍	❍	❍	❍	
4. Drooling causes me to be frustrated or embarrassed.	❍	❍	❍	❍	❍	
5. In the morning I notice saliva on my pillow.	❍	❍	❍	❍	❍	
6. My mouth needs to be dabbed to prevent drooling.	❍	❍	❍	❍	❍	
7. My secretions are not manageable.	❍	❍	❍	❍	❍	
				TOTAL Sialorrhea Score: _____		
SPEECH	**Does Not Apply (1)**	**Applies Rarely (2)**	**Applies Occasionally (3)**	**Applies Frequently (4)**	**Applies Most of the Time (5)**	**Unable to Communicate by Speaking (6)**
1. My speech is difficult to understand.	❍	❍	❍	❍	❍	❍
2. To be understood I repeat myself.	❍	❍	❍	❍	❍	❍
3. People who understand me tell other people what I said.	❍	❍	❍	❍	❍	❍
4. To communicate I write things down or use devices such as a computer.	❍	❍	❍	❍	❍	❍
5. I am talking less because it takes so much effort to speak.	❍	❍	❍	❍	❍	❍

6. My speech is slower than usual.	❍	❍	❍	❍	❍	❍
7. It is hard for people to hear me.	❍	❍	❍	❍	❍	❍
				TOTAL Speech Score: _____		
SWALLOWING	**Does Not Apply (1)**	**Applies Rarely (2)**	**Applies Occasionally (3)**	**Applies Frequently (4)**	**Applies Most of the Time (5)**	
☐ Feeding tube is in place						
1. Swallowing is a problem.	❍	❍	❍	❍	❍	
2. Cutting my food makes it easier to chew and swallow.	❍	❍	❍	❍	❍	
3. To get food down I have switched to a soft diet.	❍	❍	❍	❍	❍	
4. After swallowing I gag or choke.	❍	❍	❍	❍	❍	
5. It takes longer to eat.	❍	❍	❍	❍	❍	
6. My weight is dropping because I can't eat normally.	❍	❍	❍	❍	❍	
7. Food gets stuck in my throat.	❍	❍	❍	❍	❍	
				TOTAL Swallowing Score: _____		
				OVERALL SCORE: _____		

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